

Data visualisation and exploration

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Good Practices & Common Pitfalls

Session 2

Good Practices & Common Pitfalls

Session 2

Objectives:

- Identify common mistakes in data visualizations (e.g., misleading axes, chartjunk, poor color use).
- Choose appropriate chart types
- Evaluate and critique real-world.
- Understand how design elements (like bins, axes, and labels) affect interpretation

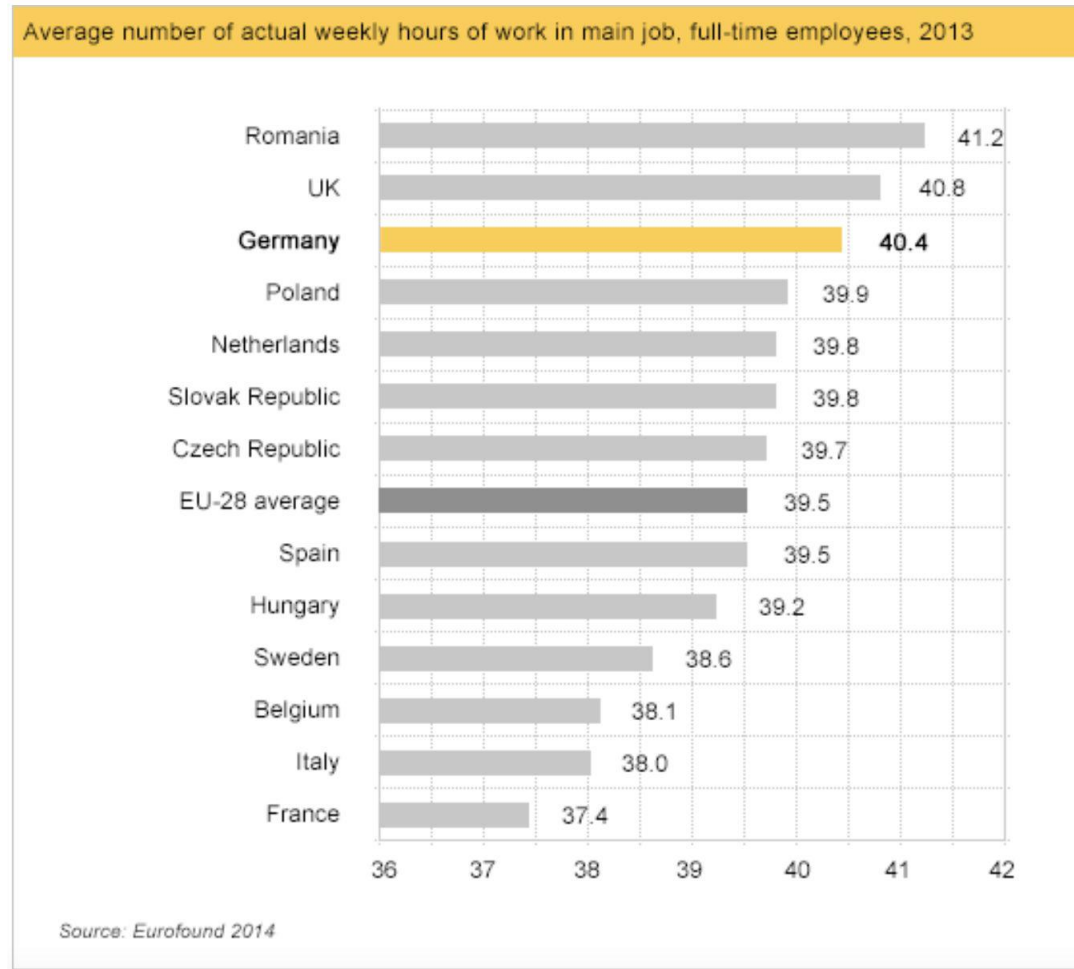
Bar charts

A bar graph plots data with the help of bars, which represent value on the y-axis and category on the x-axis. The main purpose of bar charts is to provide comparisons between categories

A bar chart is used when you want to show a distribution of data points or perform a comparison of metric values across different subgroups of your data. From a bar chart, we can see which groups are highest or most common, and how other groups compare against the others.

Bar charts

What conclusions do you draw from this graph?

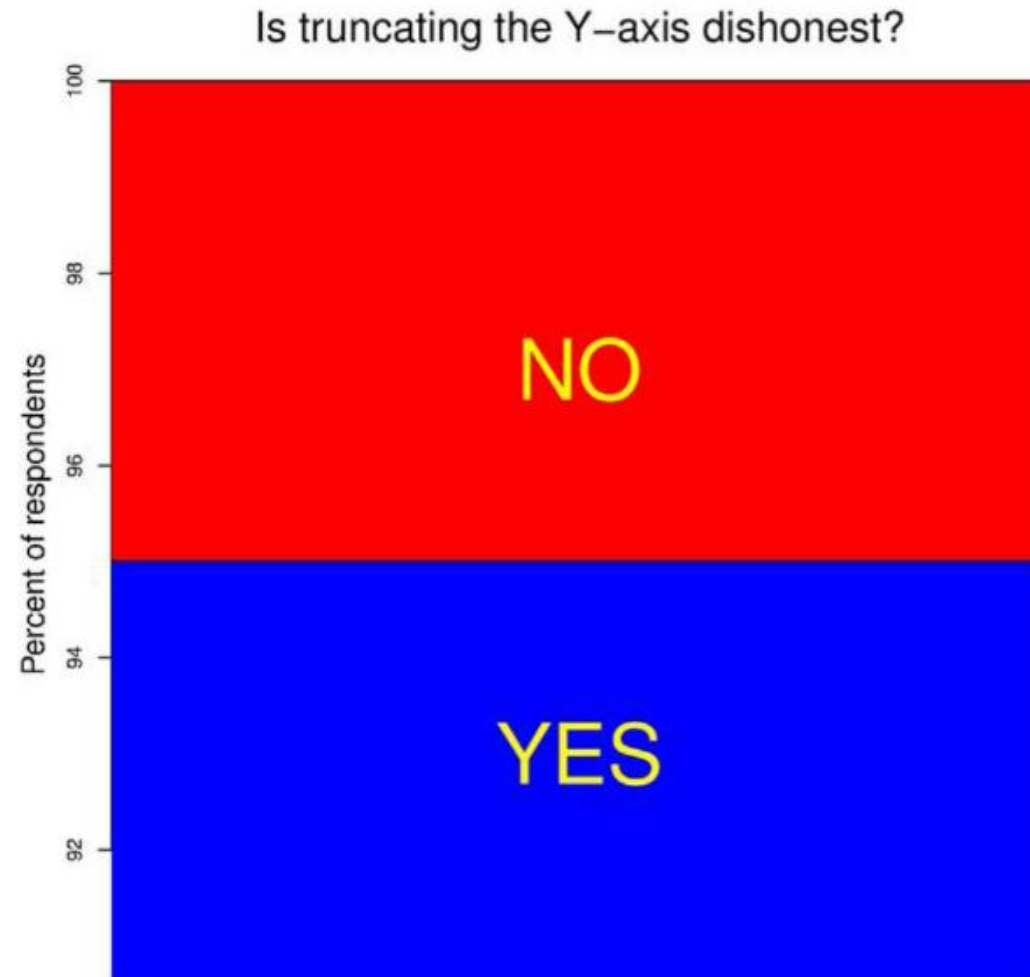


Bar charts

And from this one?



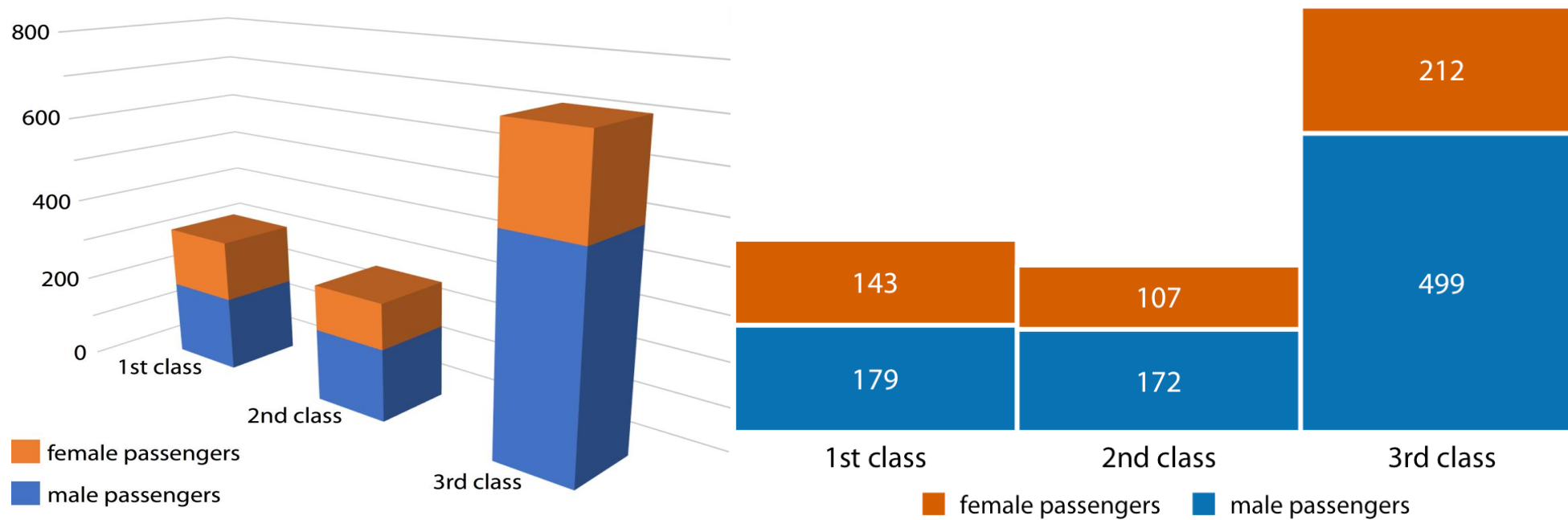
Graphical representation



Bar charts

1) Always start y axis from zero: bar chart emphasizes the absolute **magnitude of the values** associated with each category

Bar charts

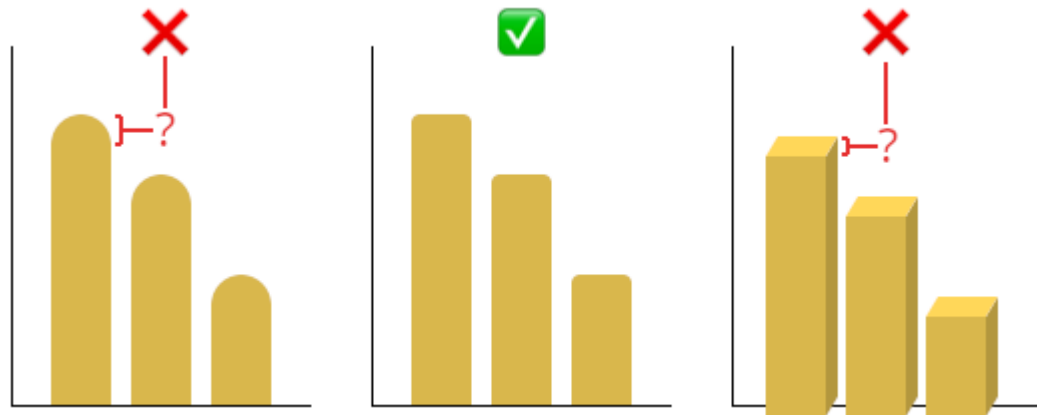


Bar charts

- 1) **Always start y axis from zero:** bar chart emphasizes the absolute **magnitude of the values** associated with each category
- 2) **Do not use 3D:** Difficult to read the right values

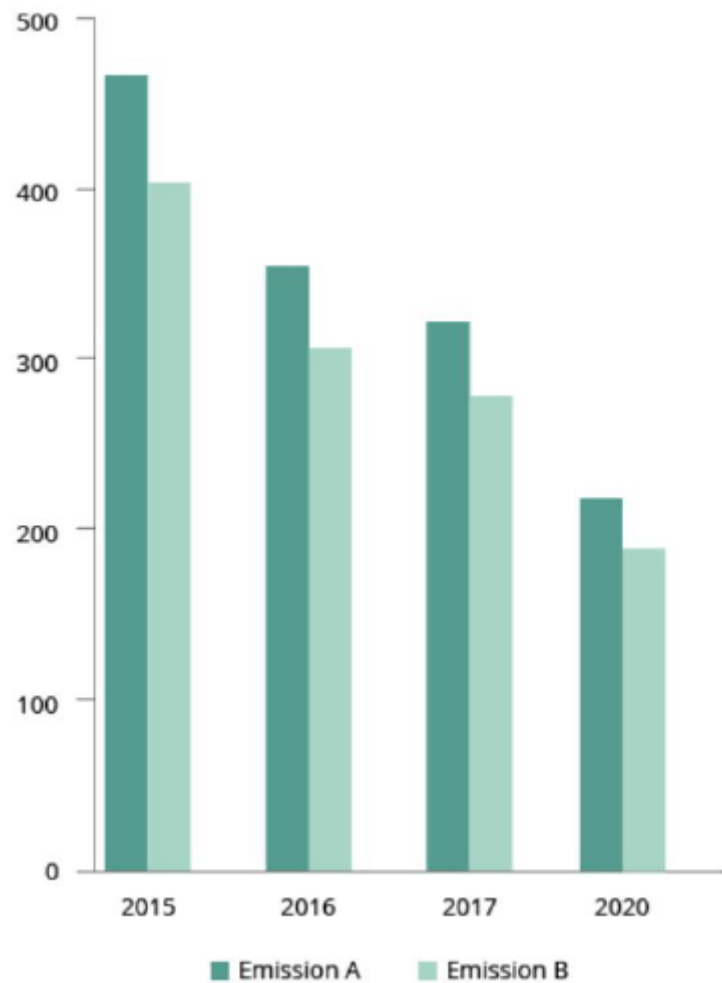
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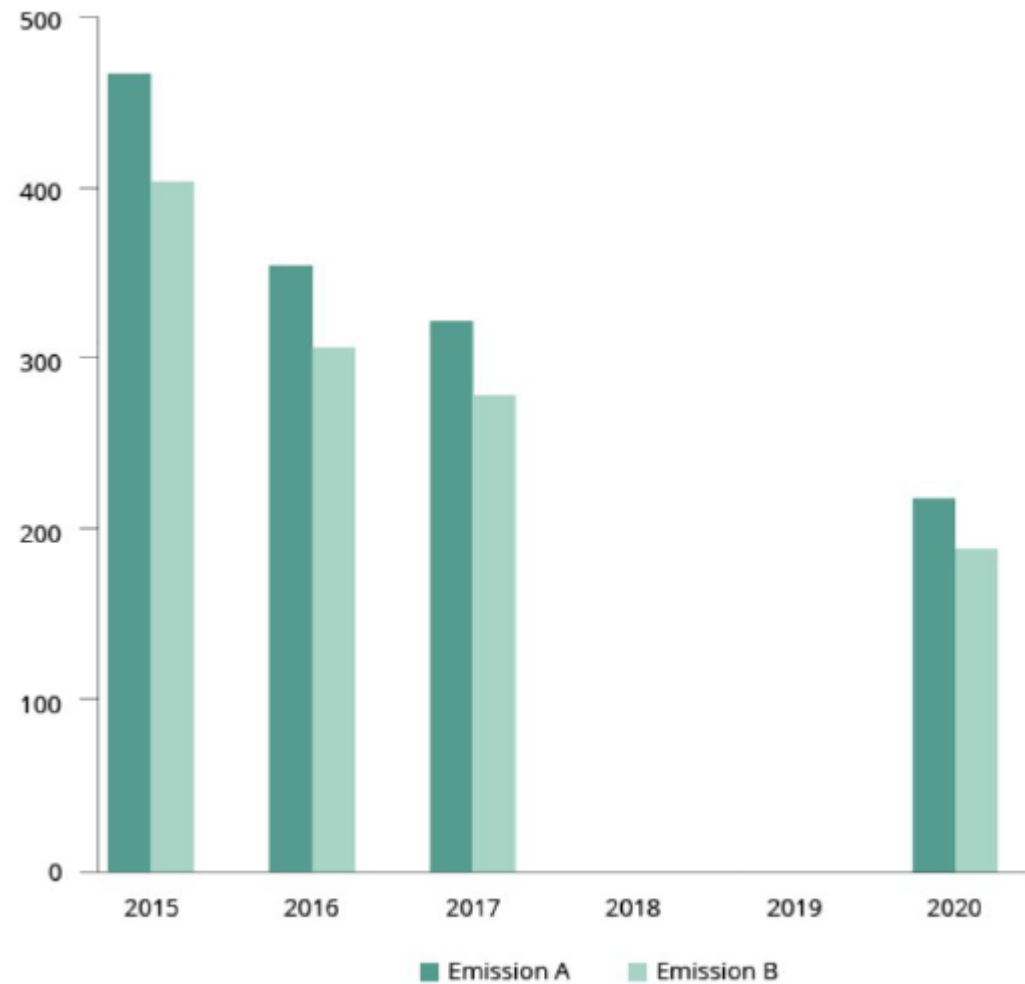


Bar charts

What do you see?



Bar charts

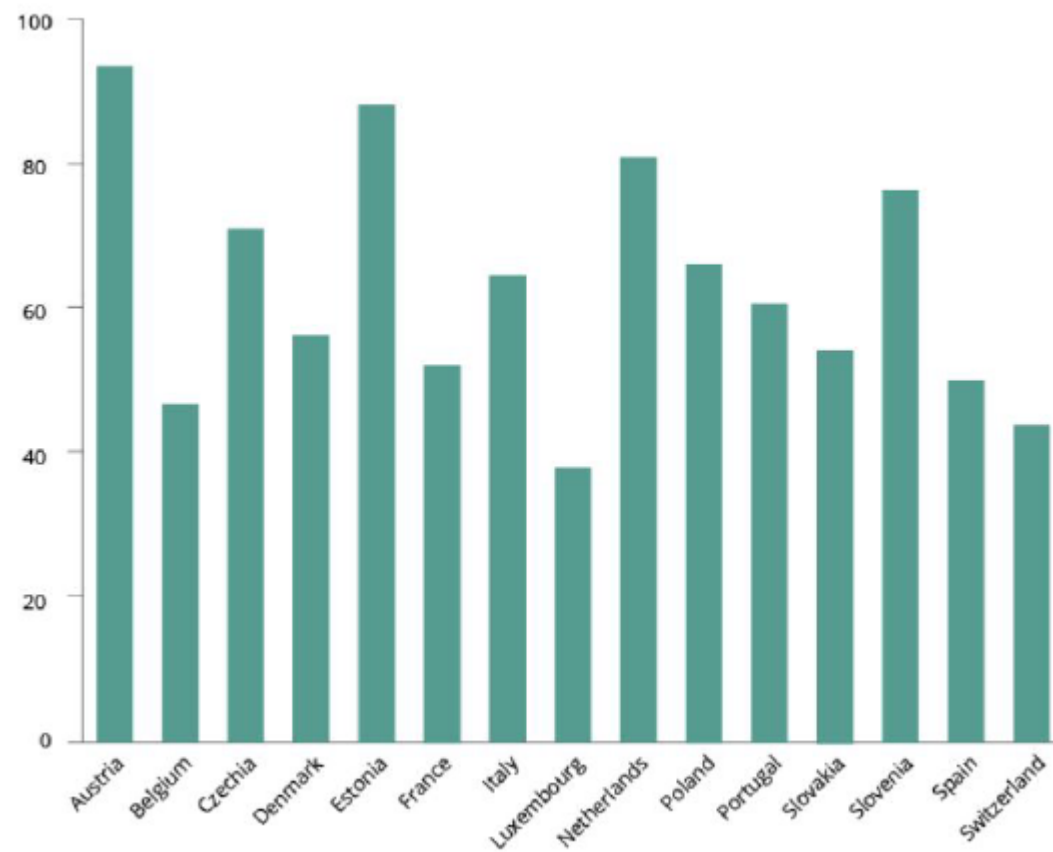


Note: No data were reported in 2018 and 2019

Bar charts

You have 5 seconds to find the position of France in the ranking

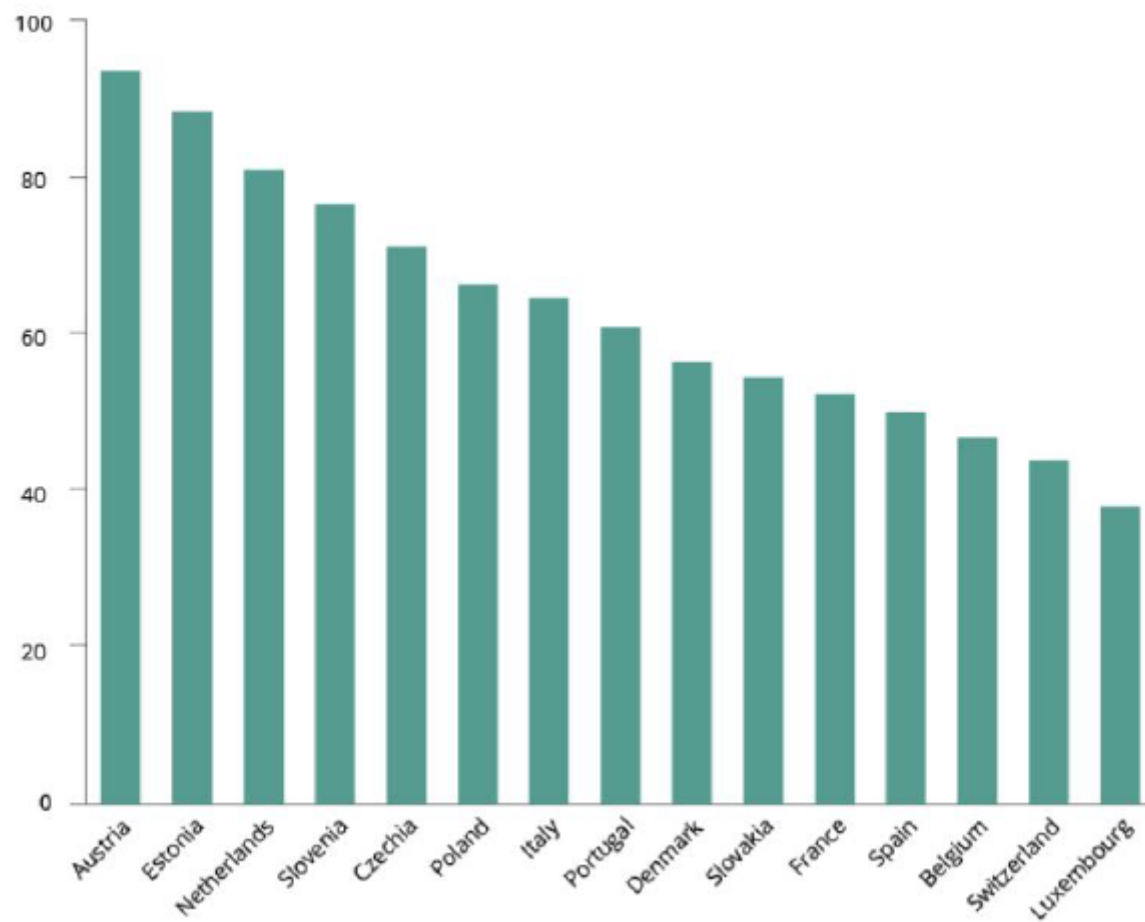
Bar charts



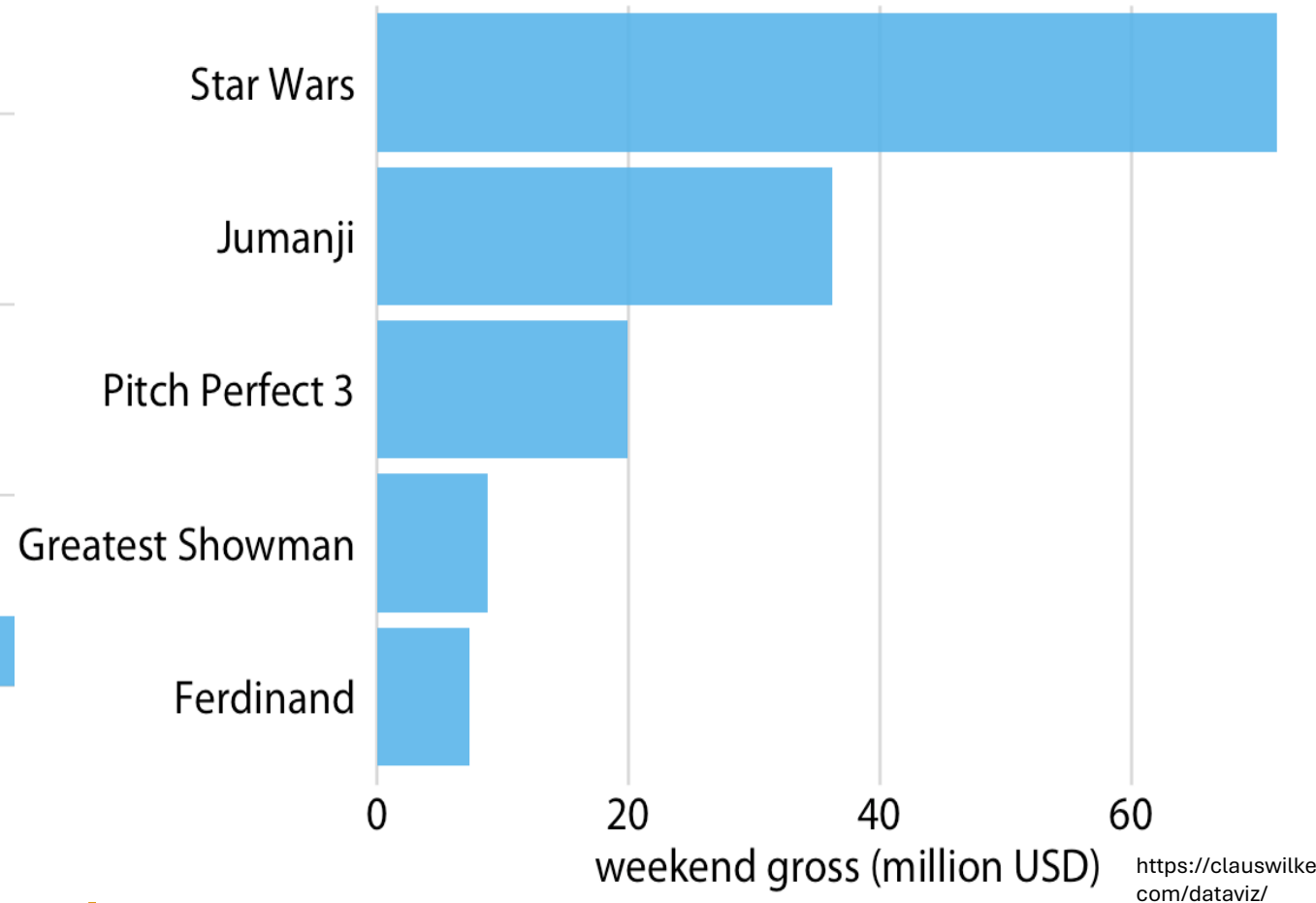
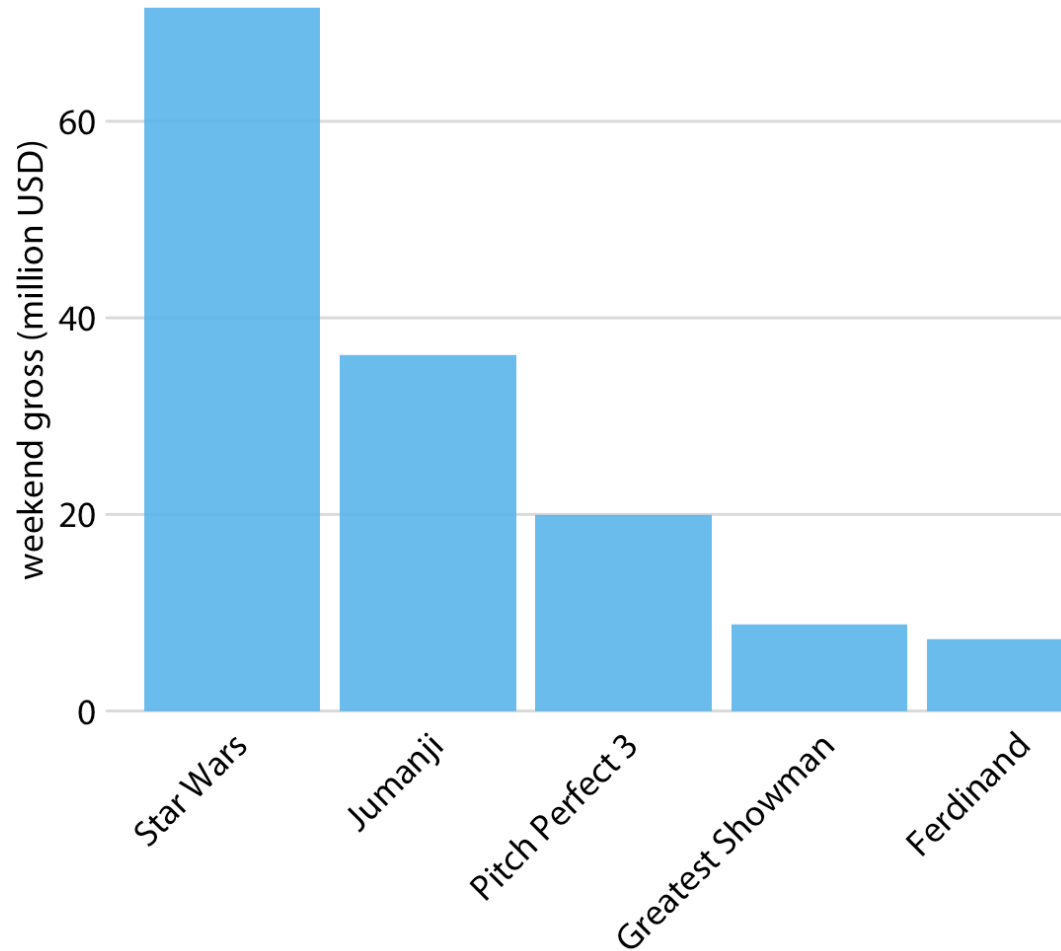
Bar charts

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Bar charts



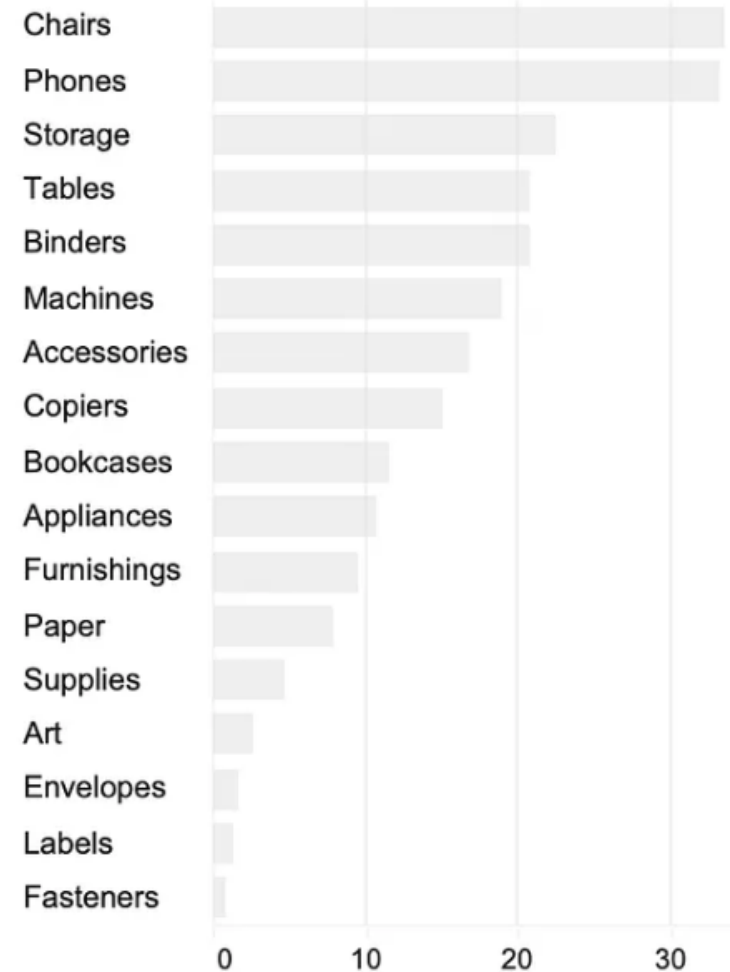
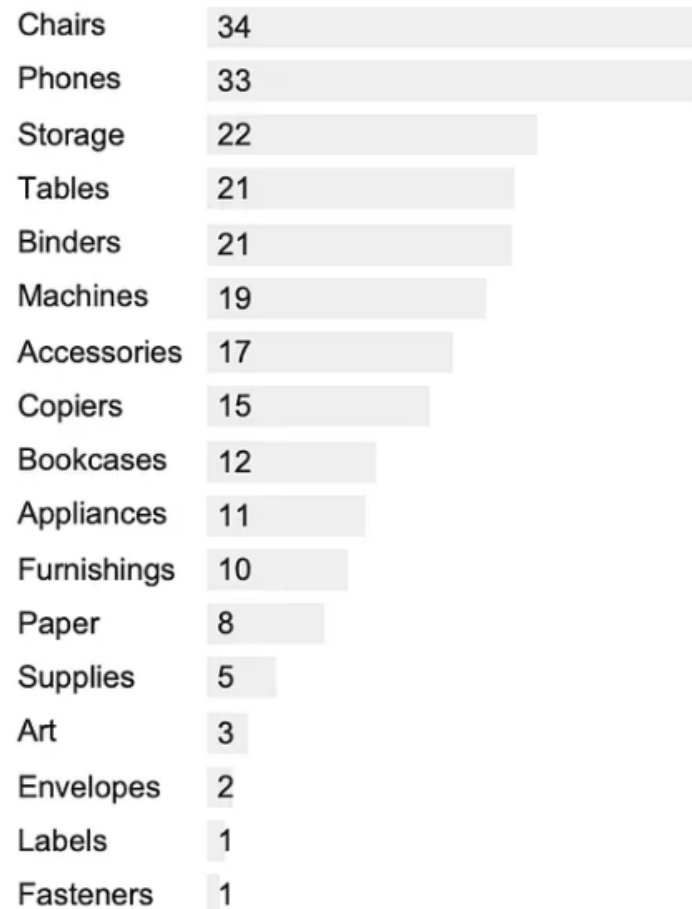
Bar charts



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- 4) **Sort your data:** Allow to make the comparison faster
- 5) **Horizontal orientation for long label**

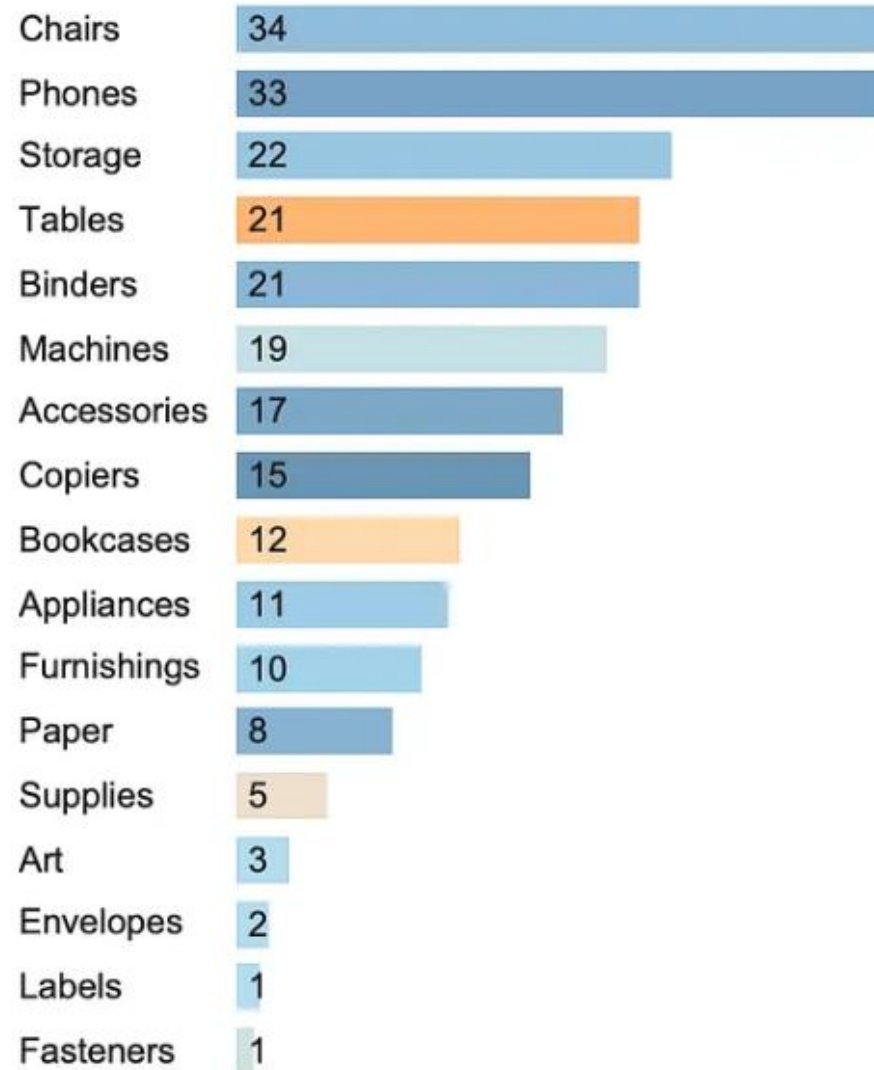
Bar charts



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Bar charts

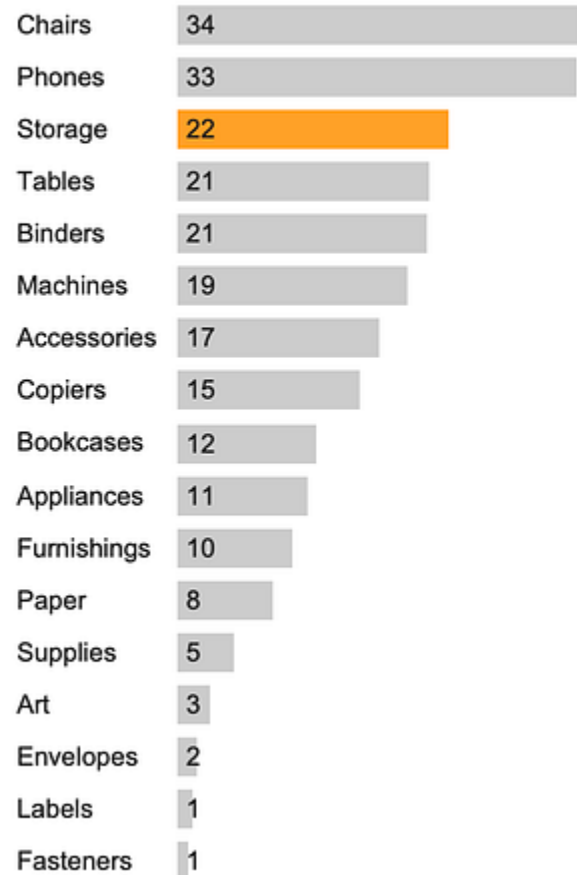


Bar charts

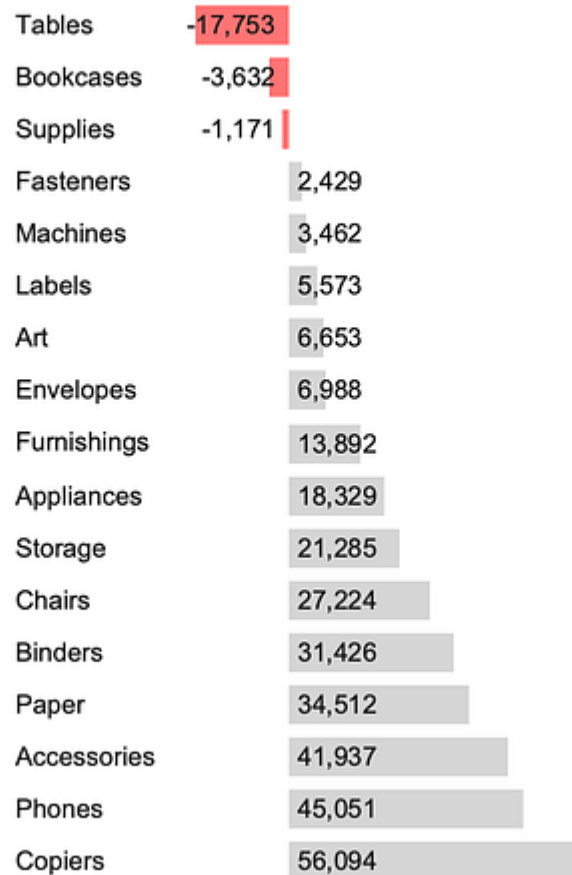
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- 7) **Do not use color if they do not add additional information**

Bar charts

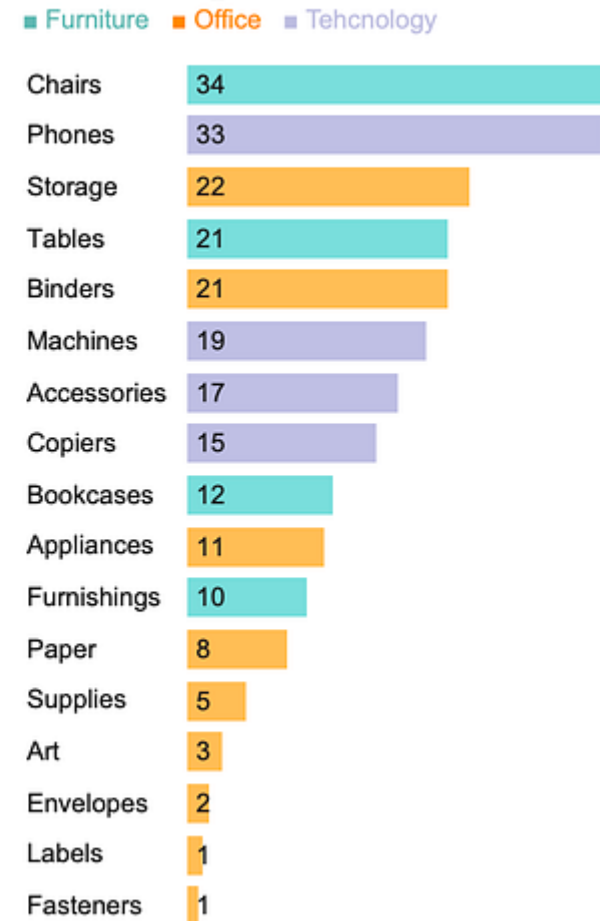
✓ Highlight Storage



✓ Alert: Not generating profit



✓ Distinguish

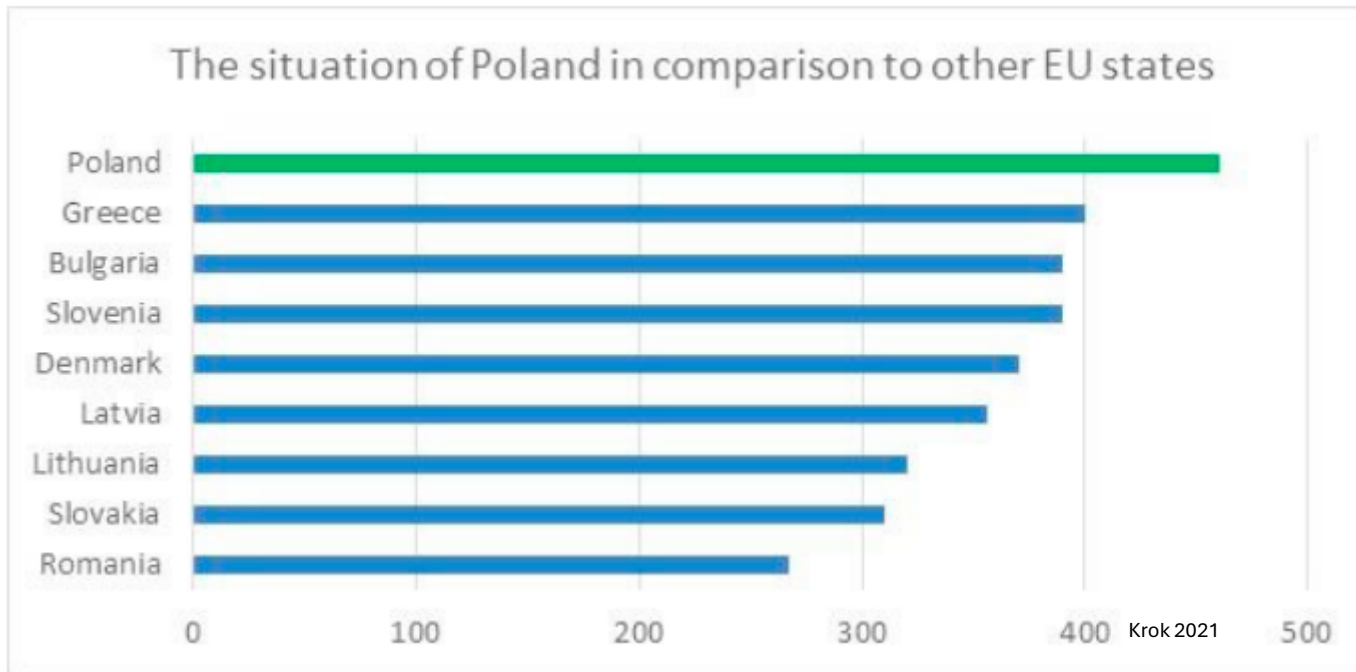


Bar charts

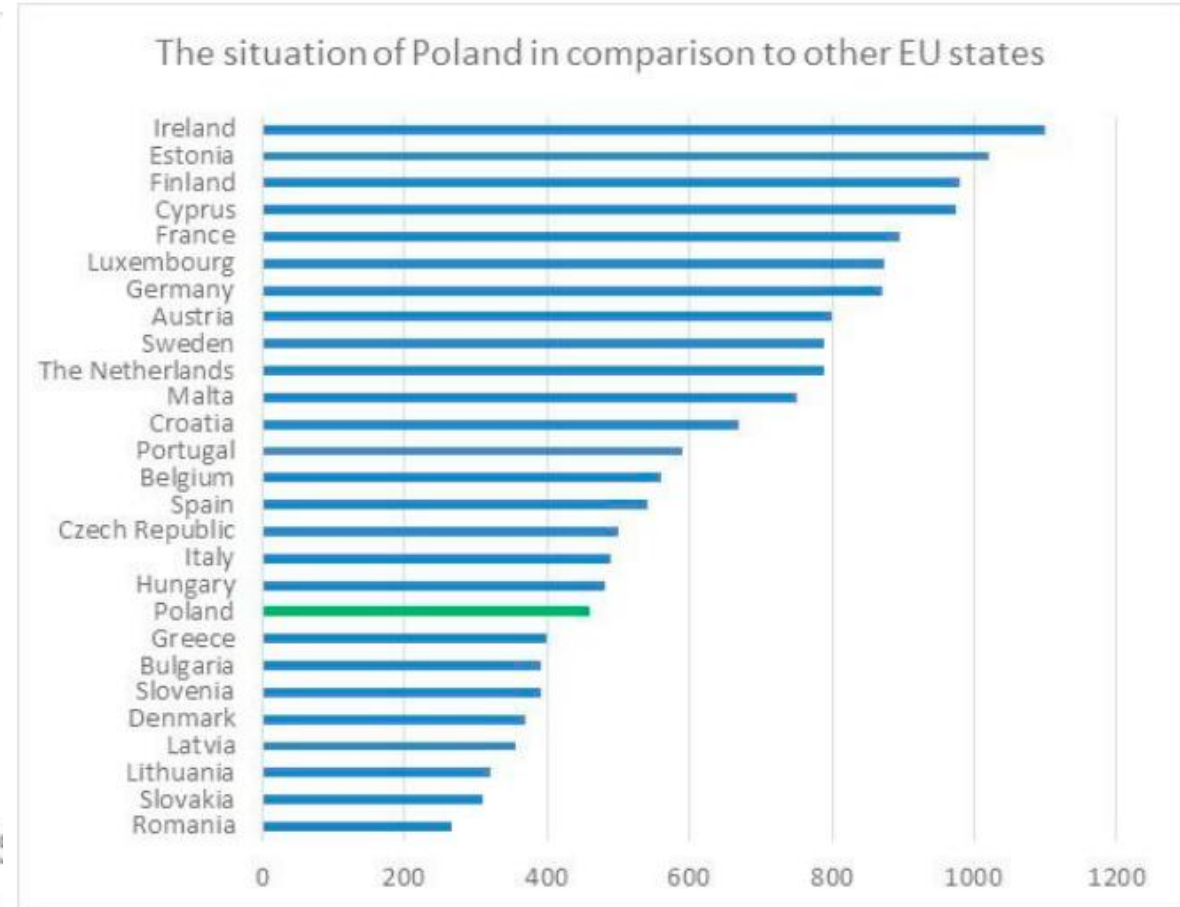
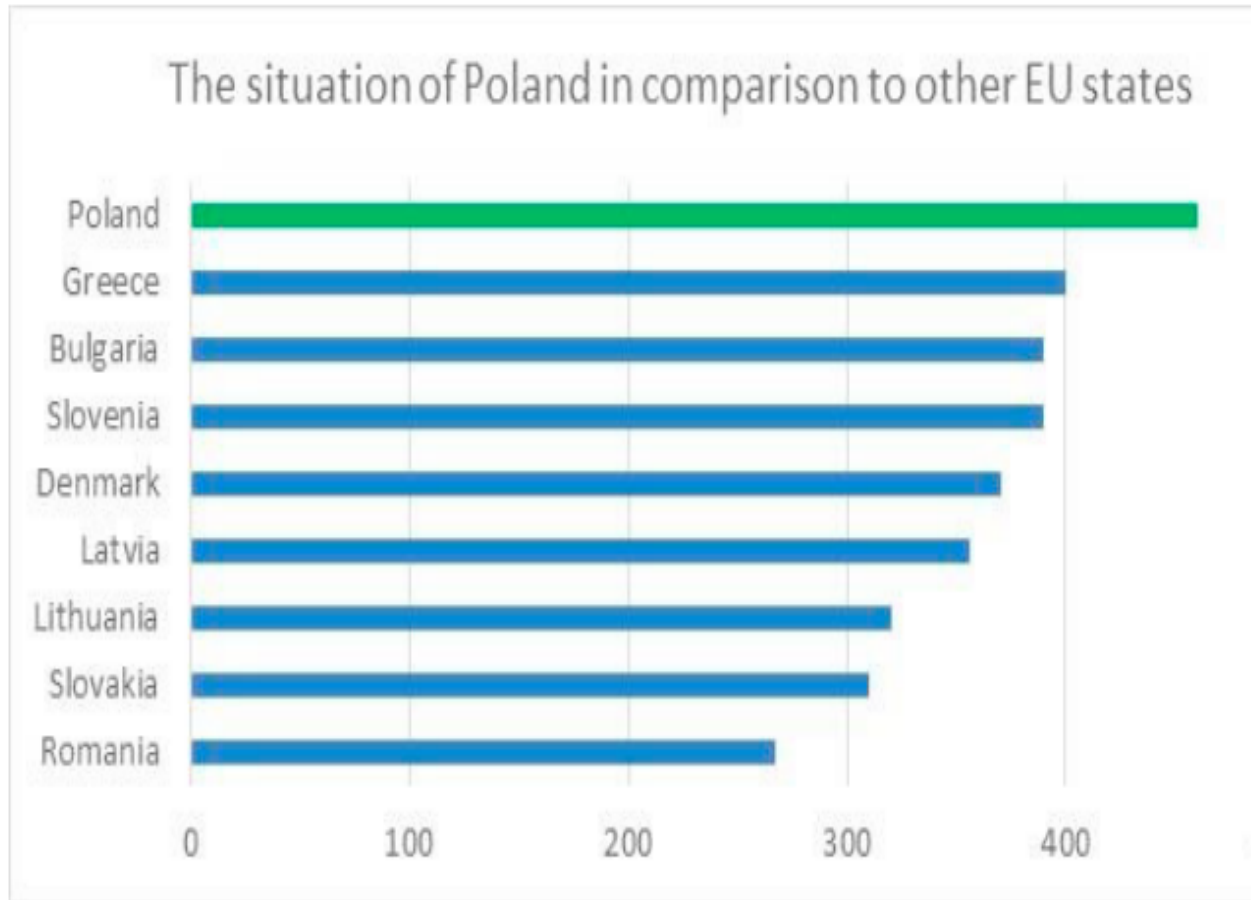
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Bar charts

What do you
conclude?



Bar charts

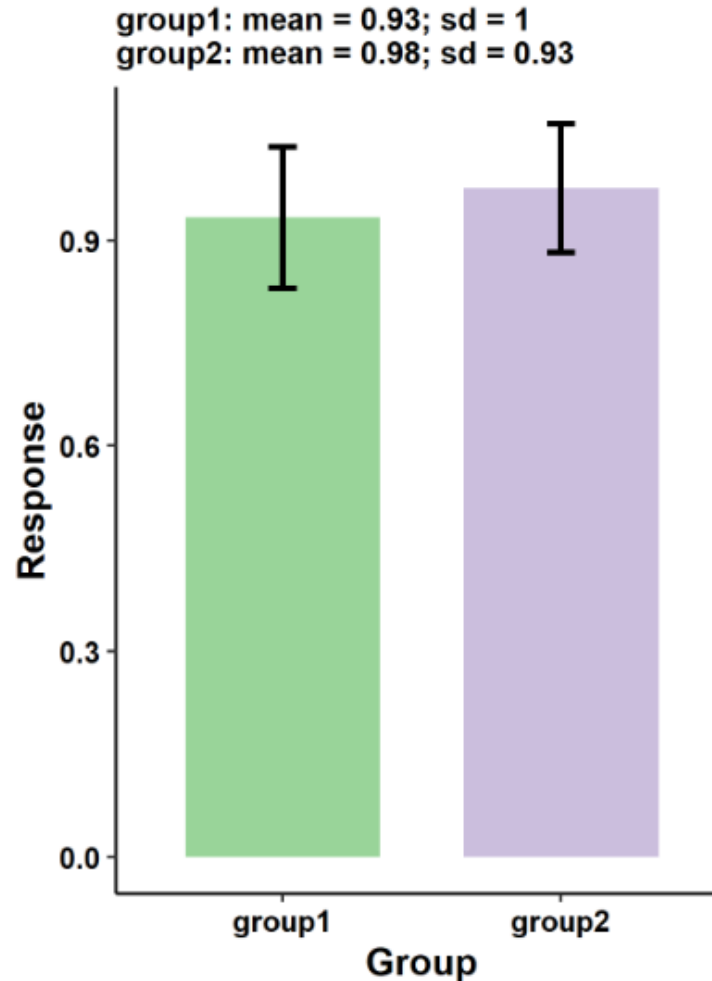


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- 8) **Do not apply cherry-picking it is not ethical**

Bar charts

What do you
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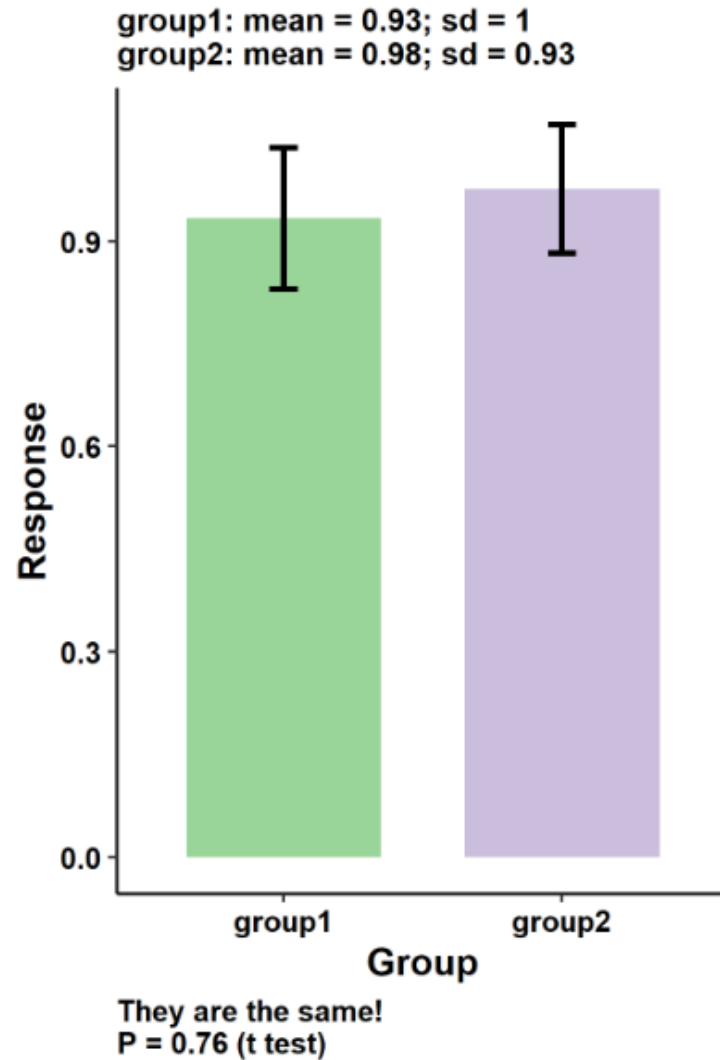


They are the same!
 $P = 0.76$ (t test)

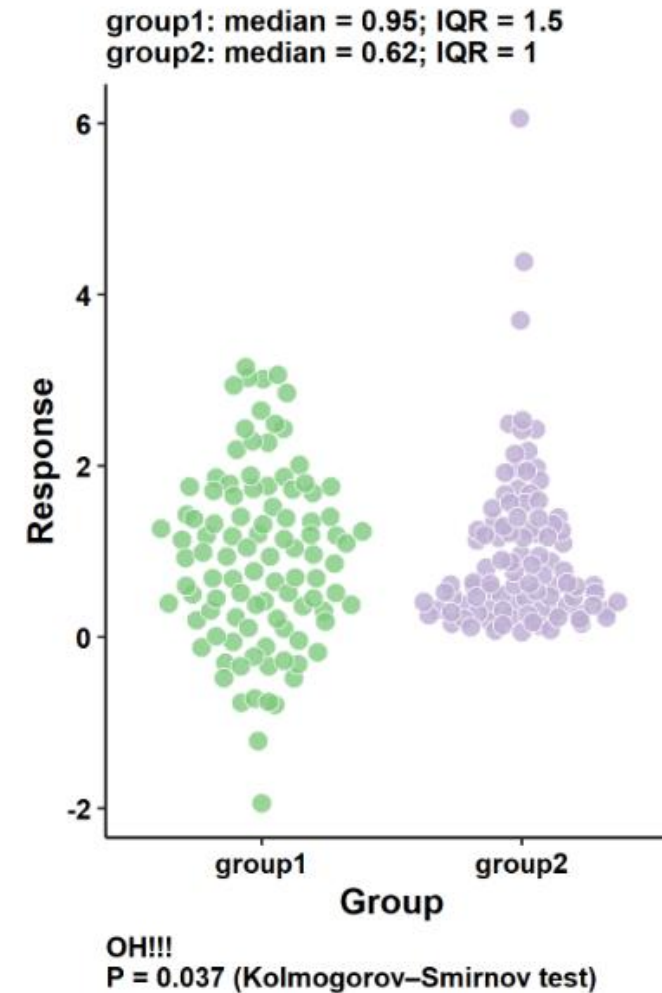
[Source.](#)

Bar charts

What do you conclude?



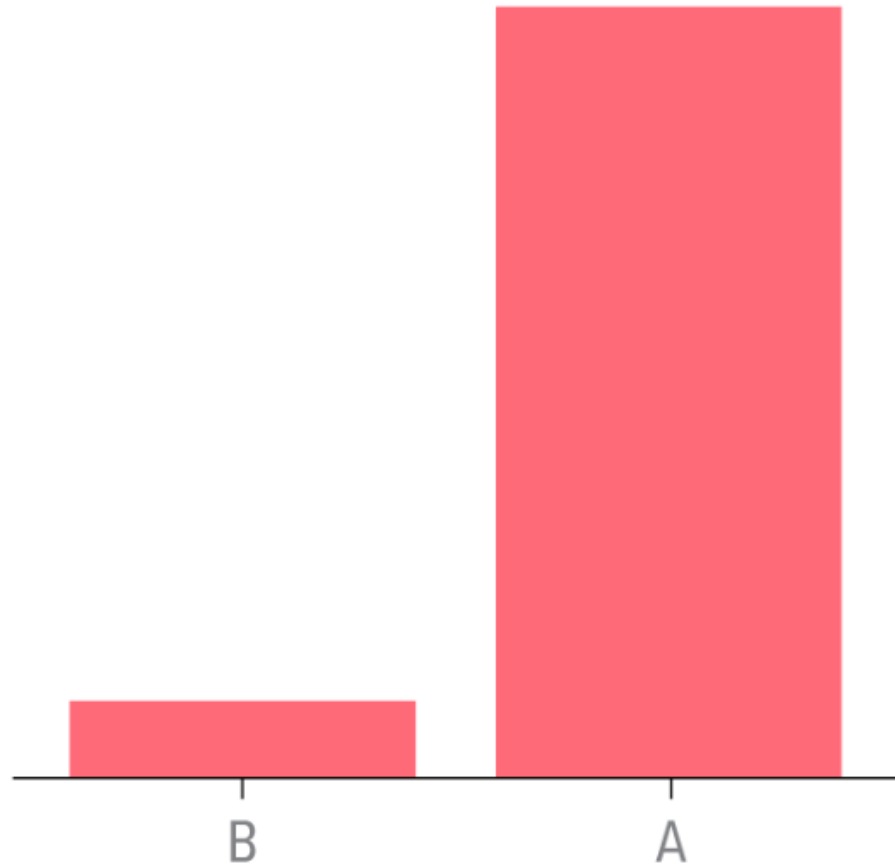
[Source.](#)



Bar charts

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- 7) **Do not use color if they do not add additional information**
- 8) **Do not apply cherry-picking it is not ethical**
- 9) **Do not make Bar Plots for Means Separation**

Bar charts



The value of A is how many times as large as the value of B?

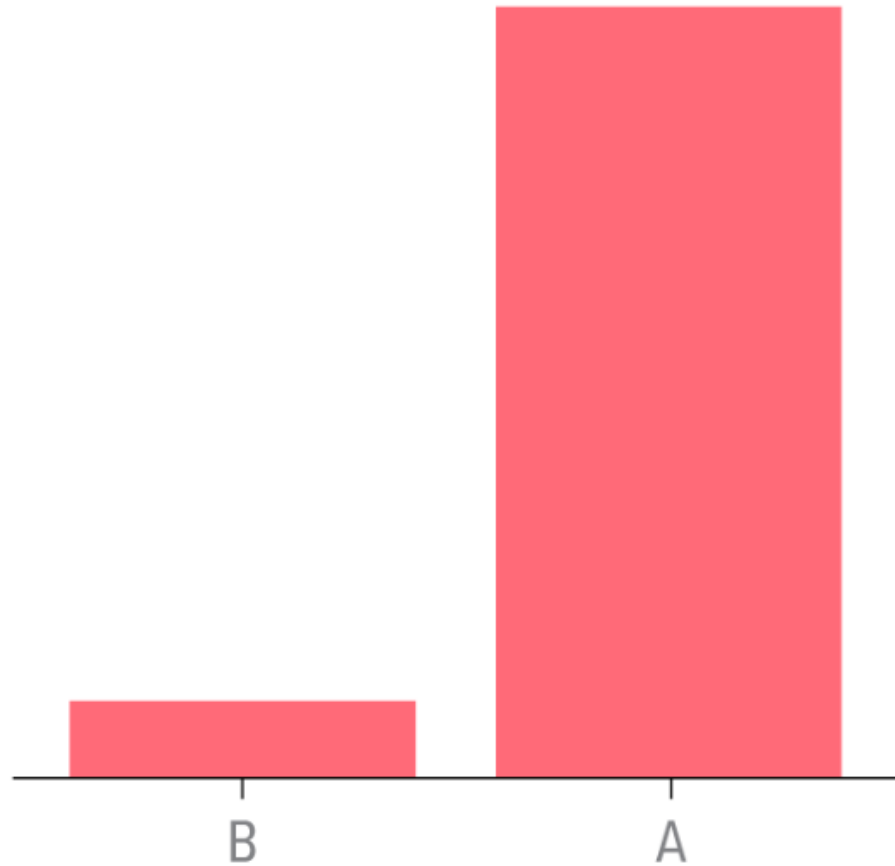
9

8

10

5

Bar charts



The value of A is how many times as large as the value of B?

9

8

10

5

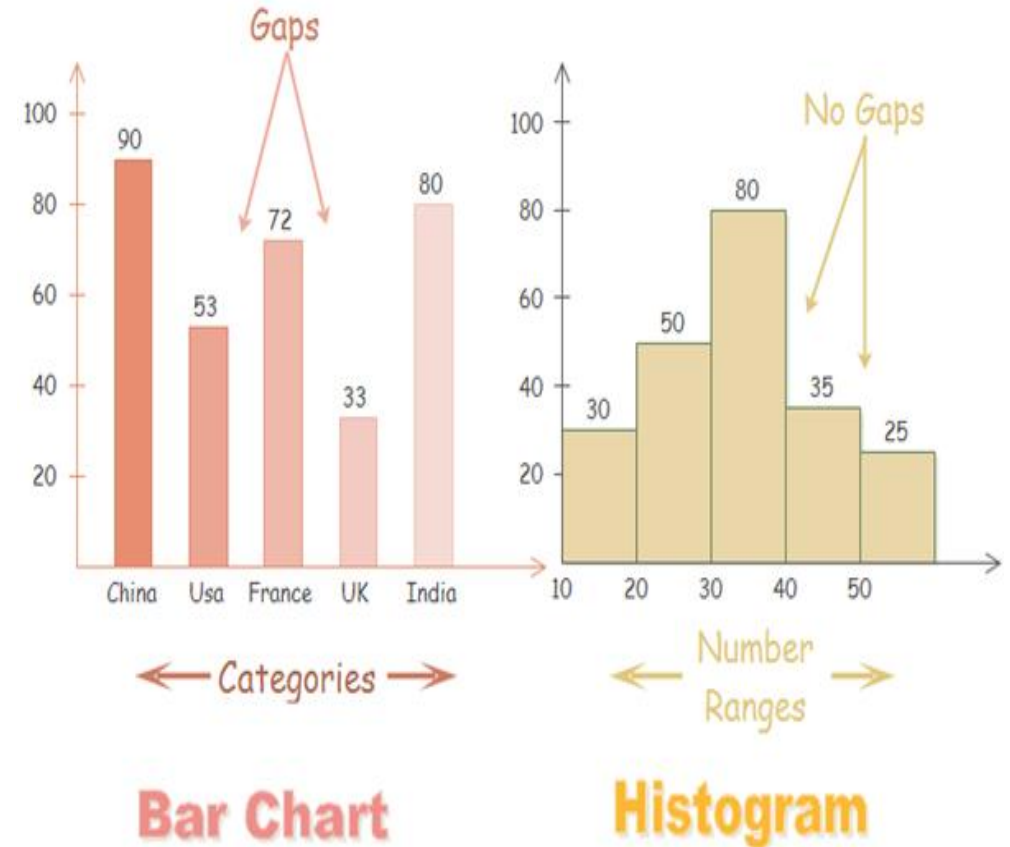
Histogram charts

- We frequently encounter the situation where we would like to understand how a particular variable is distributed in a dataset.
- **A histogram displays the distribution of numerical data by grouping values into intervals (bins) and showing their frequency as bars. It helps reveal the shape, spread and patterns in the data.**
- Histogram use **Non-discrete quantitative data** (discrete numerical or categorical data for bar chart)

Useful for:

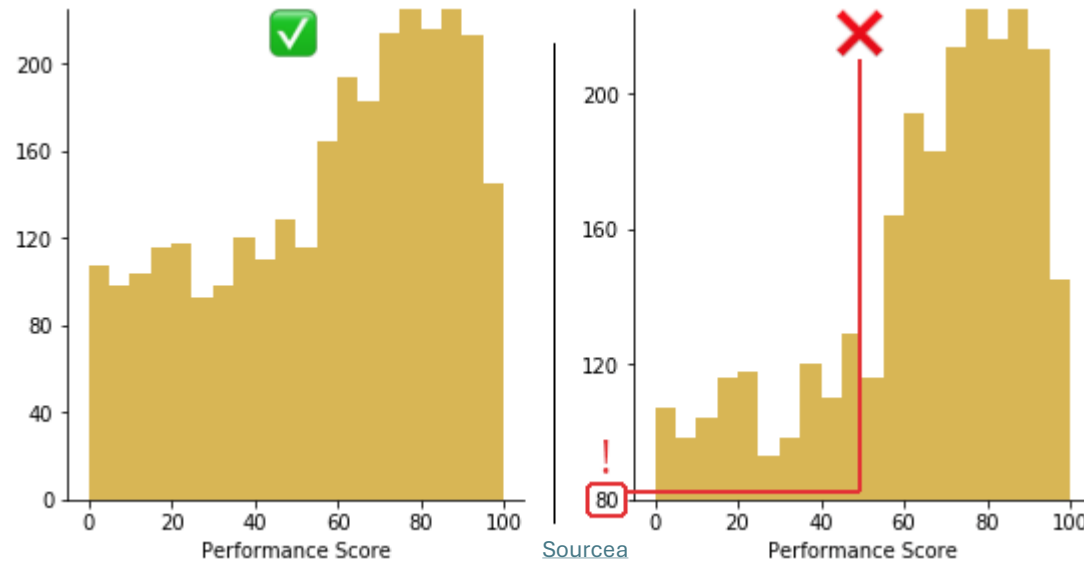
- To visualize variability
- To identify outliers
- To identify multimodal distributions

Example from [edraw](#)



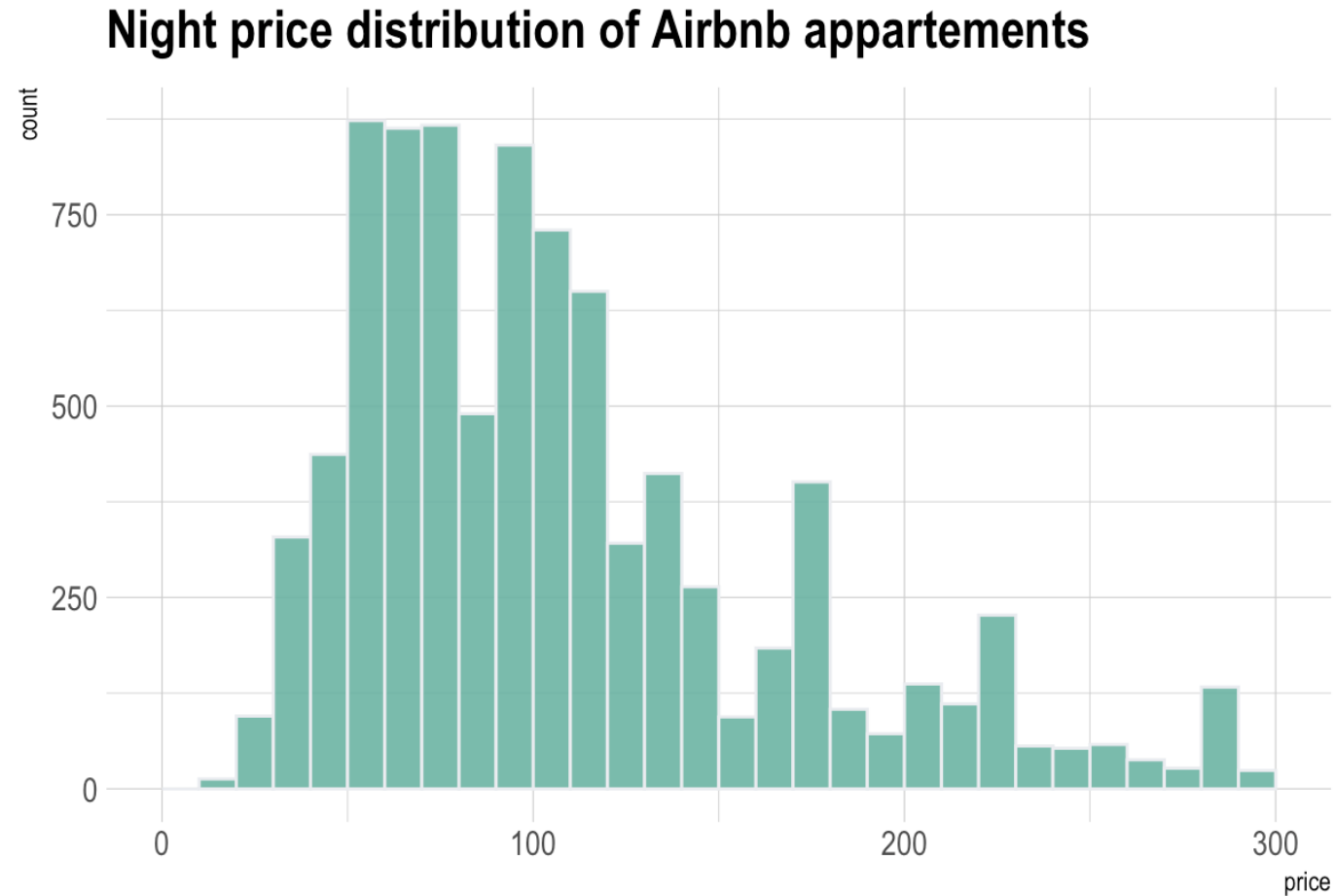
Histogram charts

1) The base value must always be zero



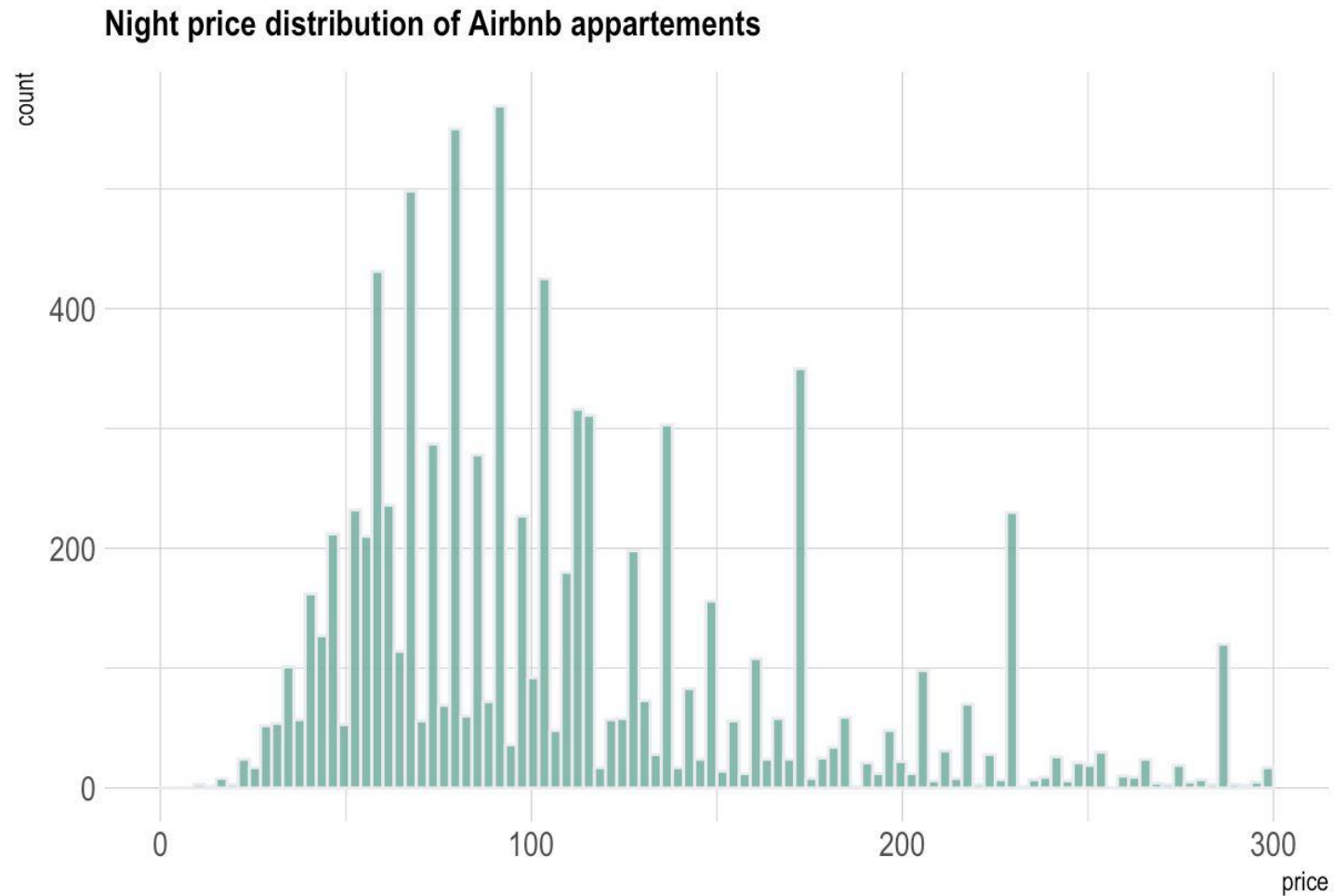
Histogram charts

What do
you see?



Histogram charts

What do
you see?



Histogram charts

1) The base value must always be zero

2) Try several bin size!!: If bin width is too small, then the histogram becomes overly peaky and visually busy and the main trends in the data may be obscured. If the bin width is too large, then smaller features in the distribution of the data can be hidden.

Histogram charts

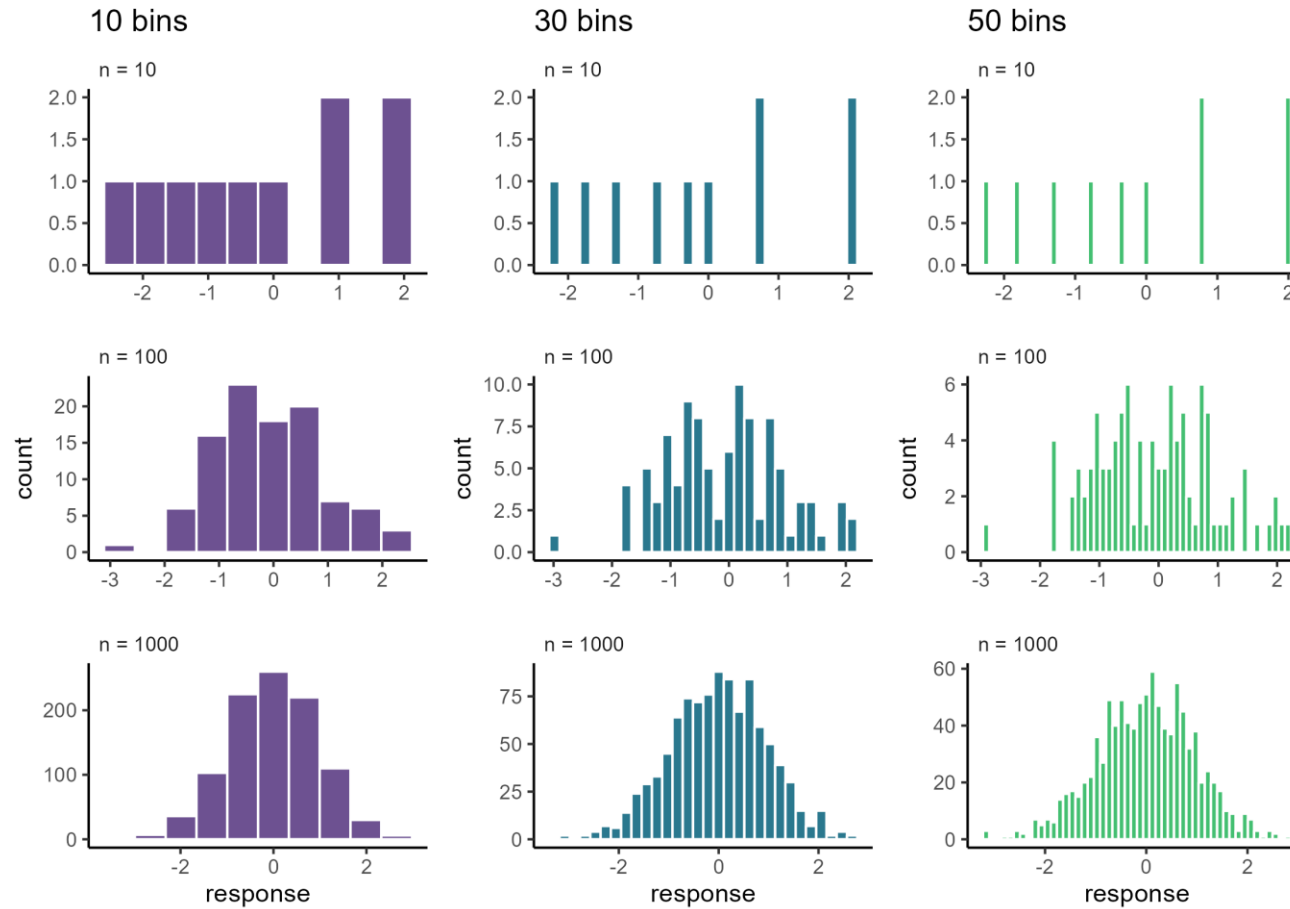
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3) Do use weird colors

Histogram charts

Random number
from the same
distribution



Wow, the appearance does change with different bin numbers.

[Source.](#)

Histogram charts

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3) Do use weird colors

4) Never use histogram on a small dataset!!

Histogram charts

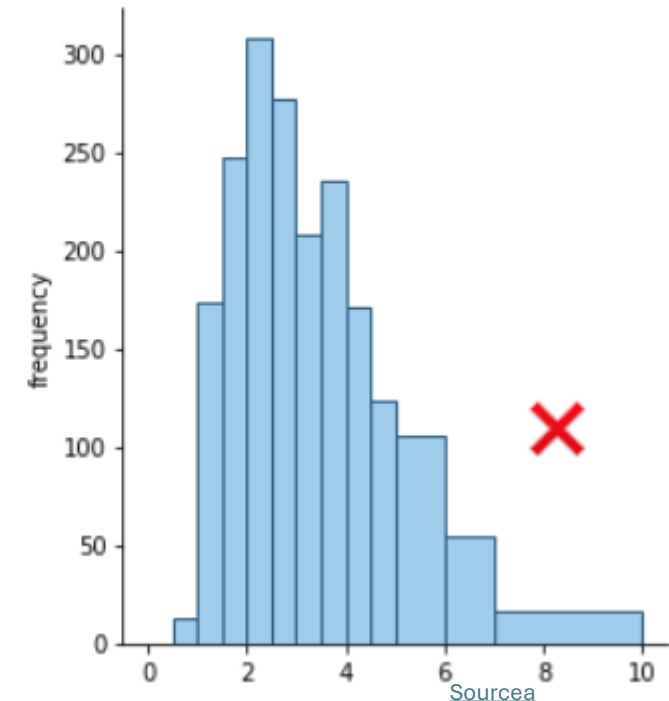
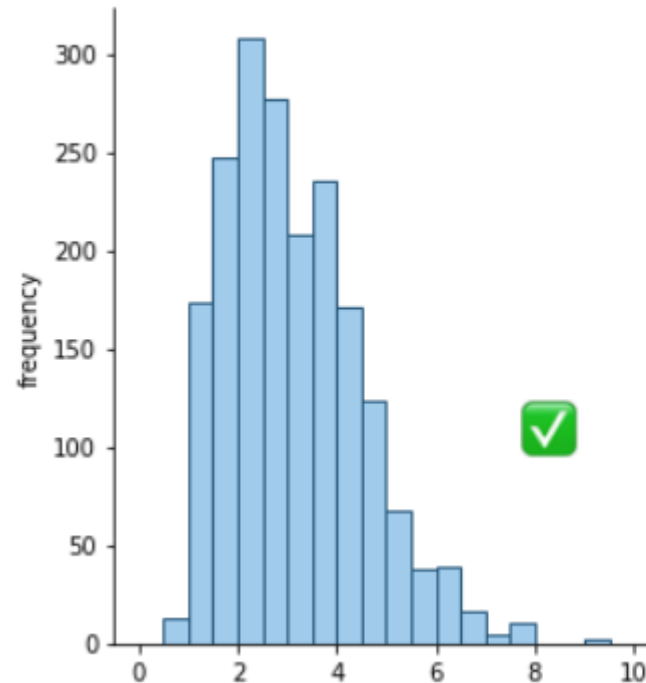
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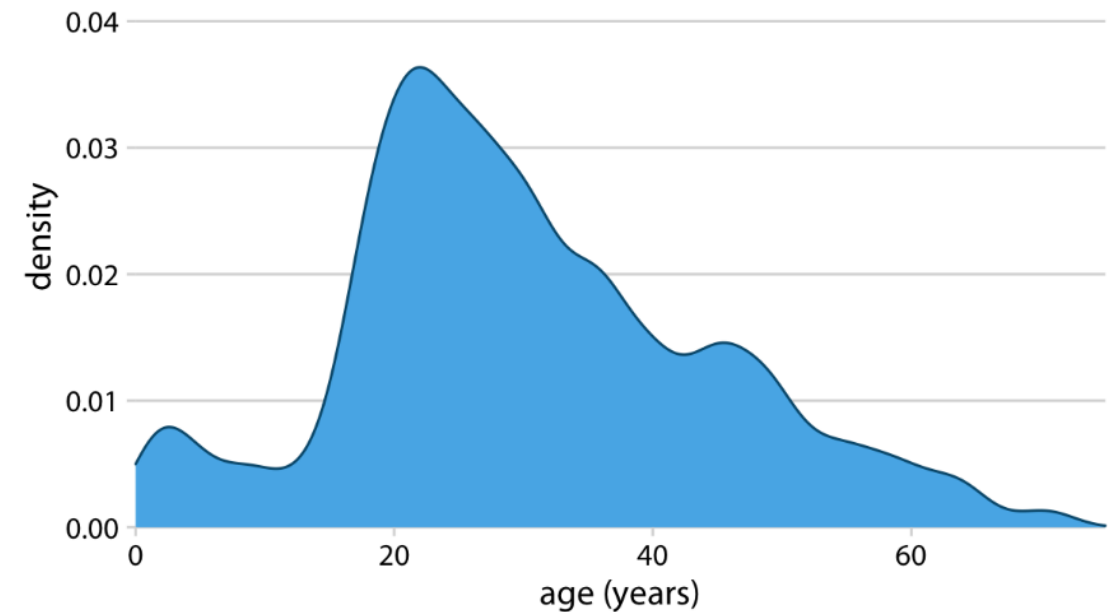
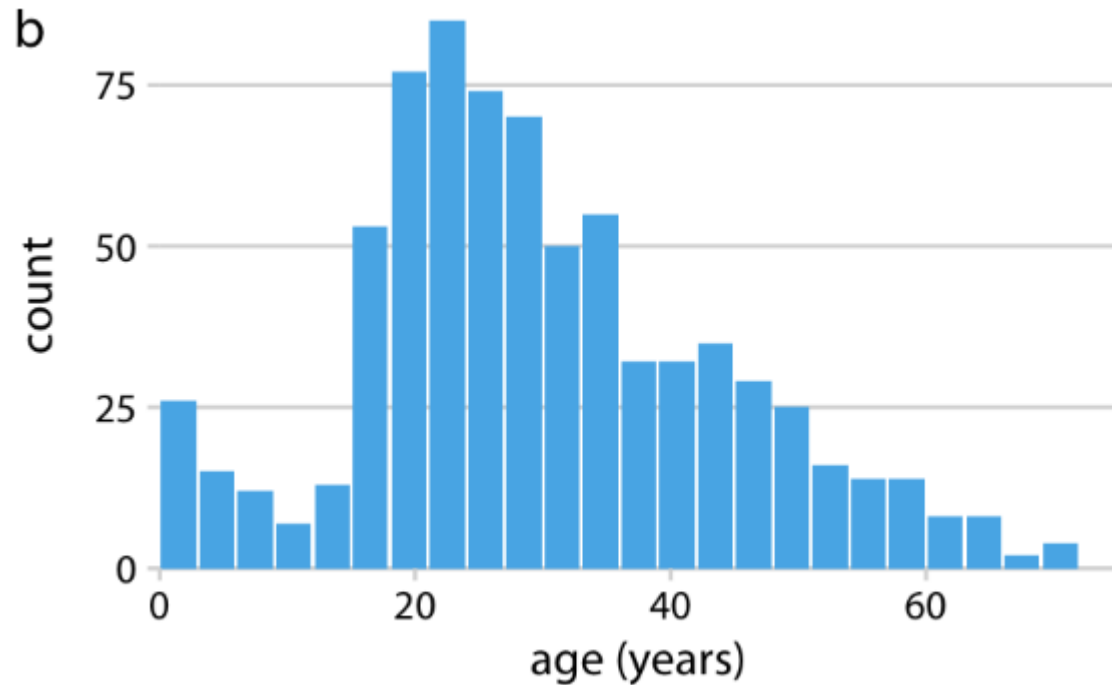
4) Never use histogram on a small dataset!!

5) Use the same bin width



Density curve

In a density plot, we attempt to visualize the underlying probability distribution of the data by drawing an appropriate continuous curve



Pie charts

A pie chart shows how a total amount is divided between levels of a categorical variable as a circle divided into radial slices.

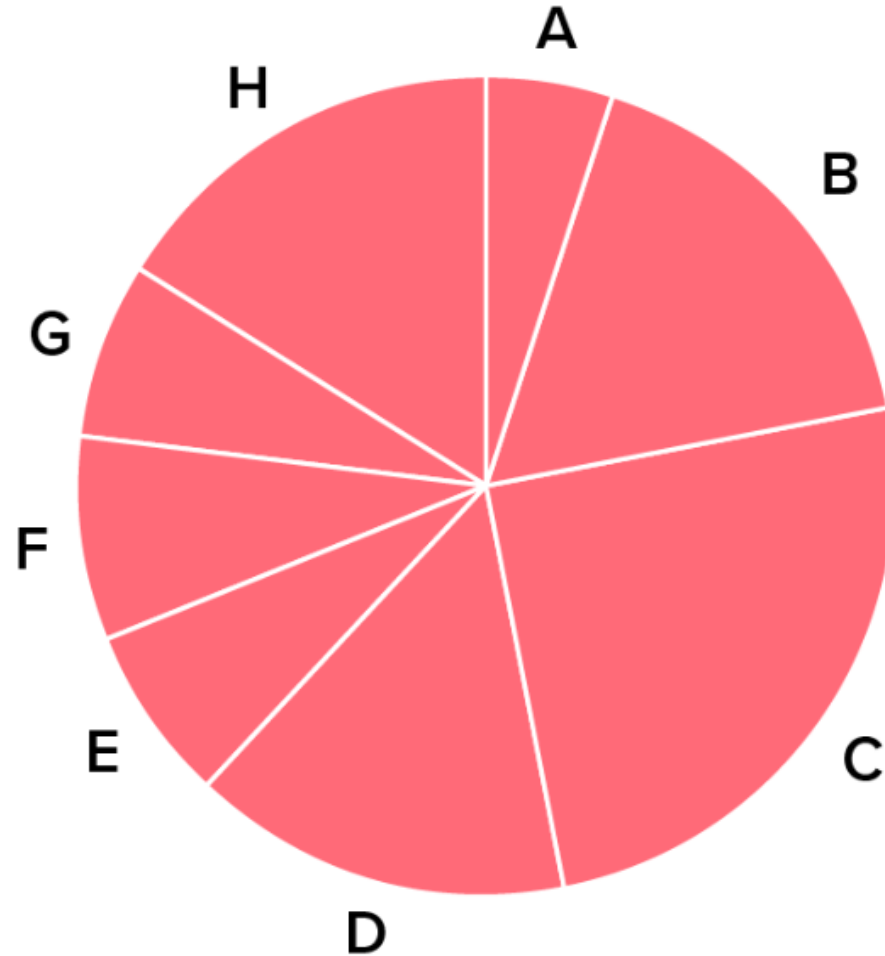
Your primary objective in a pie chart should be to compare each group's contribution to the whole, as opposed to comparing groups to each other

HOWEVER, TRY TO AVOID THEM

Table 10.1: Pros and cons of common approaches to visualizing proportions: pie charts, stacked bars, and side-by-side bars.

	Pie chart	Stacked bars	Side-by-side bars
Clearly visualizes the data as proportions of a whole	✓	✓	✗
Allows easy visual comparison of the relative proportions	✗	✗	✓
Visually emphasizes simple fractions, such as 1/2, 1/3, 1/4	✓	✗	✗
Looks visually appealing even for very small datasets	✓	✗	✓
Works well when the whole is broken into many pieces	✗	✗	✓
Works well for the visualization of many sets of proportions or time series of proportions	✗	✓	✗

Pie charts



Which is the third largest segment in the pie chart?

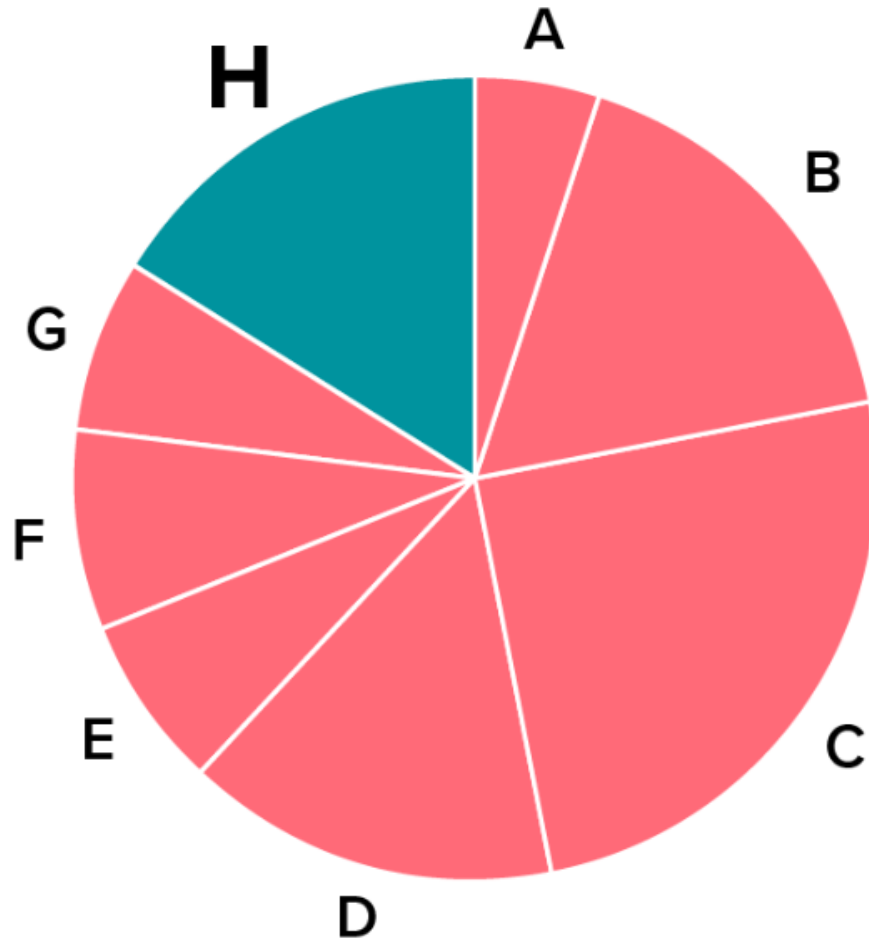
A

H

B

D

Pie charts



Which is the third largest segment in the pie chart?

A

H

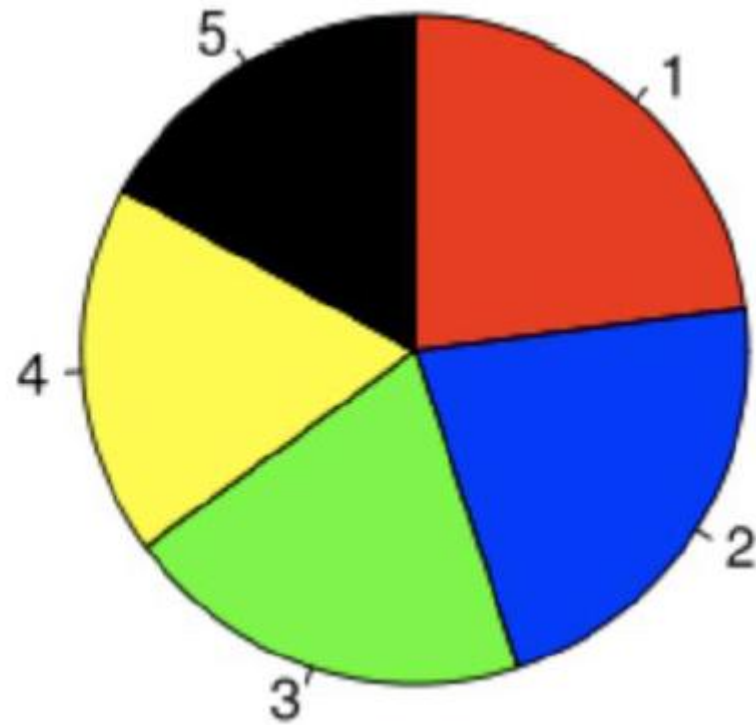
B

D

Pie charts

You have 5 seconds to find the largest part

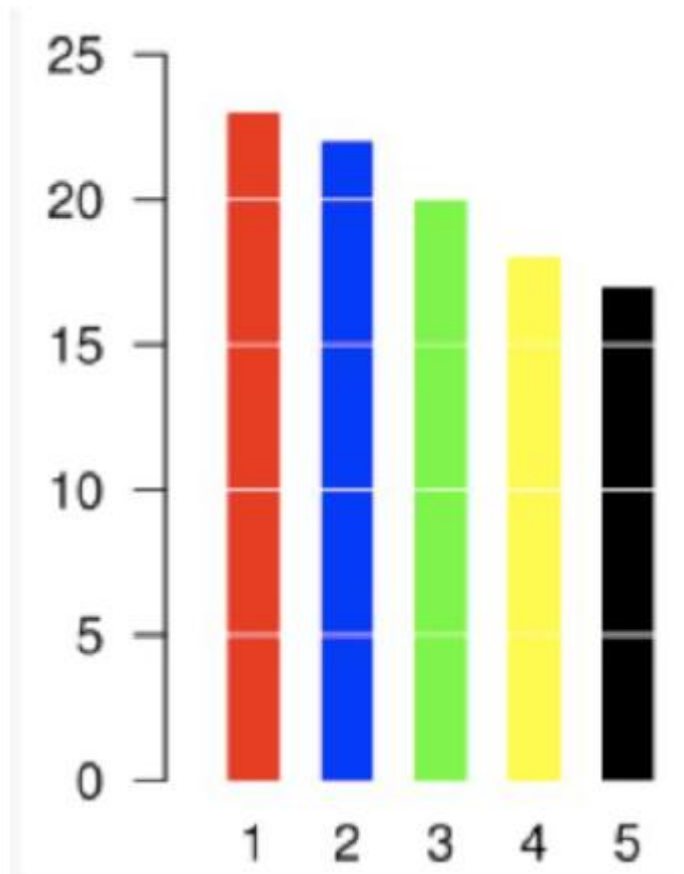
Pie charts



Pie charts

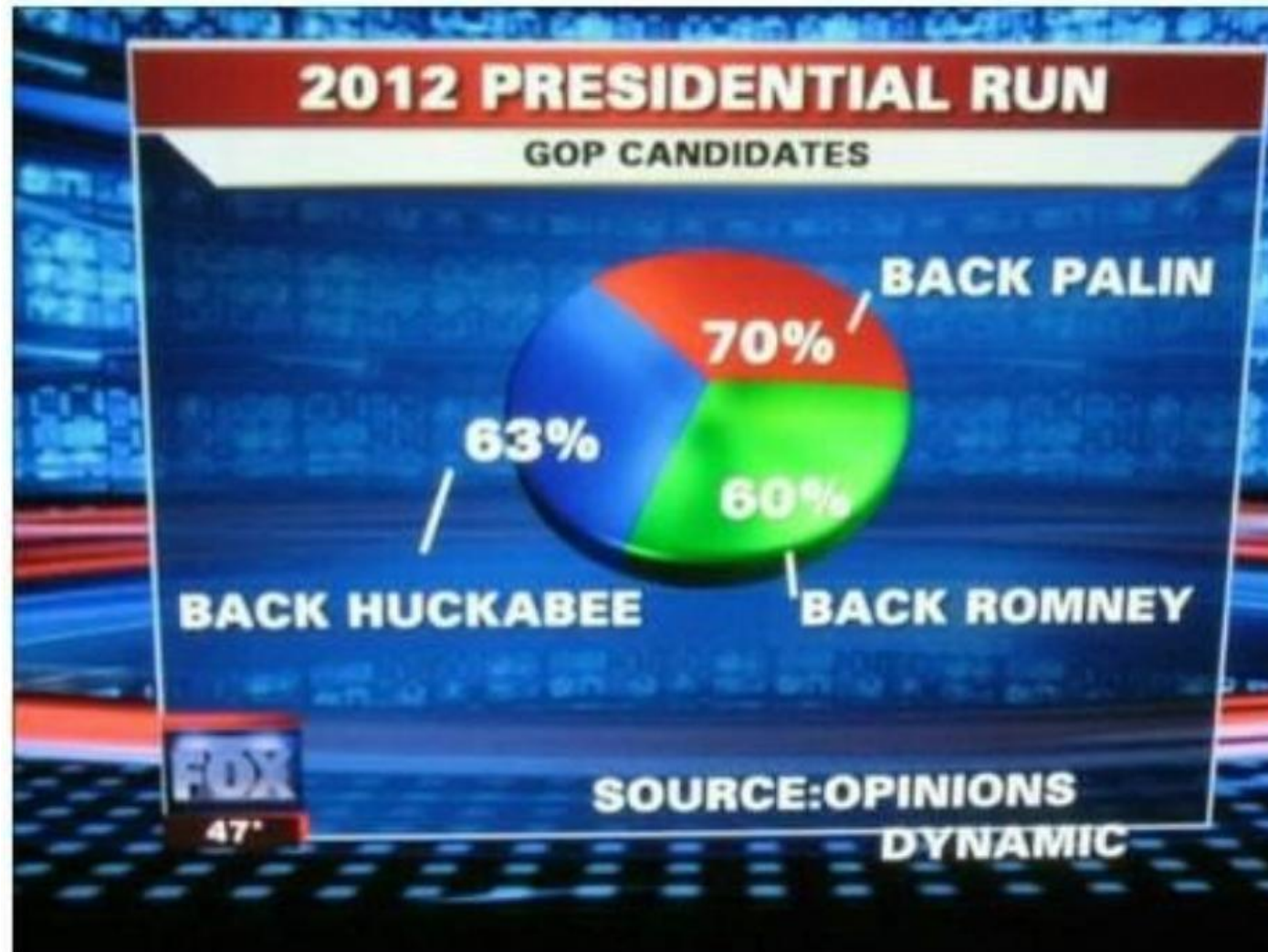
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Pie charts



Pie charts

Fox News Election 2012 Presidential Run GOP Candidates

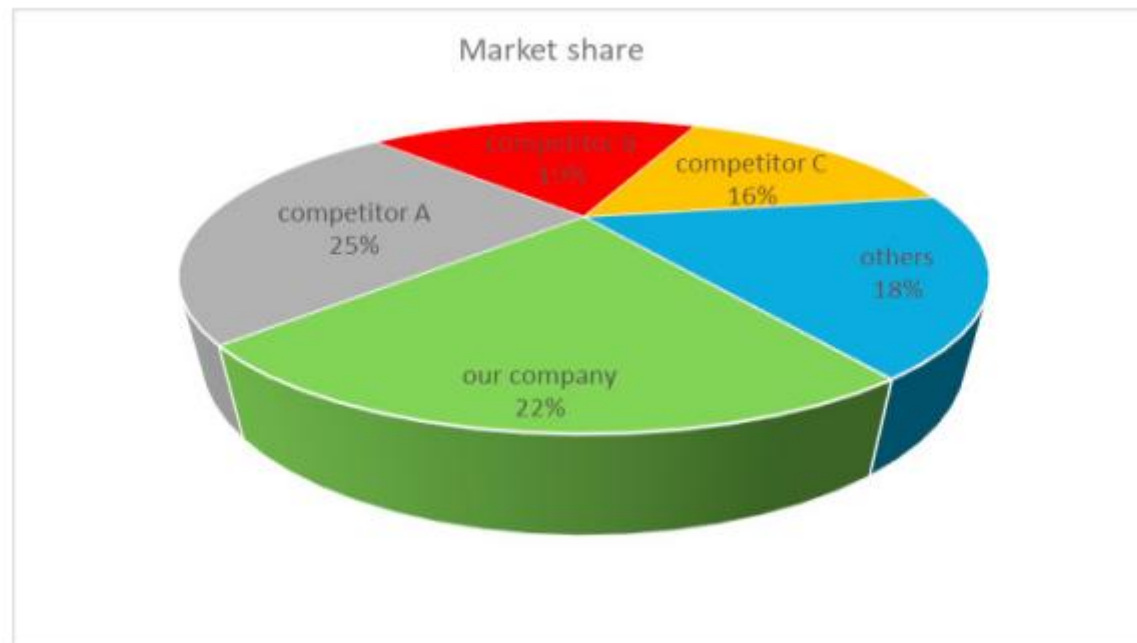


Pie charts

1) The parts of a pie chart must always be equal to 100%.

Pie charts

What do you
conclude?



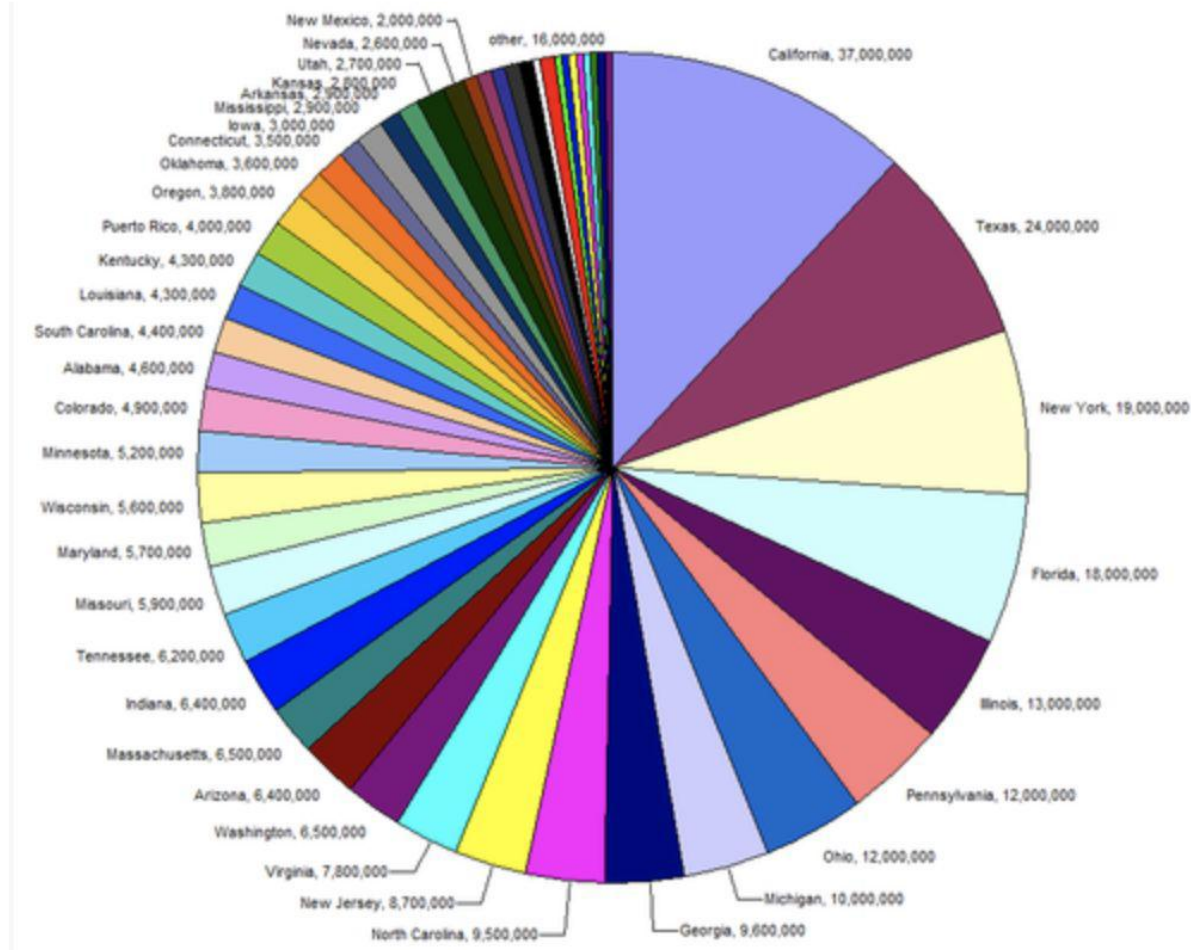
Pie charts

1) The parts of a pie chart must always be equal to 100%.

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Pie charts

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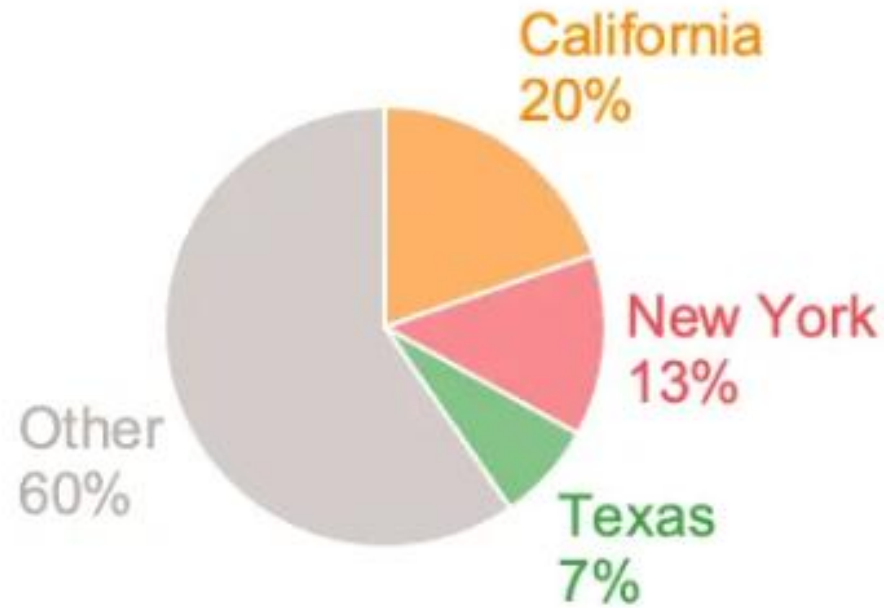


Pie charts

- 1) **The parts of a pie chart must always be equal to 100%.**
- 2) **Do not use 3D:** Difficult to read the right values
- 3) **No more than 4-5 items**

Pie charts

What do you see?

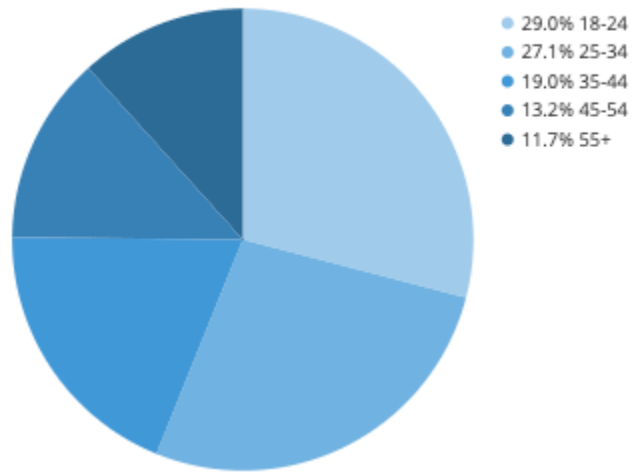


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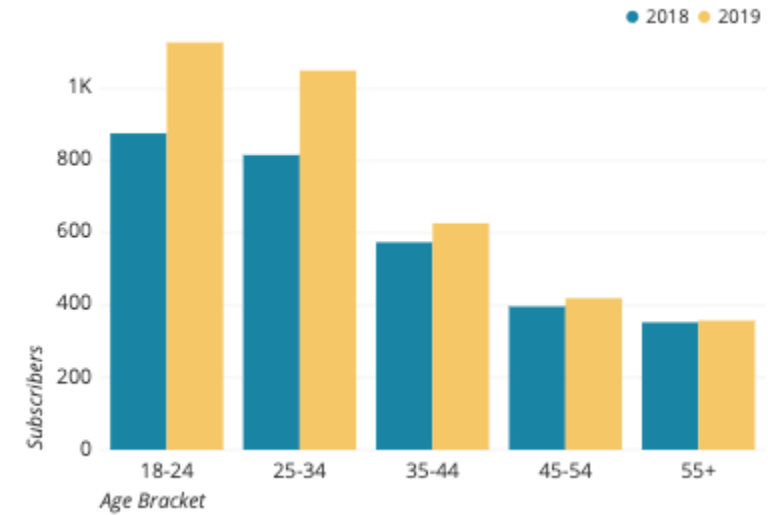
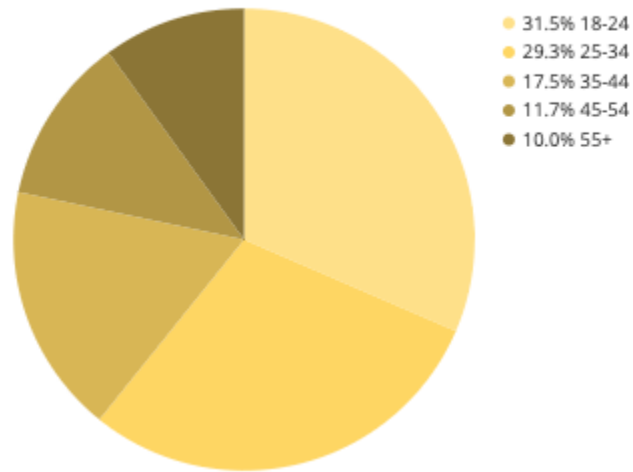
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- 2) **Do not use 3D:** Difficult to read the right values
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- 4) **Order the slices**

Pie charts

2018 Subscribers by Age



2019 Subscribers by Age



Pie charts

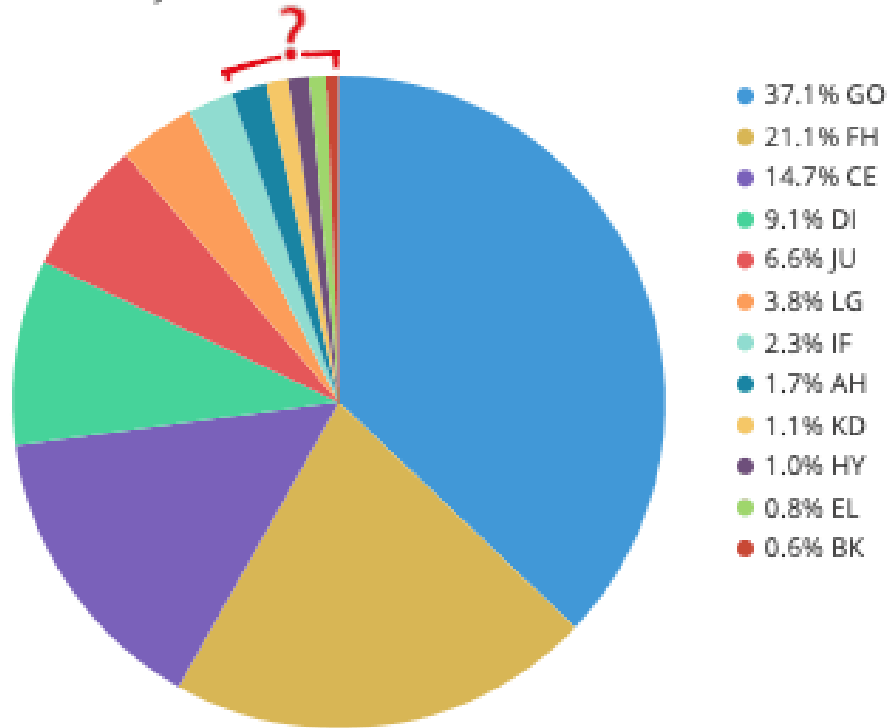
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Pie charts

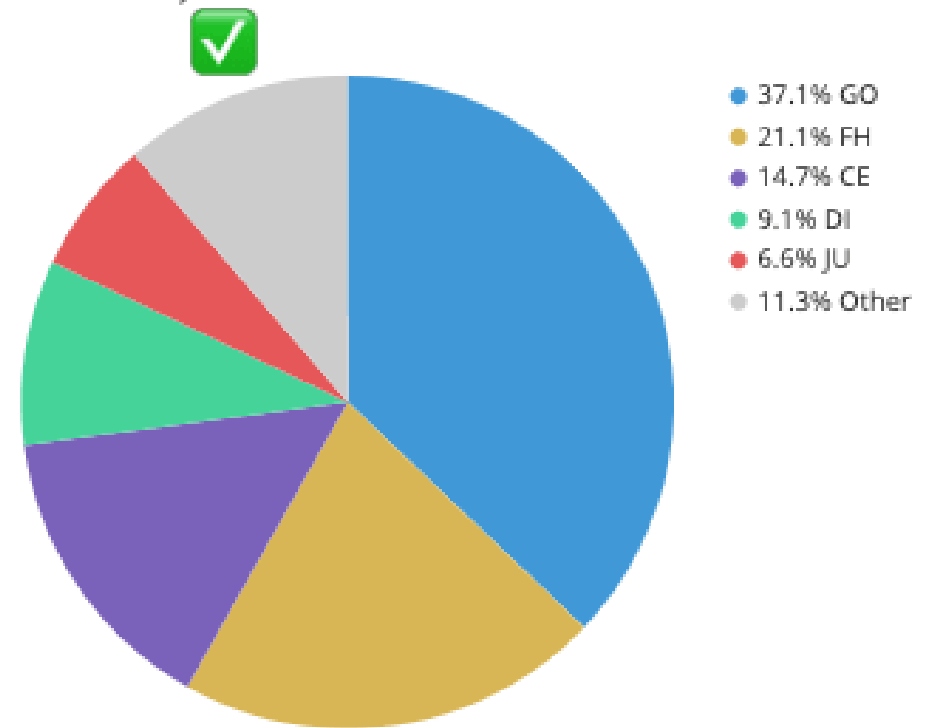
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- 6) **Label values directly; don't create an additional legend**

Pie charts

Production by district



Production by district



Pie charts

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- 6) **Label values directly; don't create an additional legend**
- 7) **Try to group small categories into 'Other'**

Line charts

A line chart uses points connected by line segments from left to right to demonstrate changes in value. The horizontal axis depicts a continuous progression, often that of time, while the vertical axis reports values for a metric of interest across that progression.

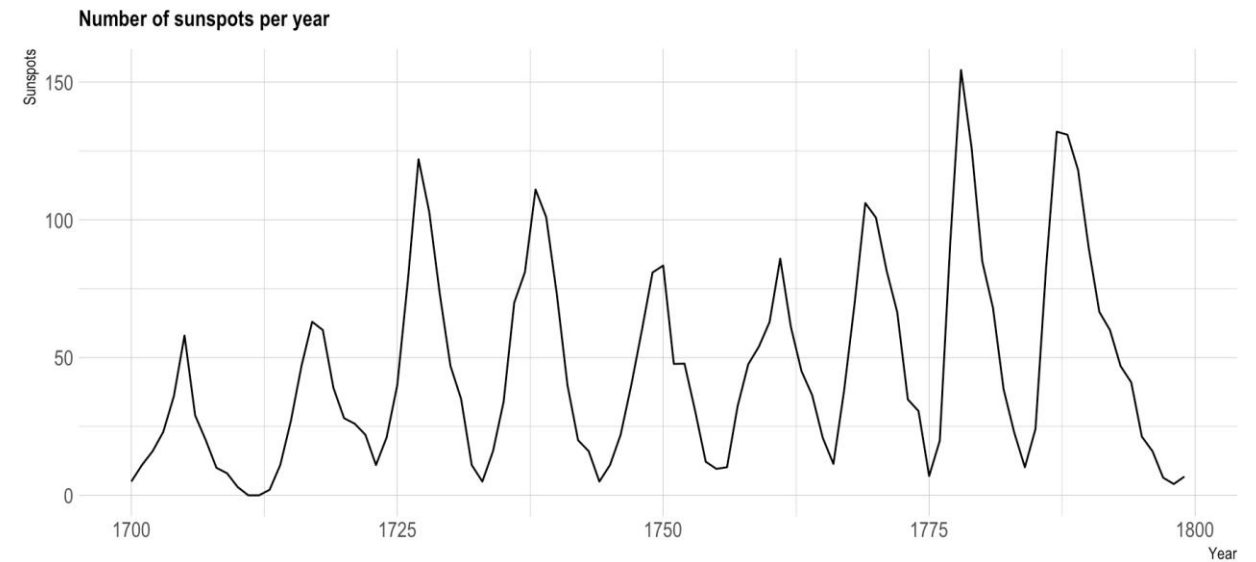
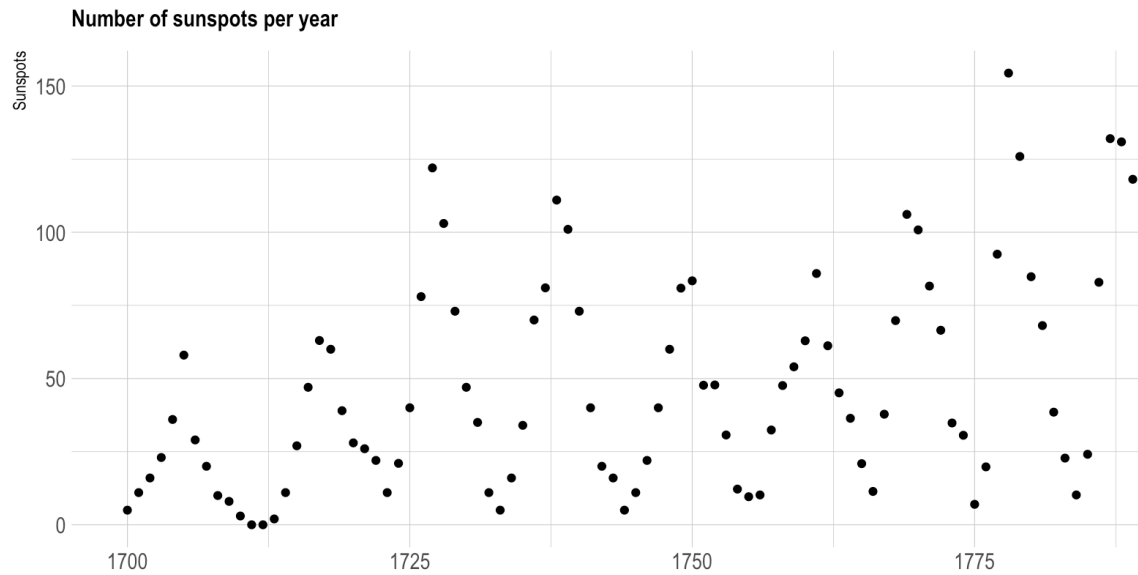
Your primary objective is to emphasize changes in values for one variable for **continuous values** of a second variable.

ZZD to QQY Exchange Rates



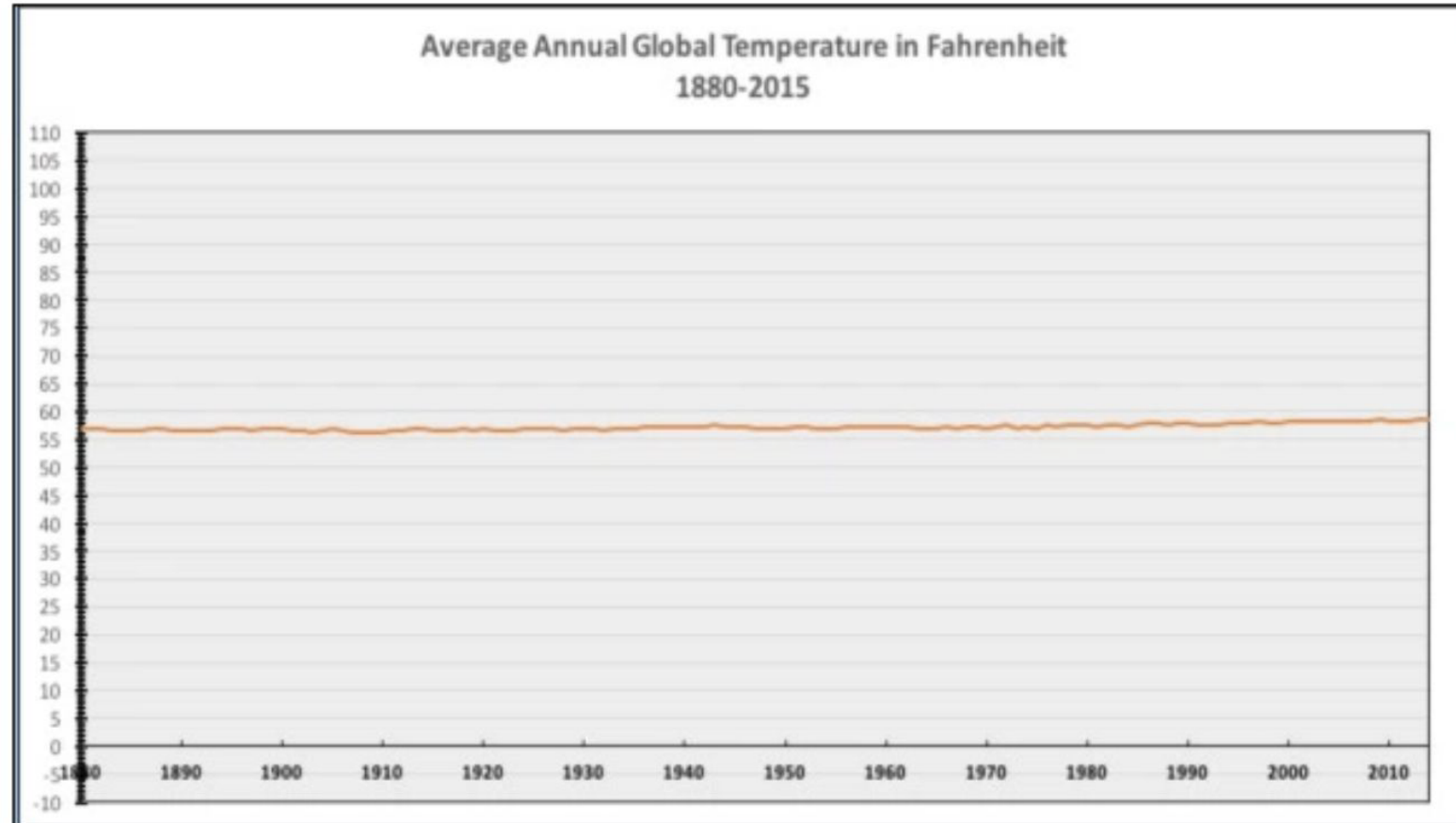
Line charts

CONNECT YOUR DOT!!



Line charts

What do you
conclude?

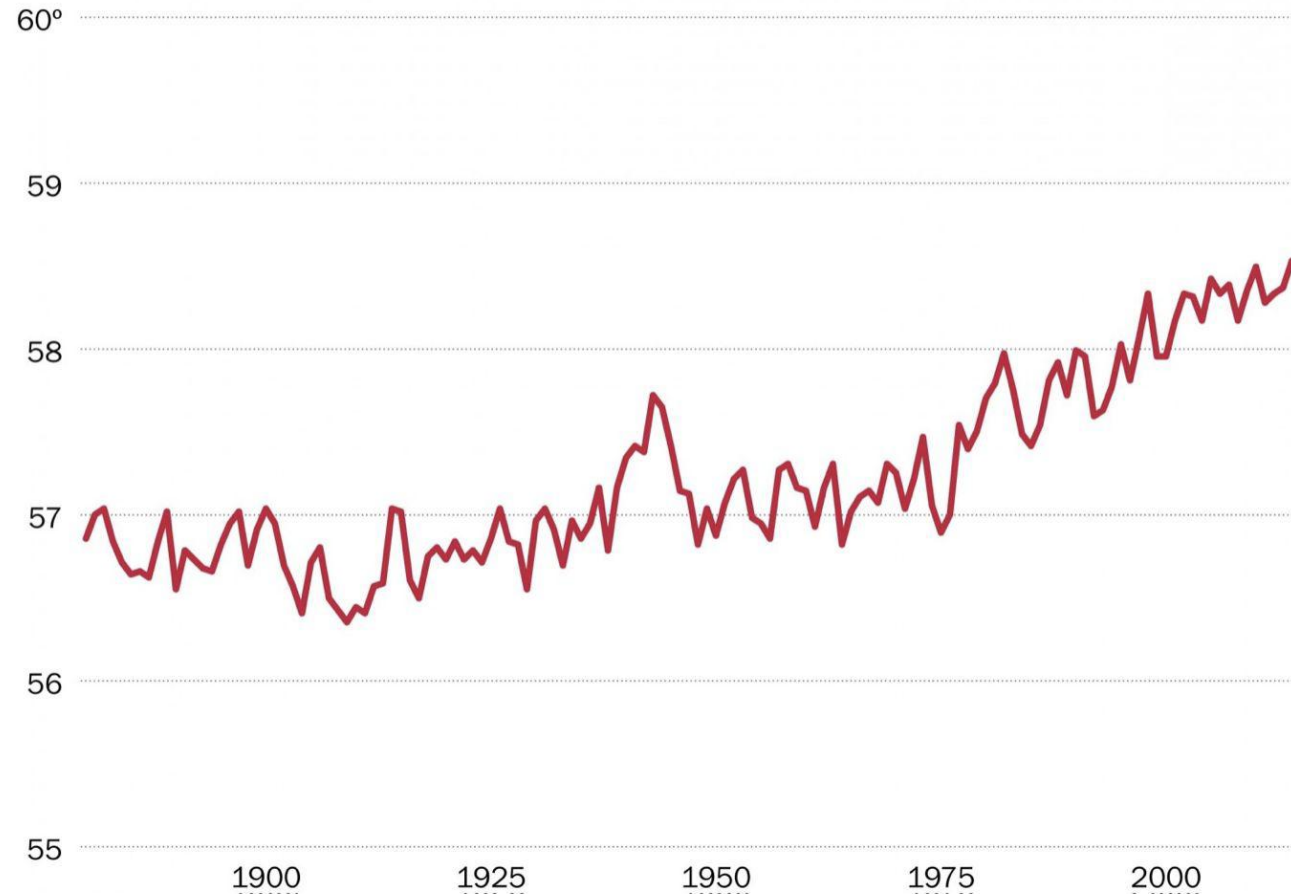


Line charts

What do you
conclude?

Average global temperature by year

Data from NASA/GISS.

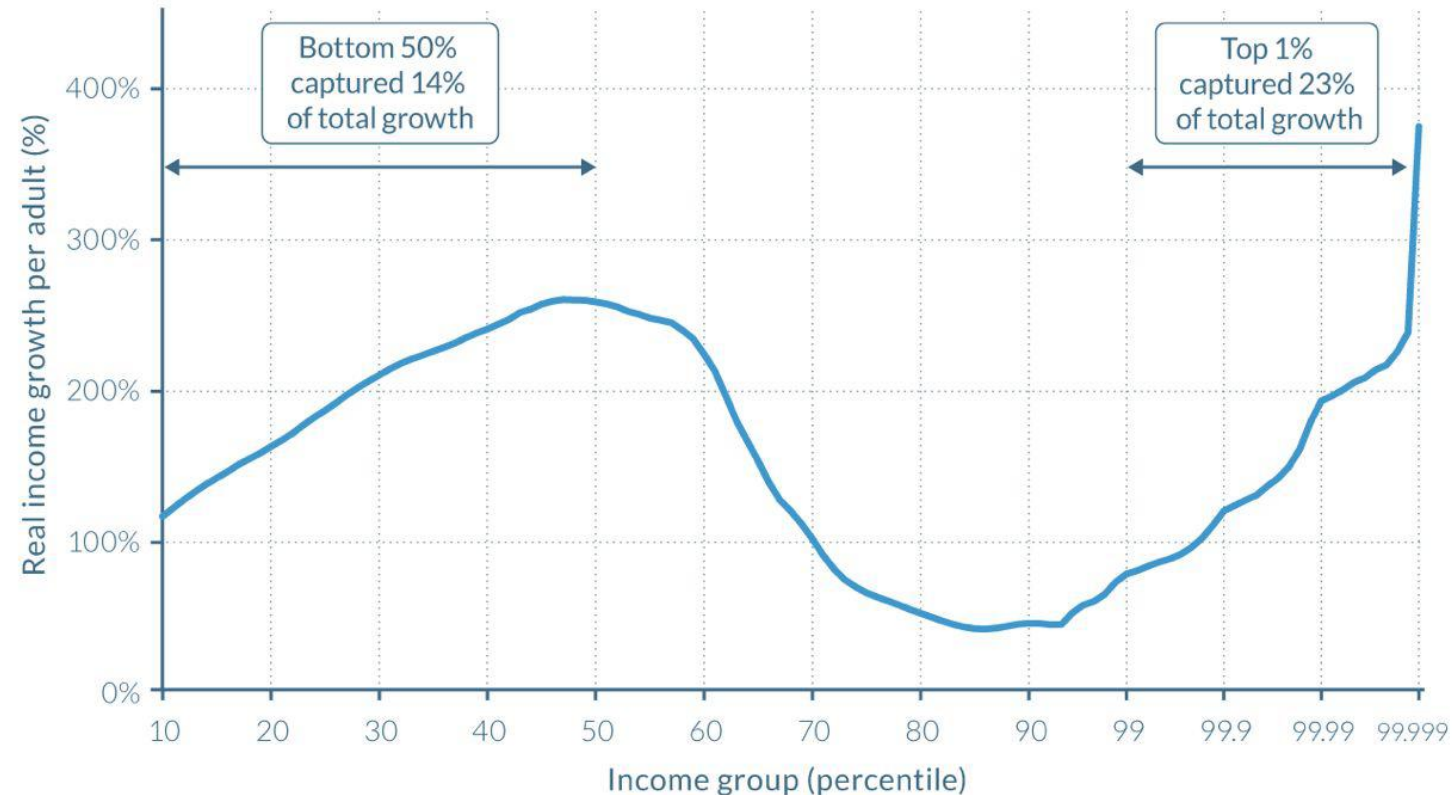


Line charts

1) no strict convention for starting the y axis at 0. By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).

Line charts

What do you see?



Source: WID.world (2017). See wir2018.wid.world/methodology.html for data series and notes.

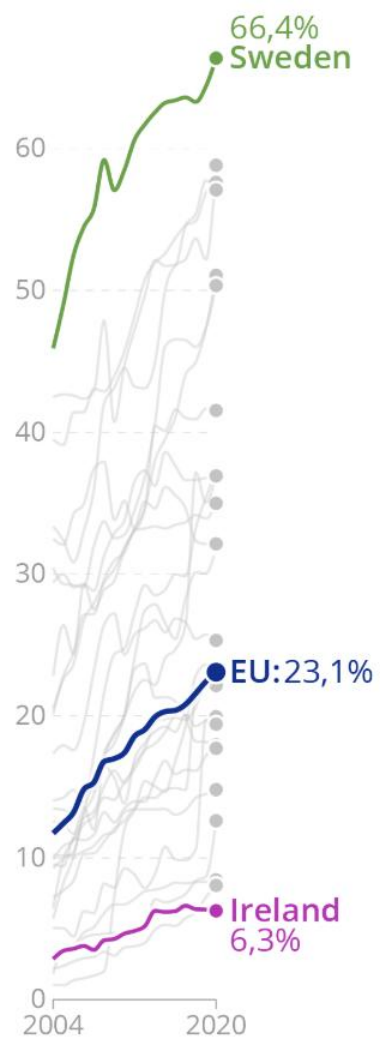
On the horizontal axis, the world population is divided into a hundred groups of equal population size and sorted in ascending order from left to right, according to each group's income level. The Top 1% group is divided into ten groups, the richest of these groups is also divided into ten groups, and the very top group is again divided into ten groups of equal population size. The vertical axis shows the total income growth of an average individual in each group between 1980 and 2016. For percentile group p99p99.1 (the poorest 10% among the world's richest 1%), growth was 77% between 1980 and 2016. The Top 1% captured 23% of total growth over this period. Income estimates account for differences in the cost of living between countries. Values are net of inflation.

Line charts

- 1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).
- 2) scale** you choose on your axes must **remain consistent**.

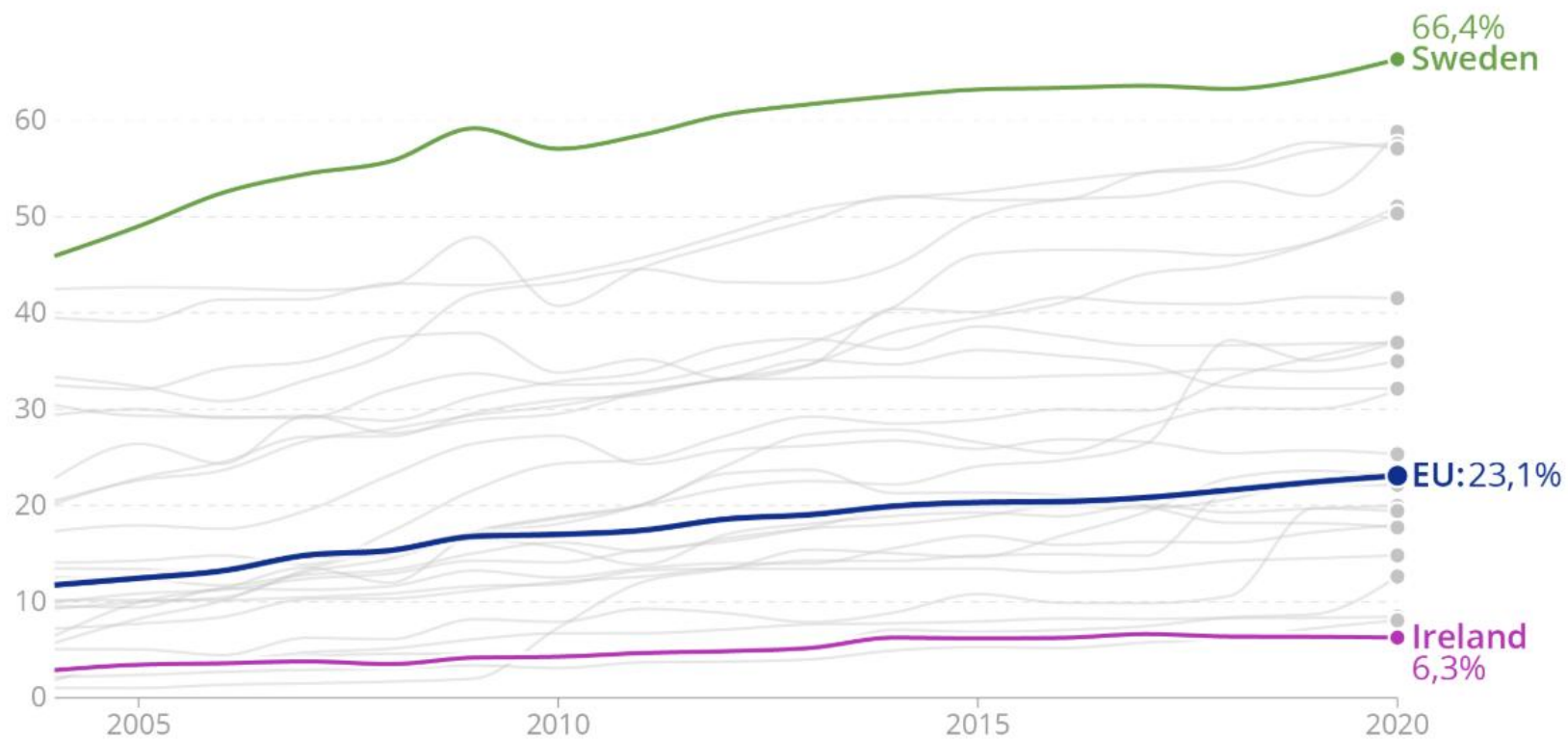
Line charts

What do you
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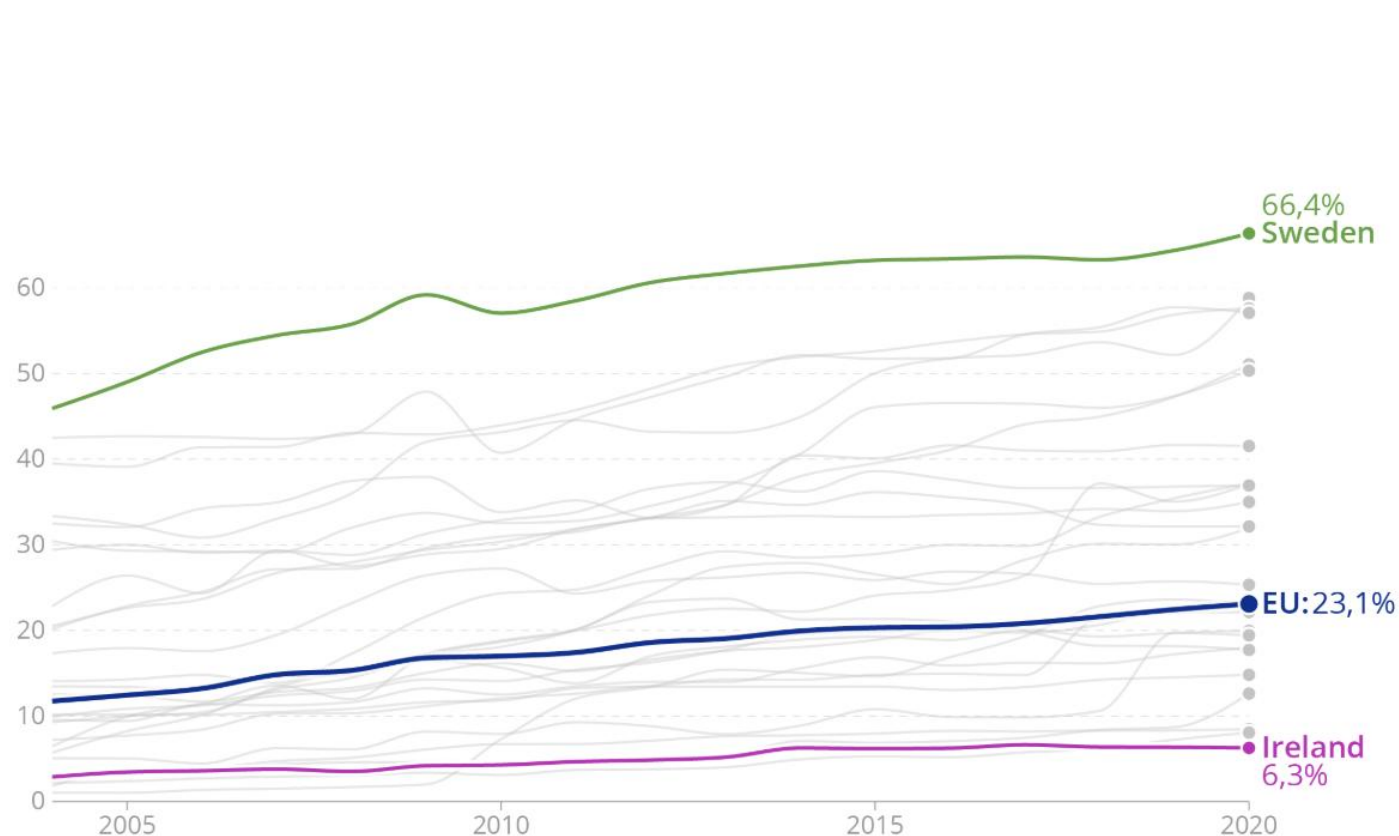


Line charts

What do you
conclude?



Line charts



A wide and flat layout suggest slow trends or even flat lines. Source: Maarten Lambrechts, CC BY 4.0



Narrow and high designs exaggerate trends. Source: Maarten Lambrechts, CC BY 4.0

Line charts

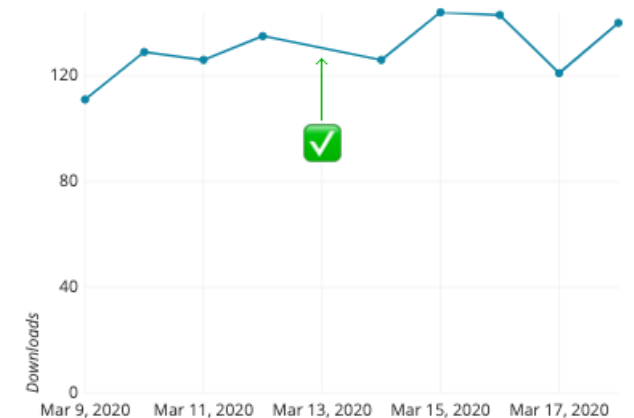
- 1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).
- 2) scale** you choose on your axes must **remain consistent**.
- 3) Respect the length/weight ratio of your graph**

Line charts

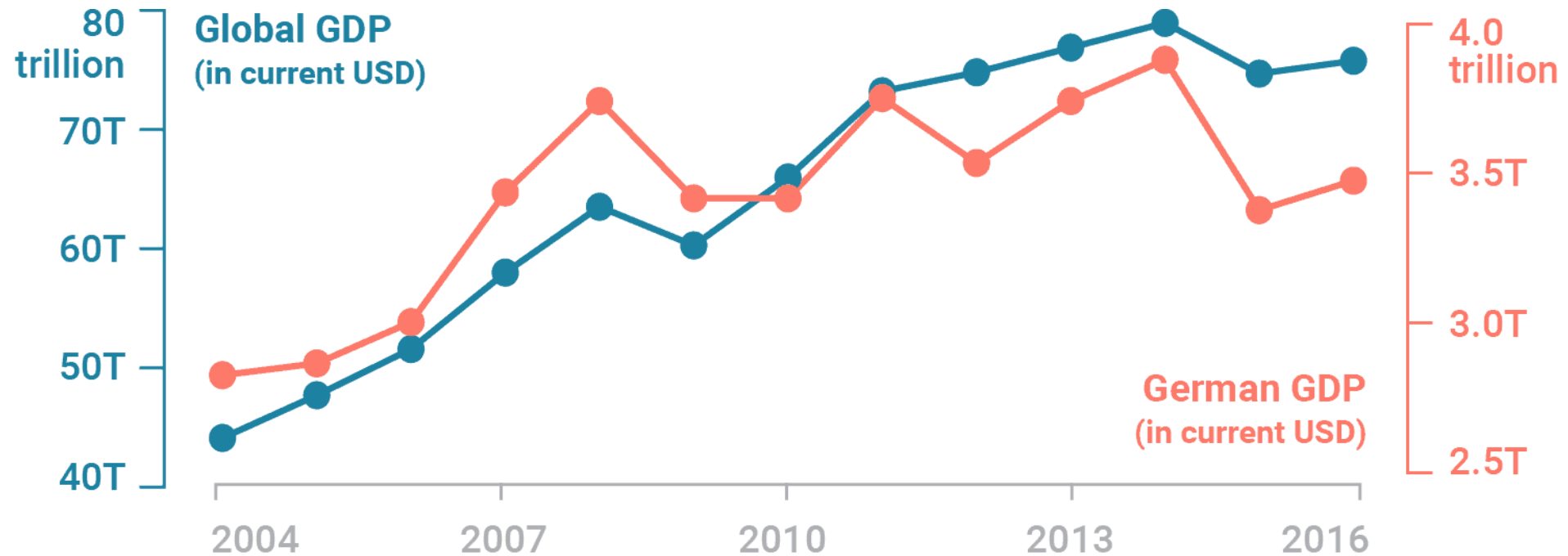
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Line charts

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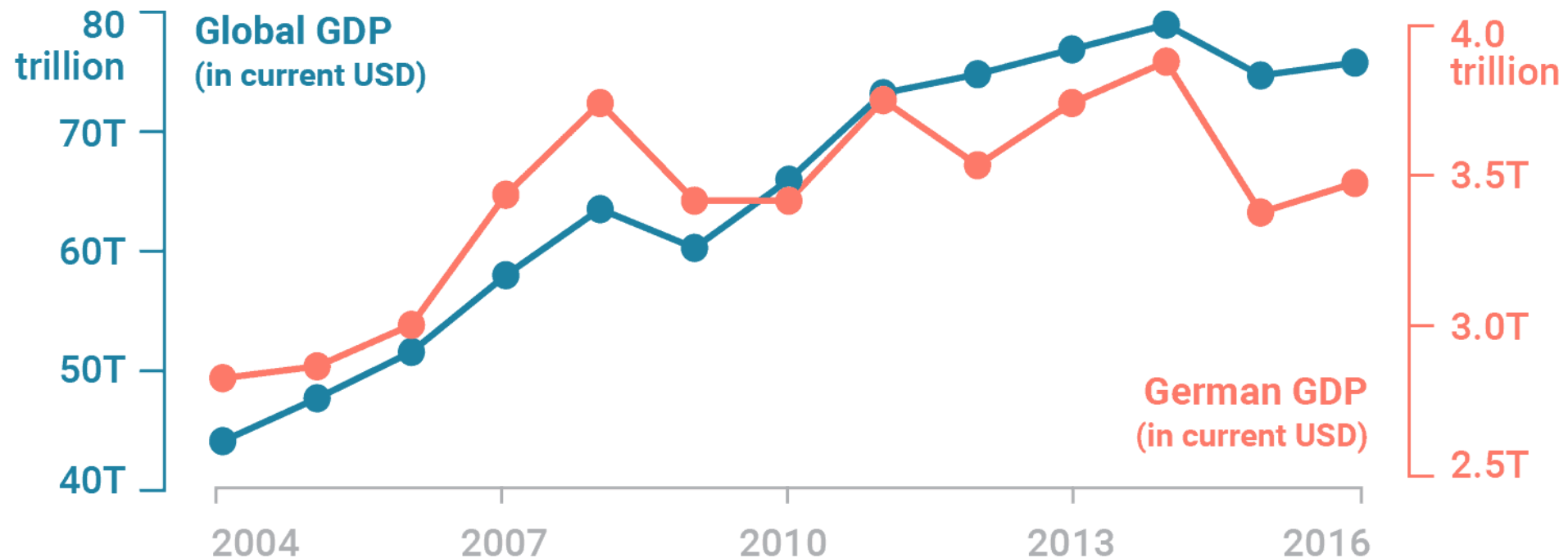


Line charts



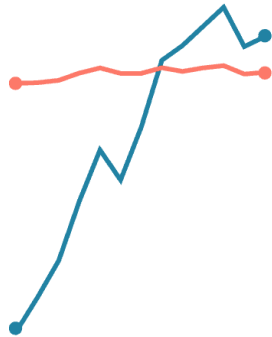
Line charts

AVOID DUAL SCALE: The scales of dual axis charts are arbitrary and can therefore (deliberately) mislead readers about the relationship between the two data series.

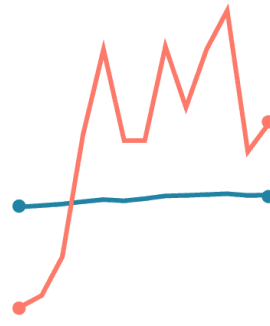


the chart looks like the German GDP and the global GDP go up at roughly the same rate (at least until 2014), they don't. The global GDP increased by 80% until 2014; the GDP of Germany by 40%

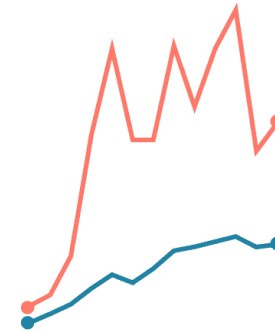
Line charts



Orange steady,
Blue massively increasing.



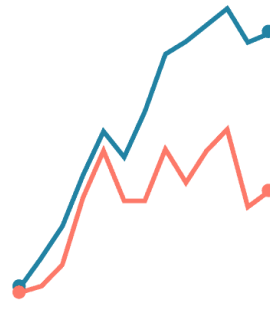
Blue steady,
Orange increasing.



Both started at the same
level, but Orange increased
far more than Blue.



Both started at the same
level, but Blue increased far
more than Orange.



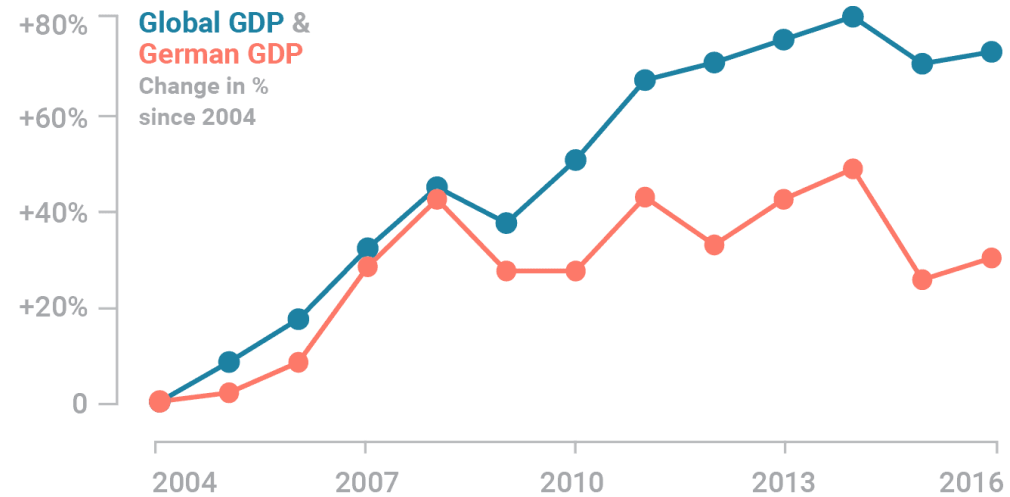
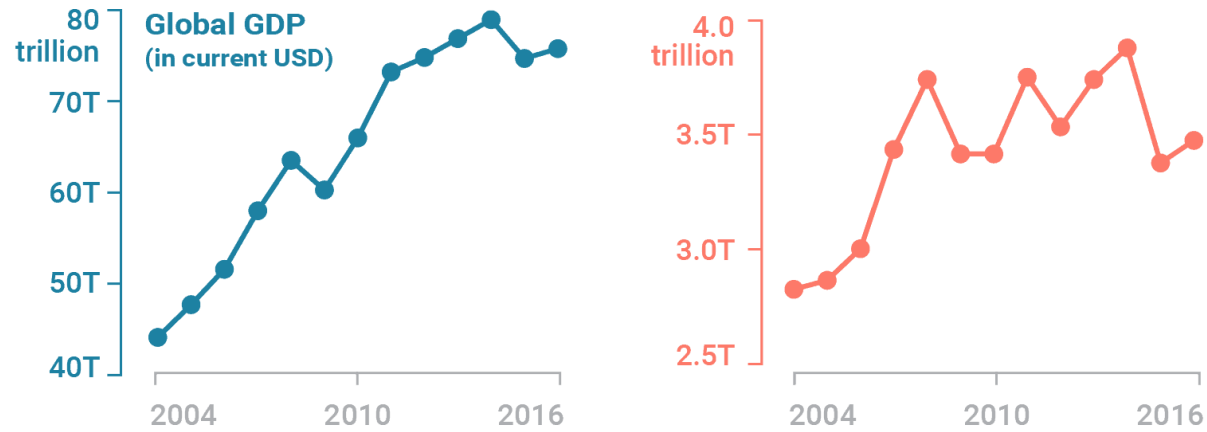
Both started with the
same increase, then Blue
raced to the top.



Both steady.

Line charts

Solution

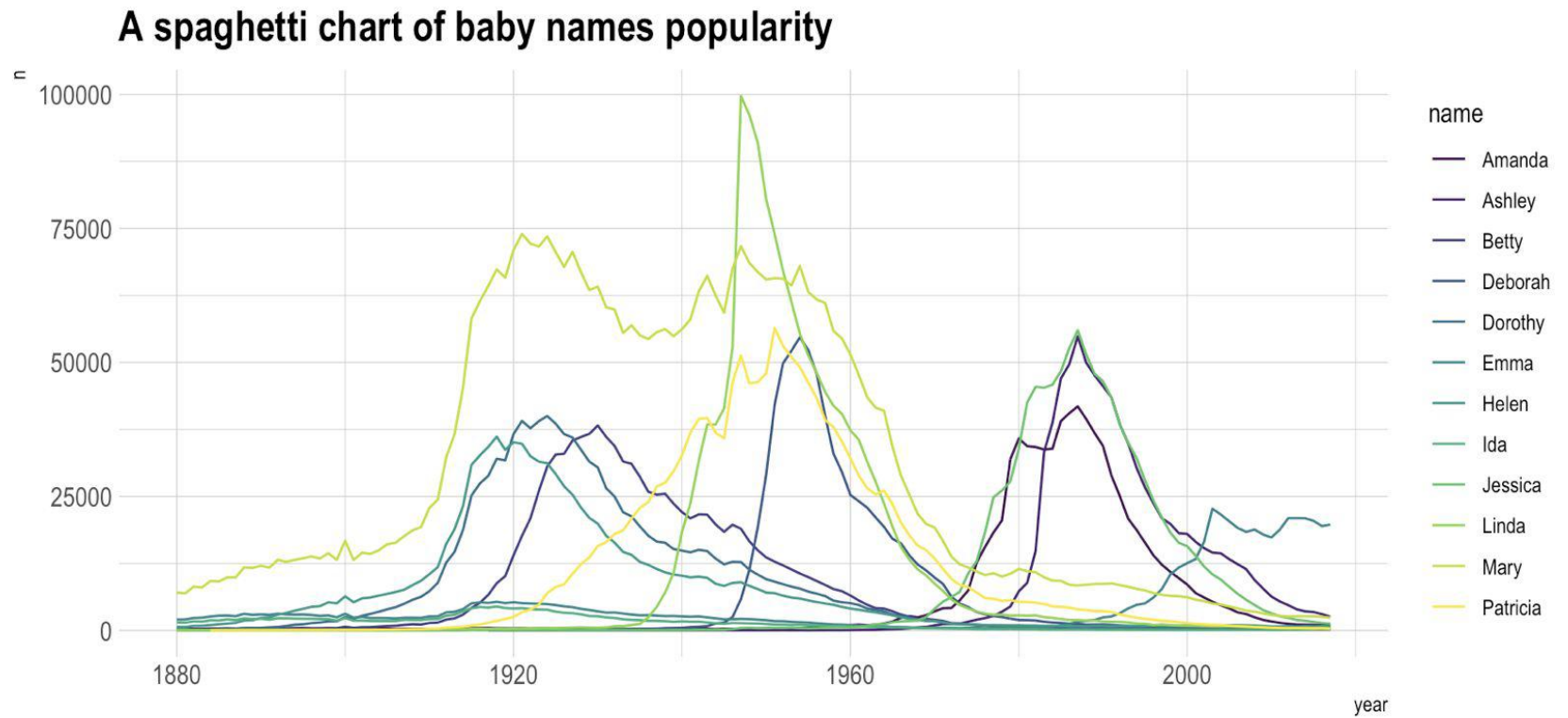


Line charts

- 1) **no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).
- 2) **scale** you choose on your axes must **remain consistent.**
- 3) **Respect the length/weight ratio of your graph**
- 4) **Time goes from Left to Right**
- 5) **If you have missing data, make it clear from the chart or show dots**
- 6) **Avoid dual scales**

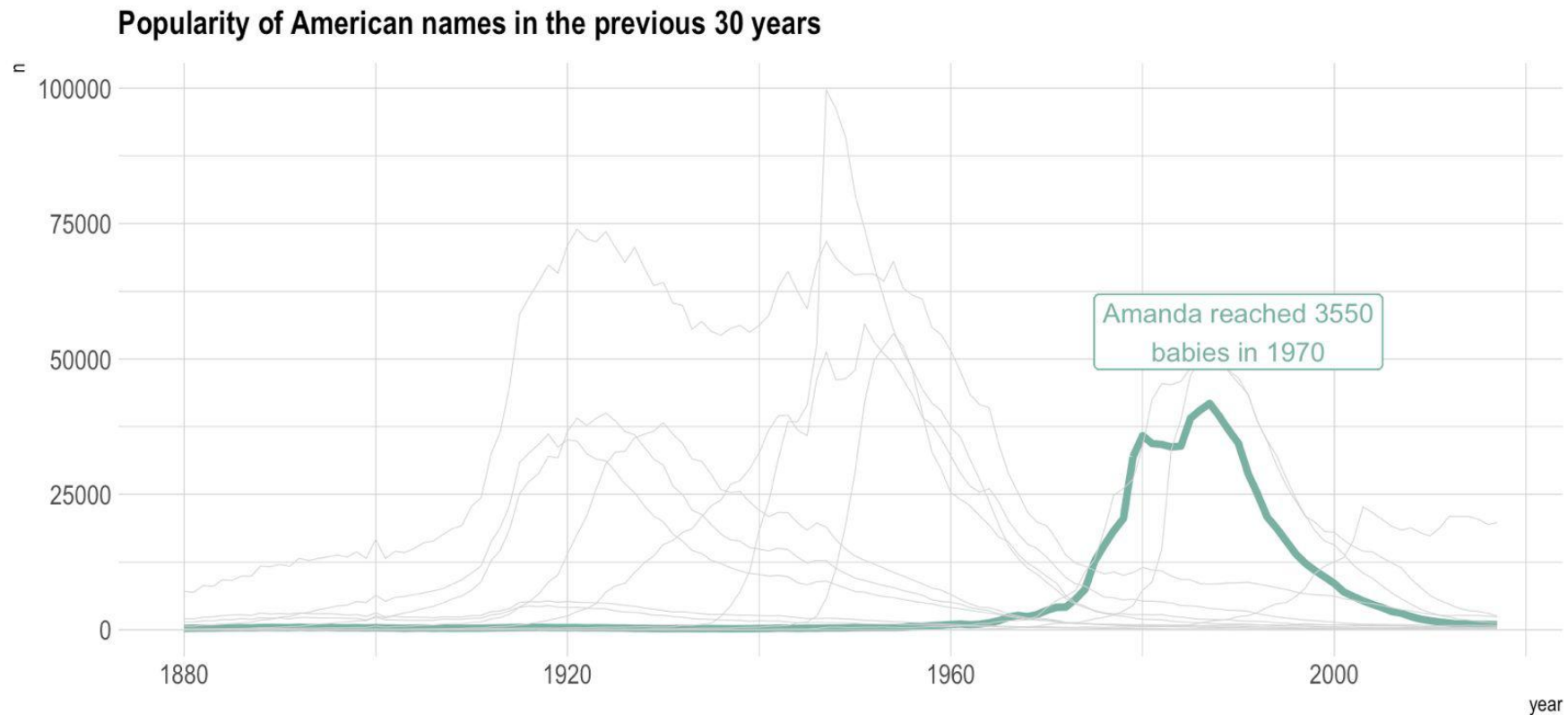
Line charts

What do you see?



Line charts

If you cannot reduce your categories, you can for example **gray** out all your curves, except the one you want to draw attention to, leaving it in color. This allows for a clear and readable narration.



Line charts

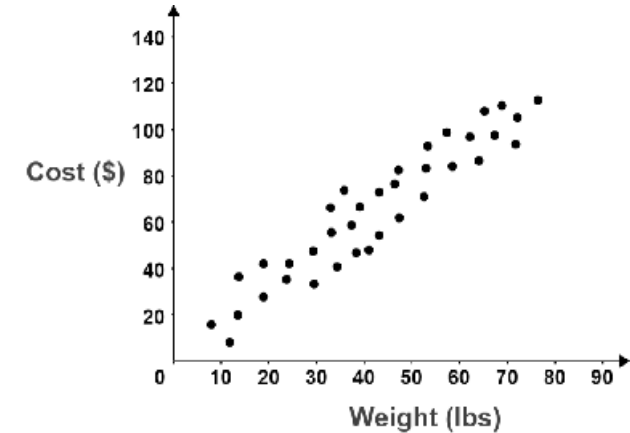
- 1) **no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).
- 2) **scale** you choose on your axes must **remain consistent.**
- 3) **Respect the length/weight ratio of your graph**
- 4) **Time goes from Left to Right**
- 5) **If you have missing data, make it clear from the chart or show dots**
- 6) **Avoid dual scales**
- 7) **Limit the number of lines, highlight the one you want to draw attention**

Scatter charts

A scatter plot (uses dots to represent values for two different numeric variables. The position of each dot on the horizontal and vertical axis indicates values for an individual data point. Scatter plots are used to observe relationships between variables.

Useful

- to observe and show relationships between two numeric variables. Identification of correlational relationships are common with scatter plots.
- identifying other patterns in data



data clusters (two)



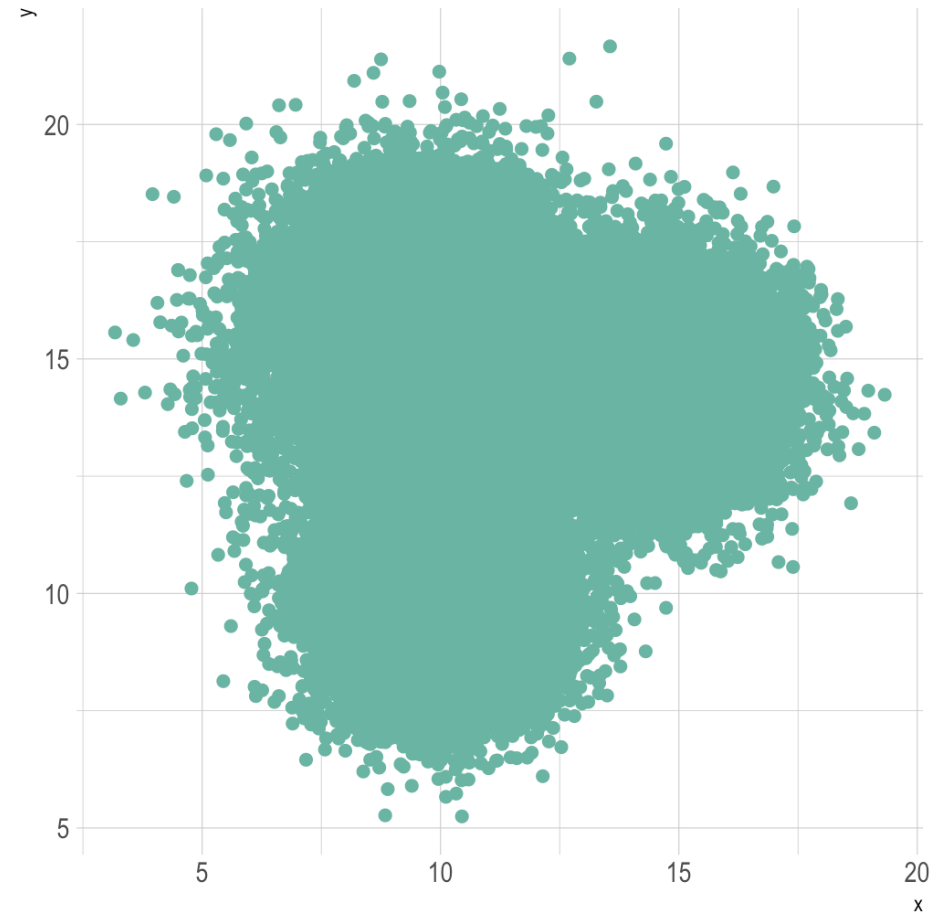
gap in values



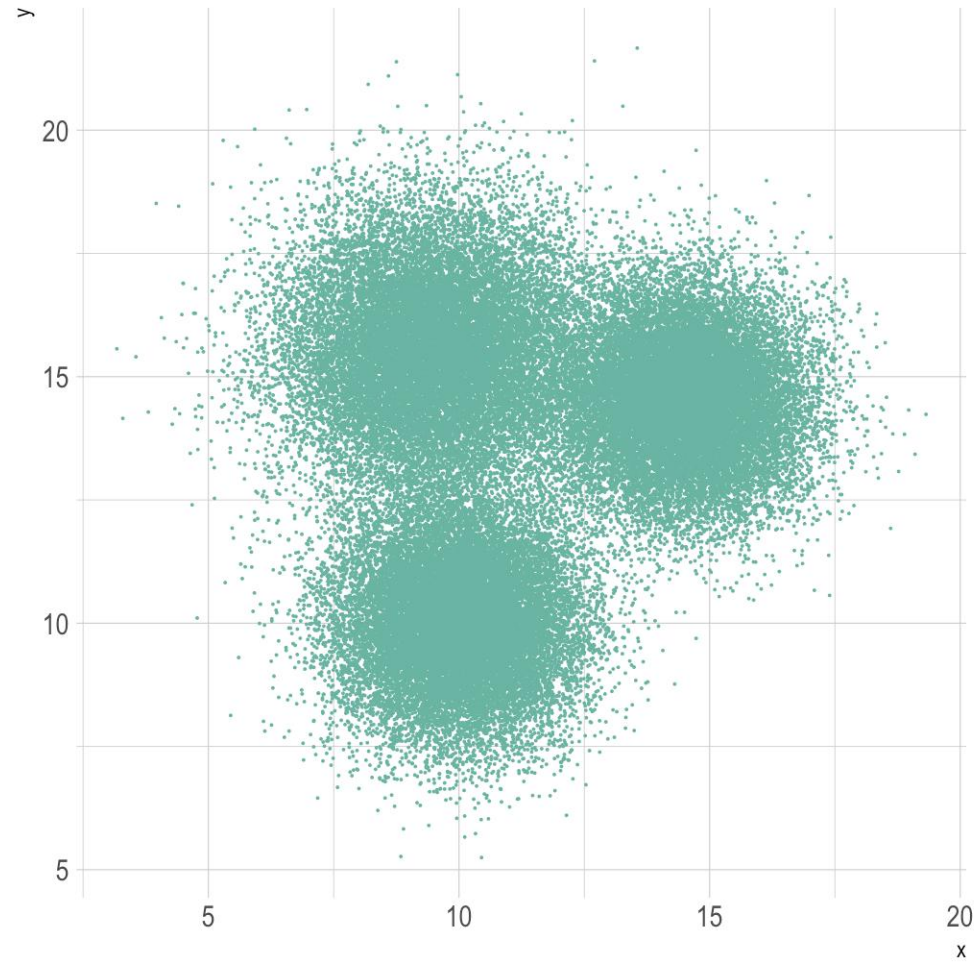
outliers (lower left)

Scatter charts

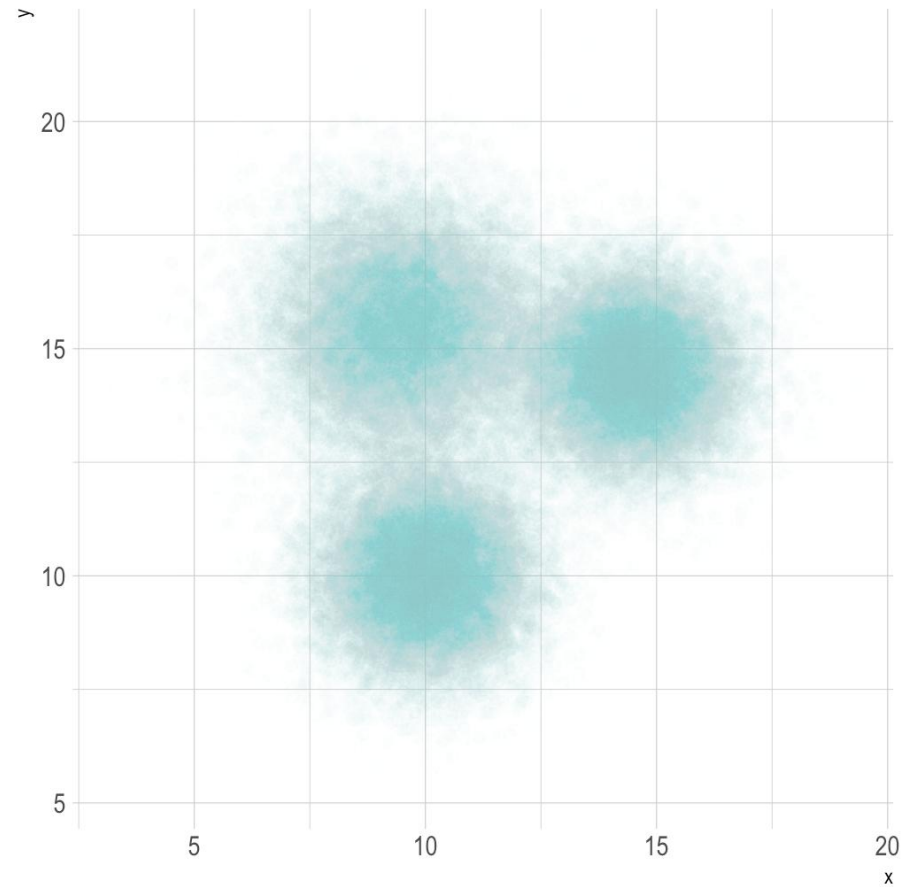
What to do?



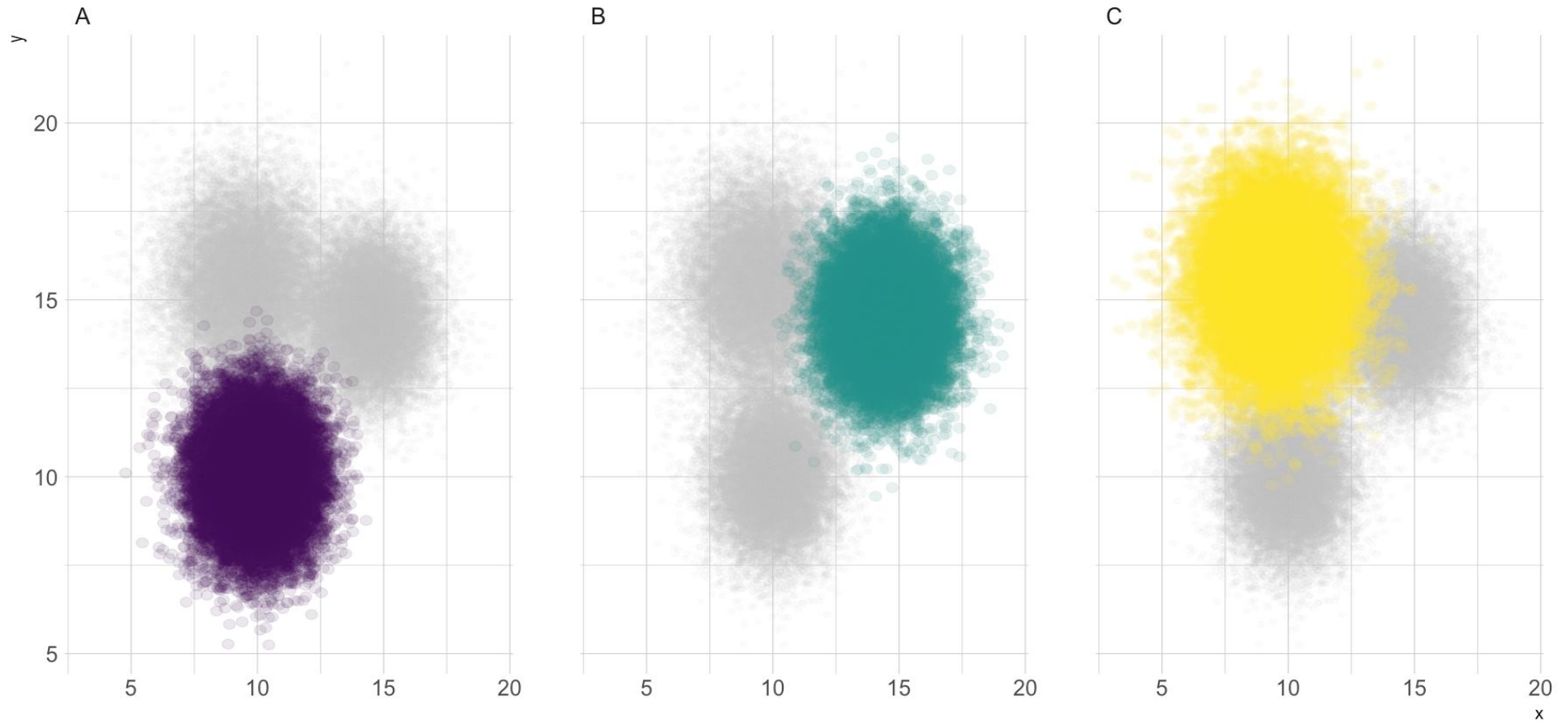
Scatter charts



Scatter charts

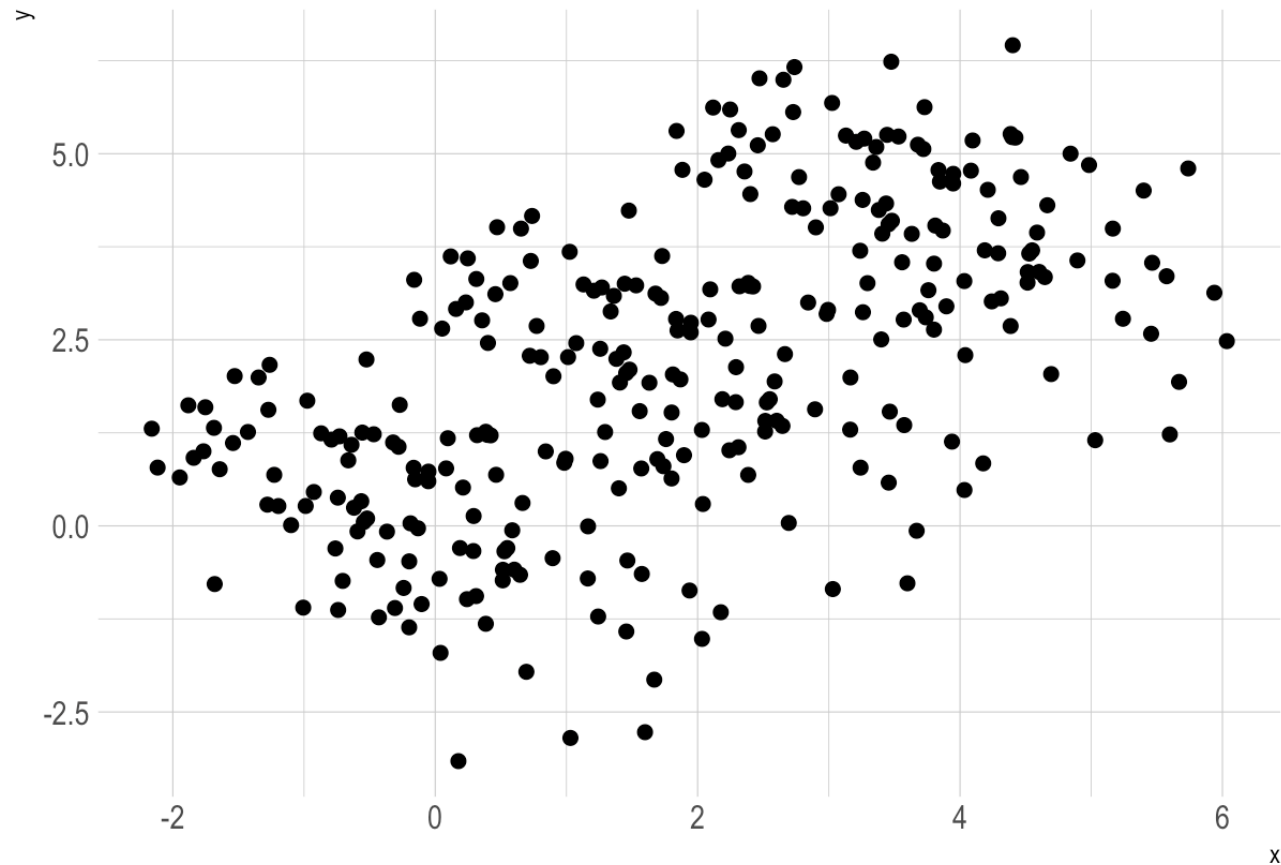


Scatter charts



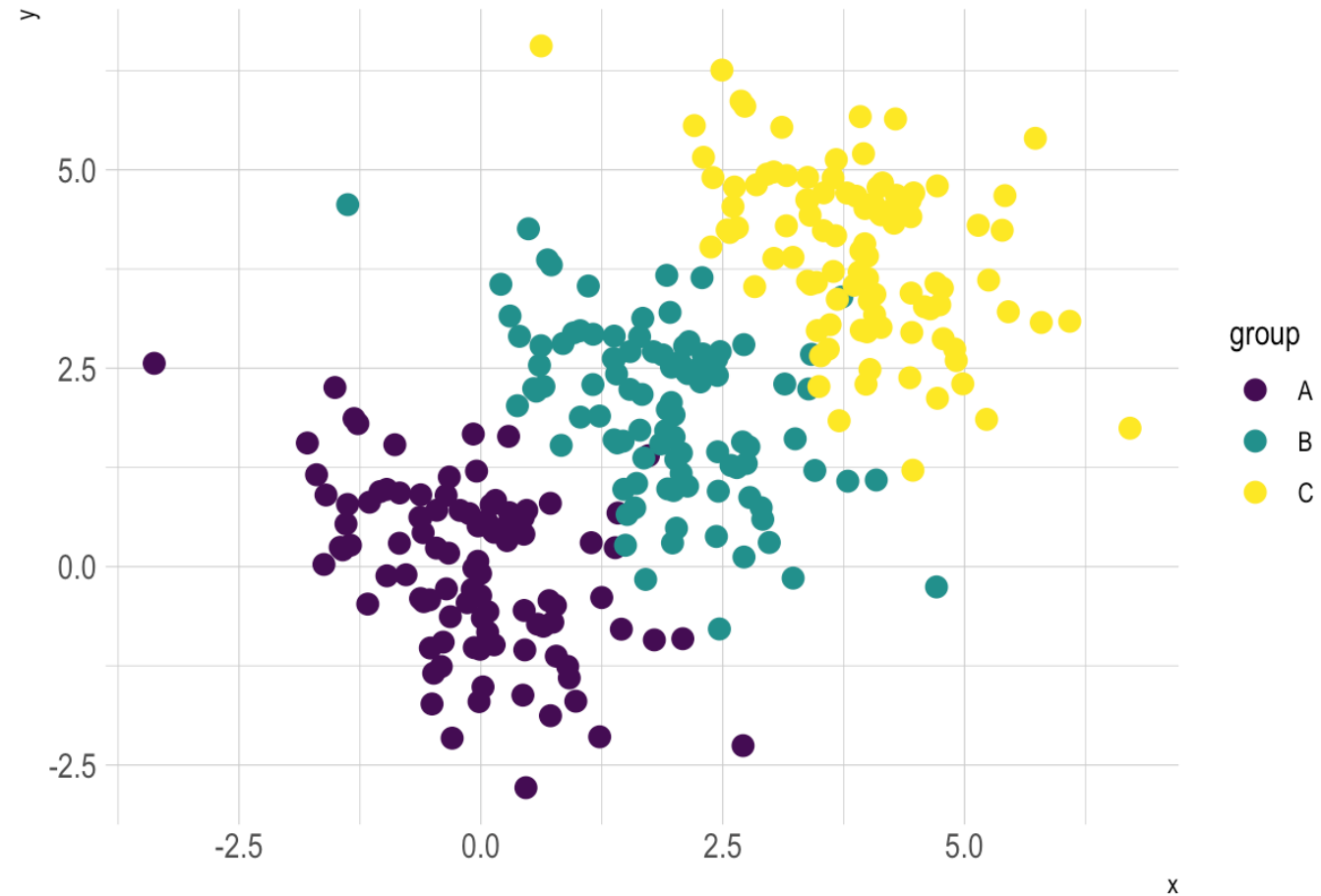
Scatter charts

Do you see a correlation?



Scatter charts

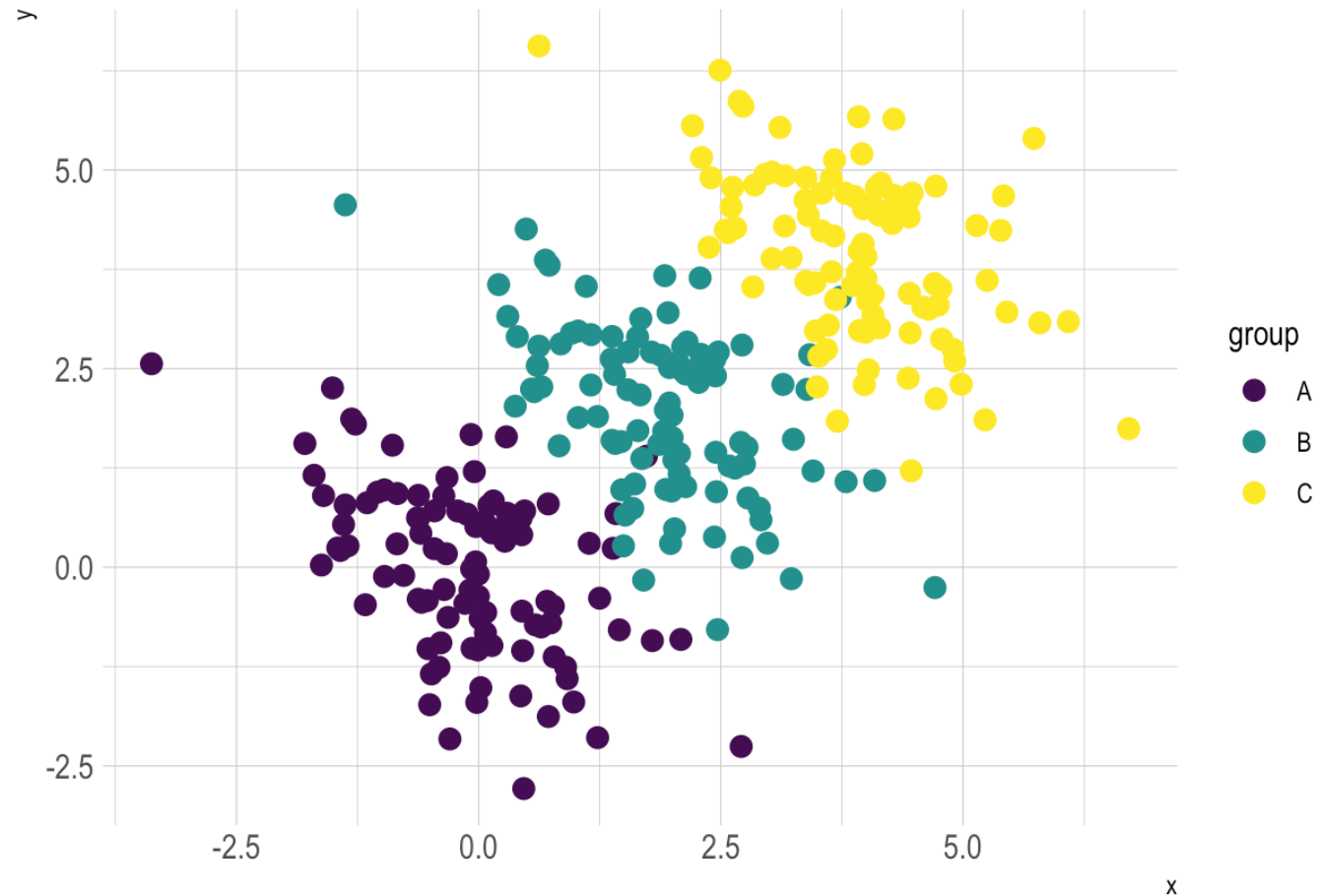
What happens if we
group?



Scatter charts

What happens if we group?

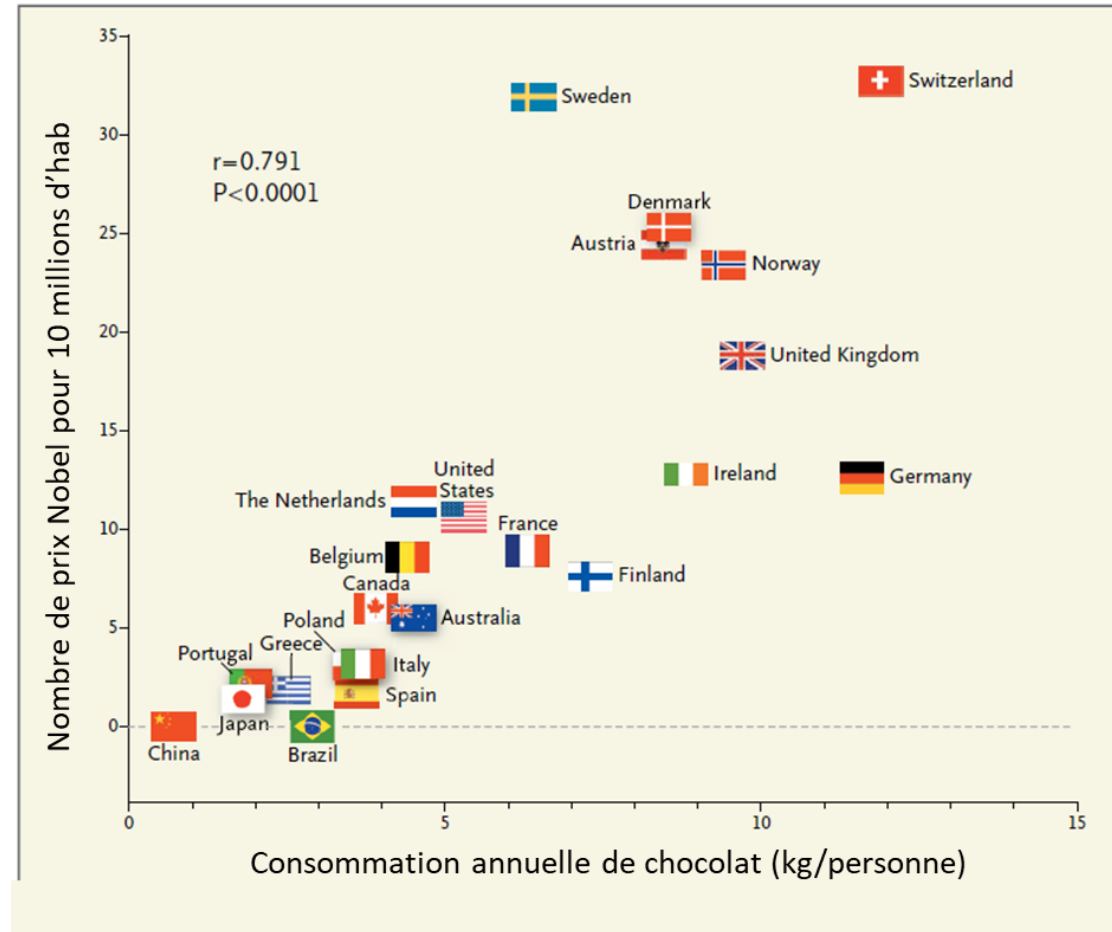
Simpson's paradox: the trend between two different variables reverses when a third variable is included



Correlation != Causality

Causality: Relationship between two events where one event is affected by the other.

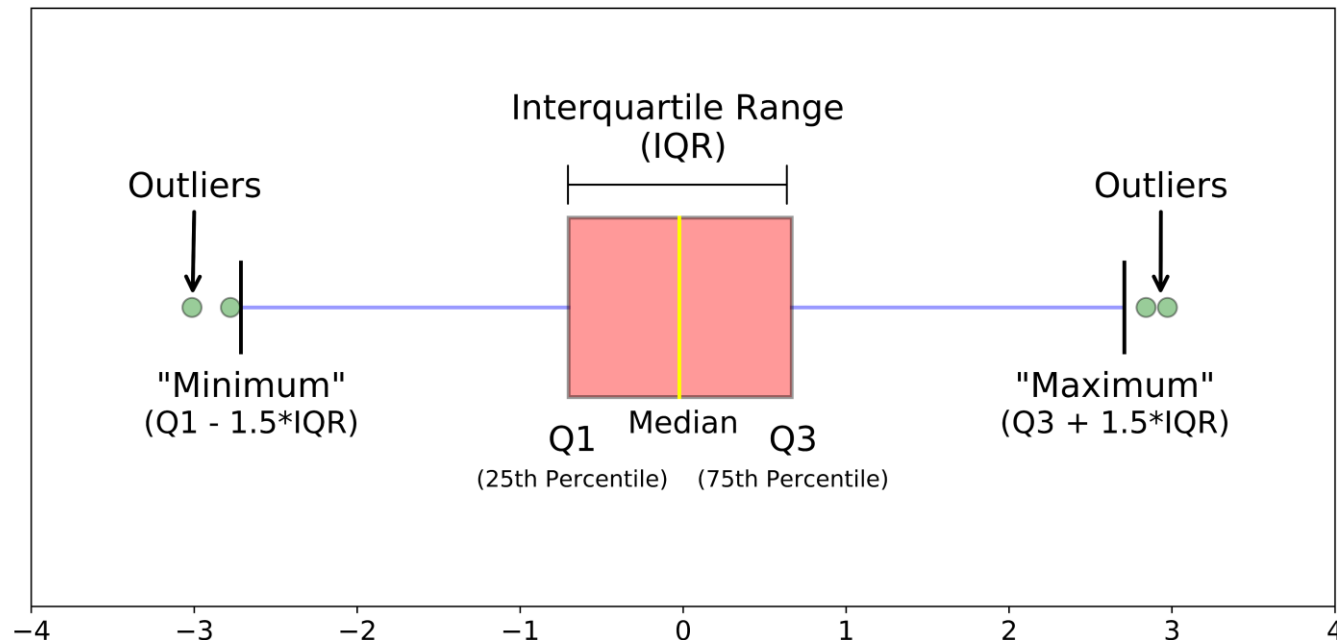
Correlation: Quantitative measure of joint variability between two or more variables. It is between -1 to 1.



Boxplot charts

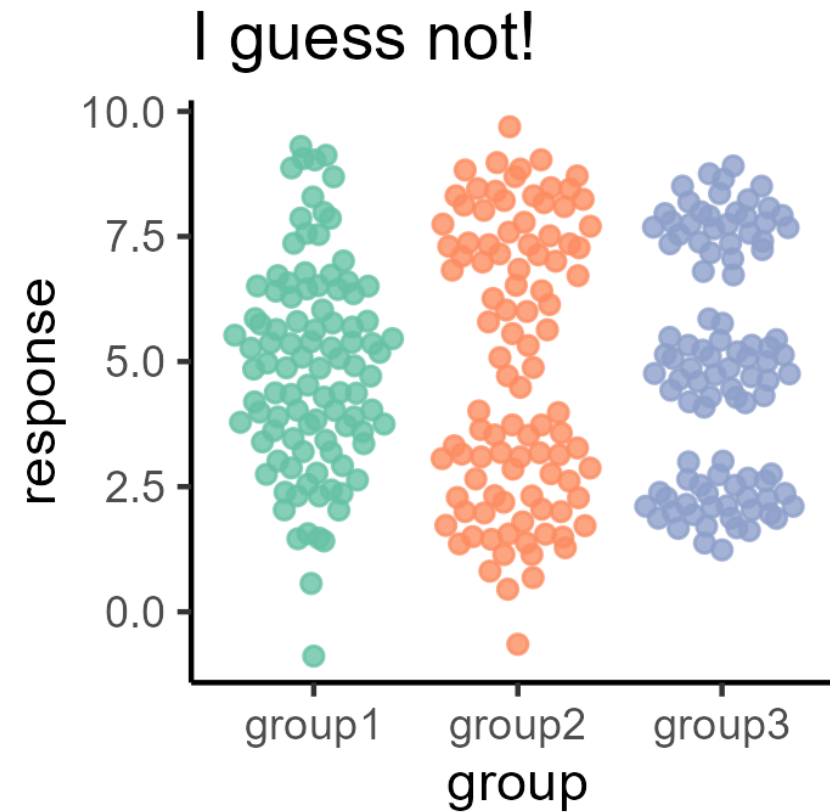
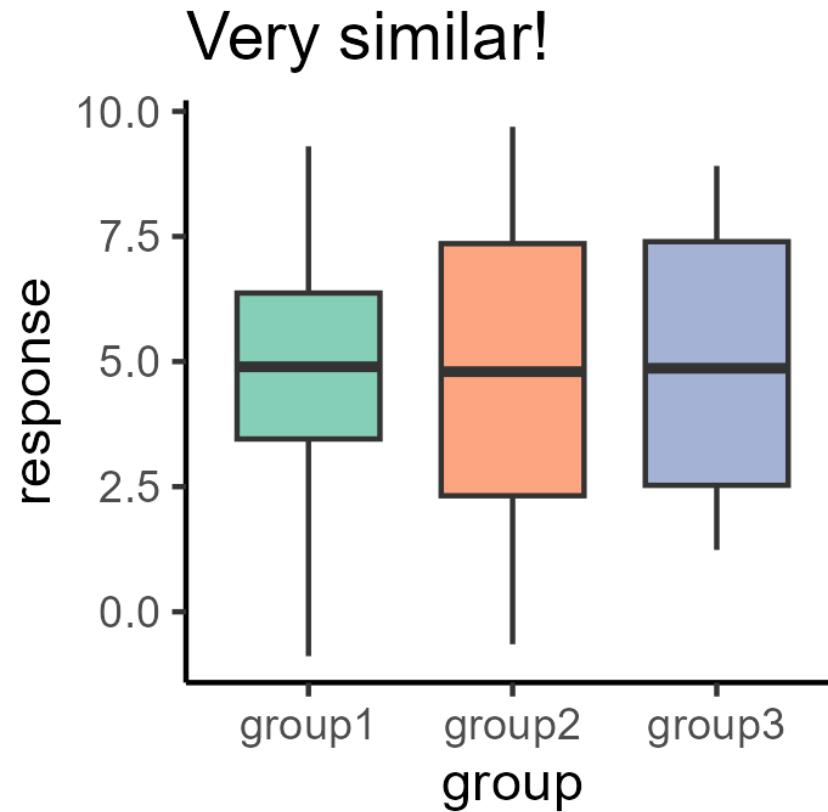
A box plot uses boxes and lines to depict the distributions of one or more groups of numeric data.

Your primary objective is to show distributions of numeric data values, especially when you want to compare them between multiple groups



Boxplot charts

If we are not comparing unimodal distributions, boxplots can be misleading and hide the real distributions. Use violin charts

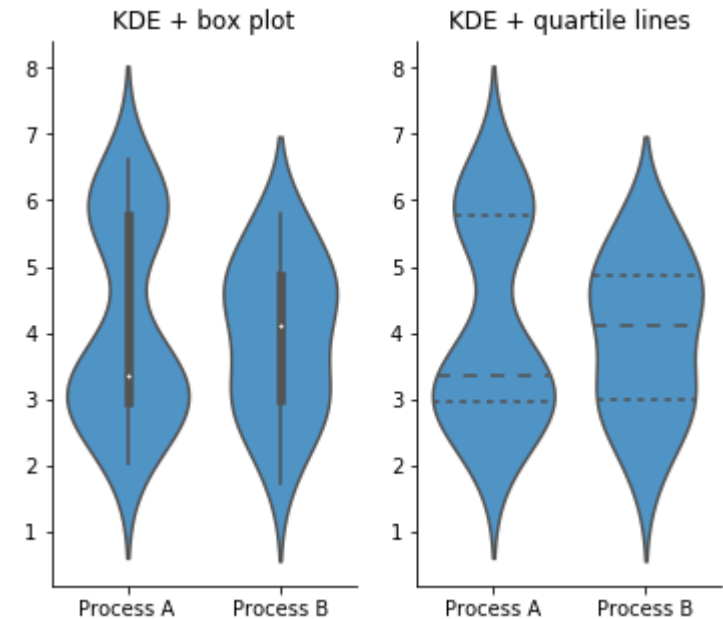


Violon charts

A violin plot depicts distributions of numeric data for one or more groups using density curves. The width of each curve corresponds with the approximate frequency of data points in each region

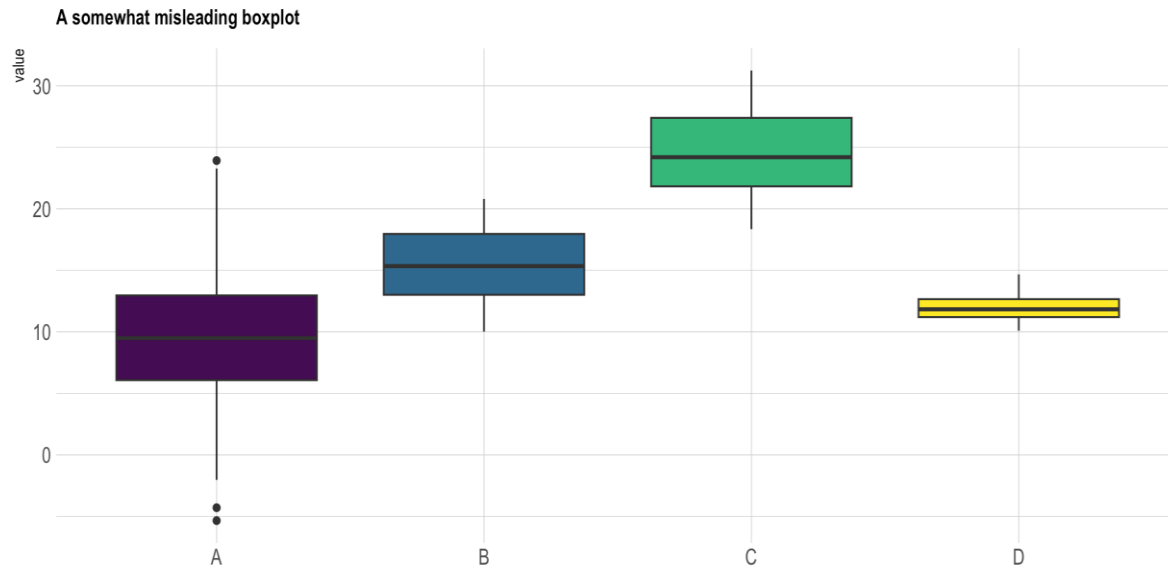
Violin plots are used when you want to observe the distribution of numeric data, and are especially useful when you want to make a comparison of distributions between multiple groups.

Distrubutions can be multimodal!!



Violon charts

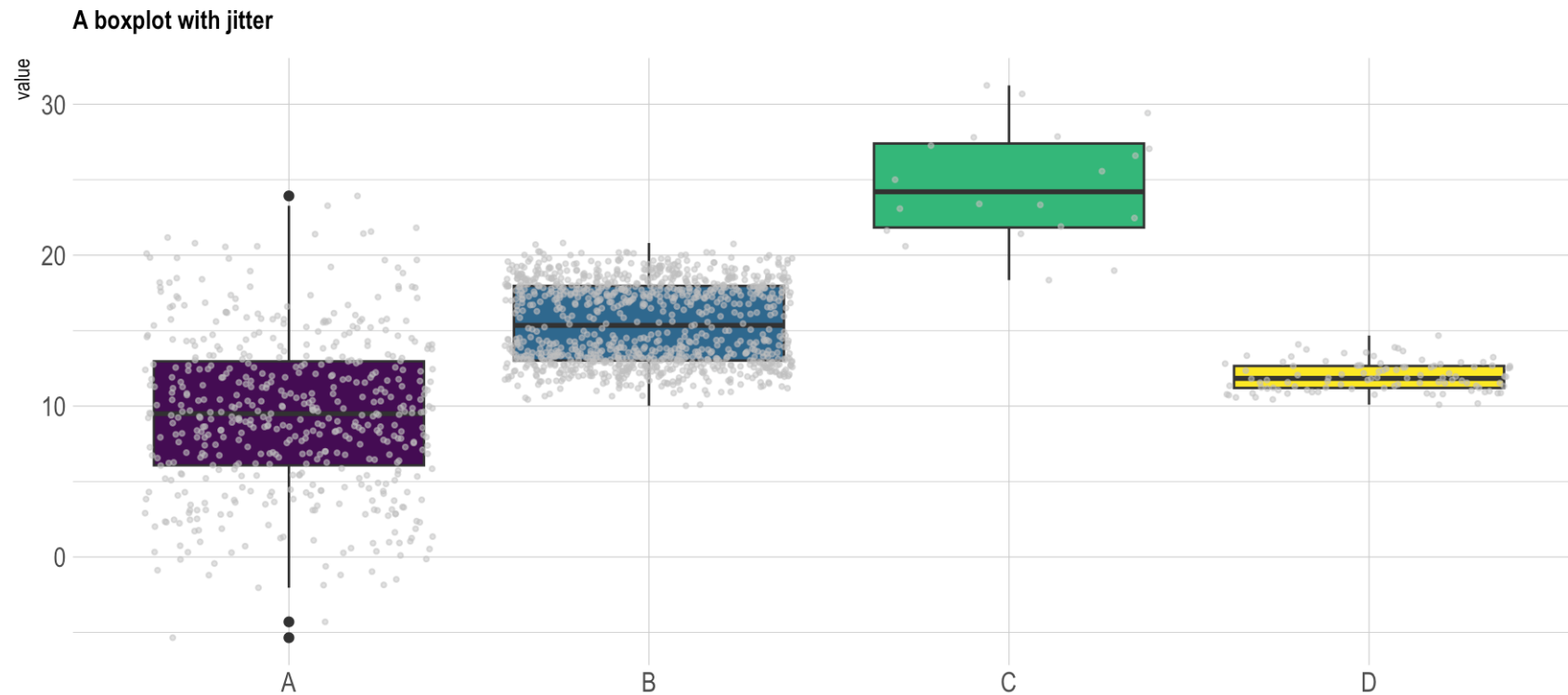
What do you see? It is obvious that group C has higher values



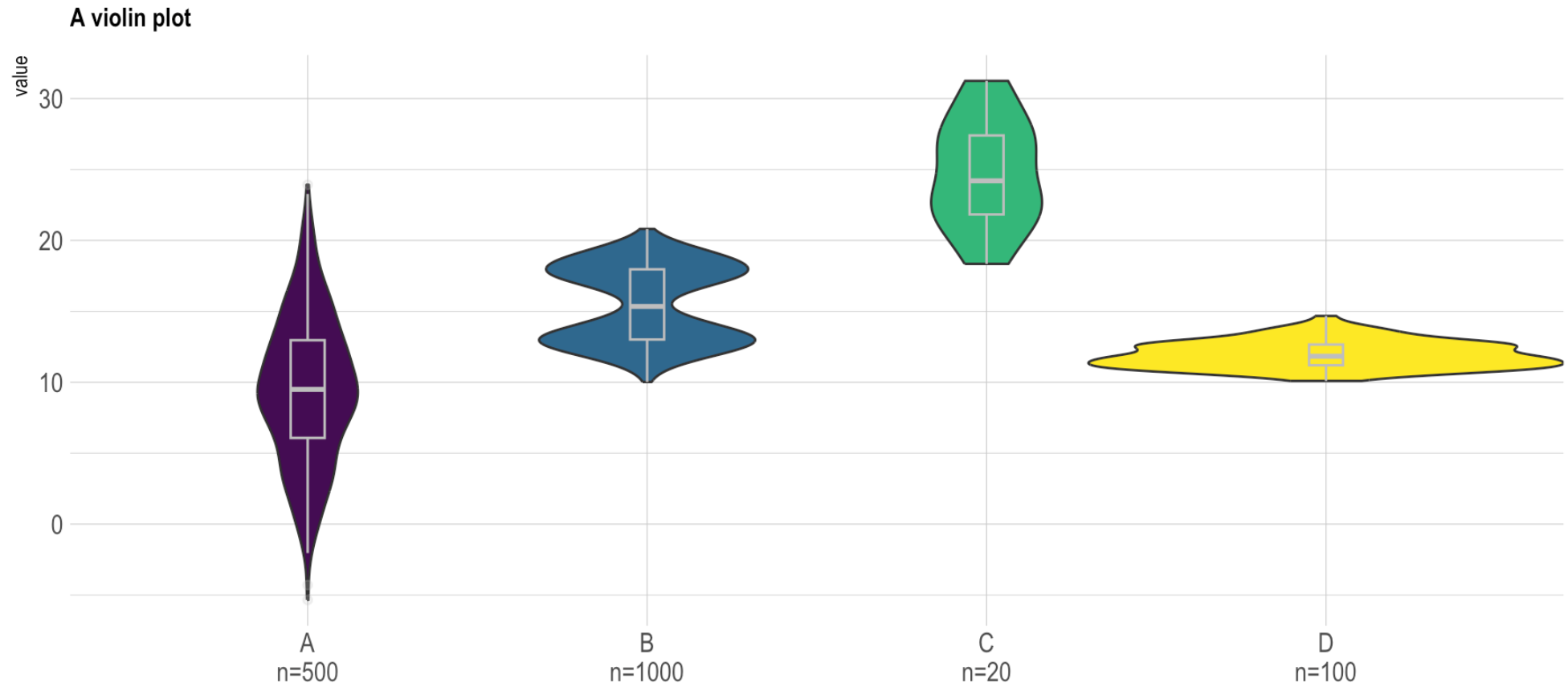
Violon charts

Group C has a smaller sample size compared to the other groups....

Group B exhibits a bimodal distribution



Violon charts



Heatmap

A heatmap displays data in a matrix format using color to represent values. It's ideal for spotting patterns, correlations and variations in large datasets.

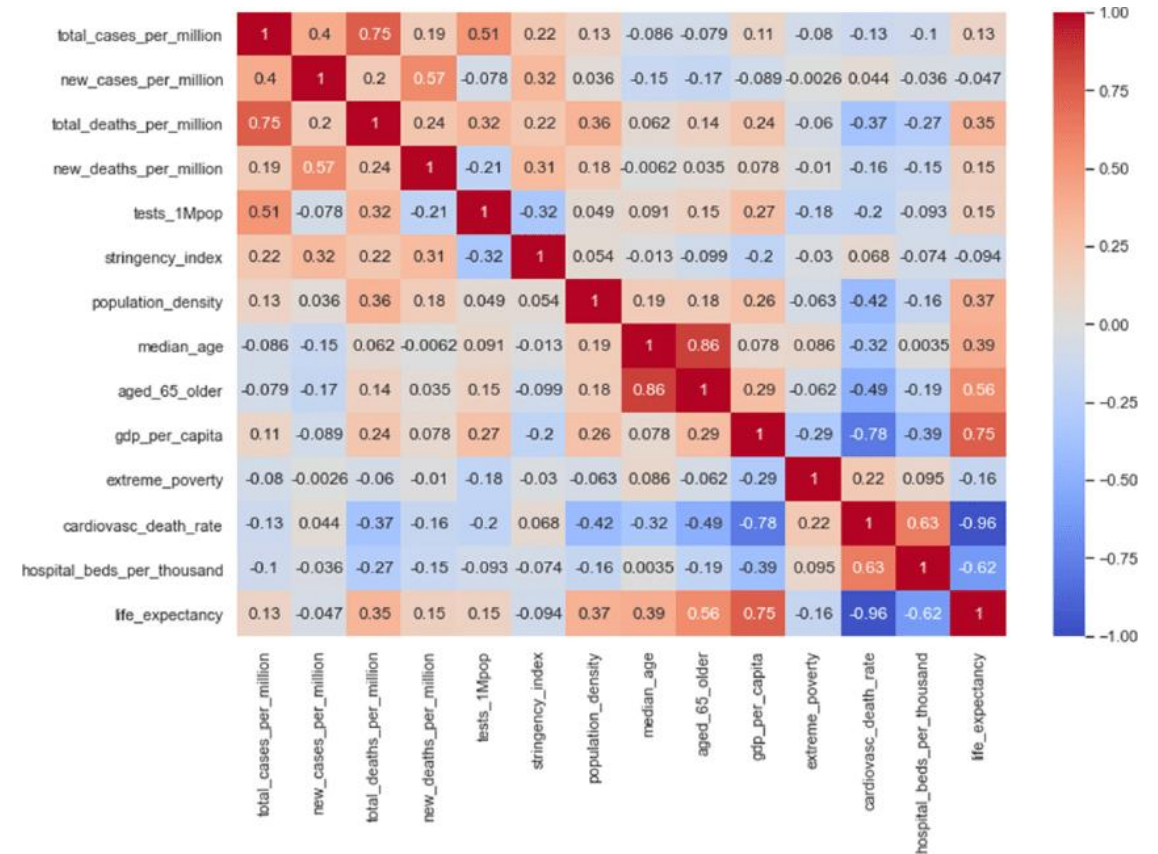
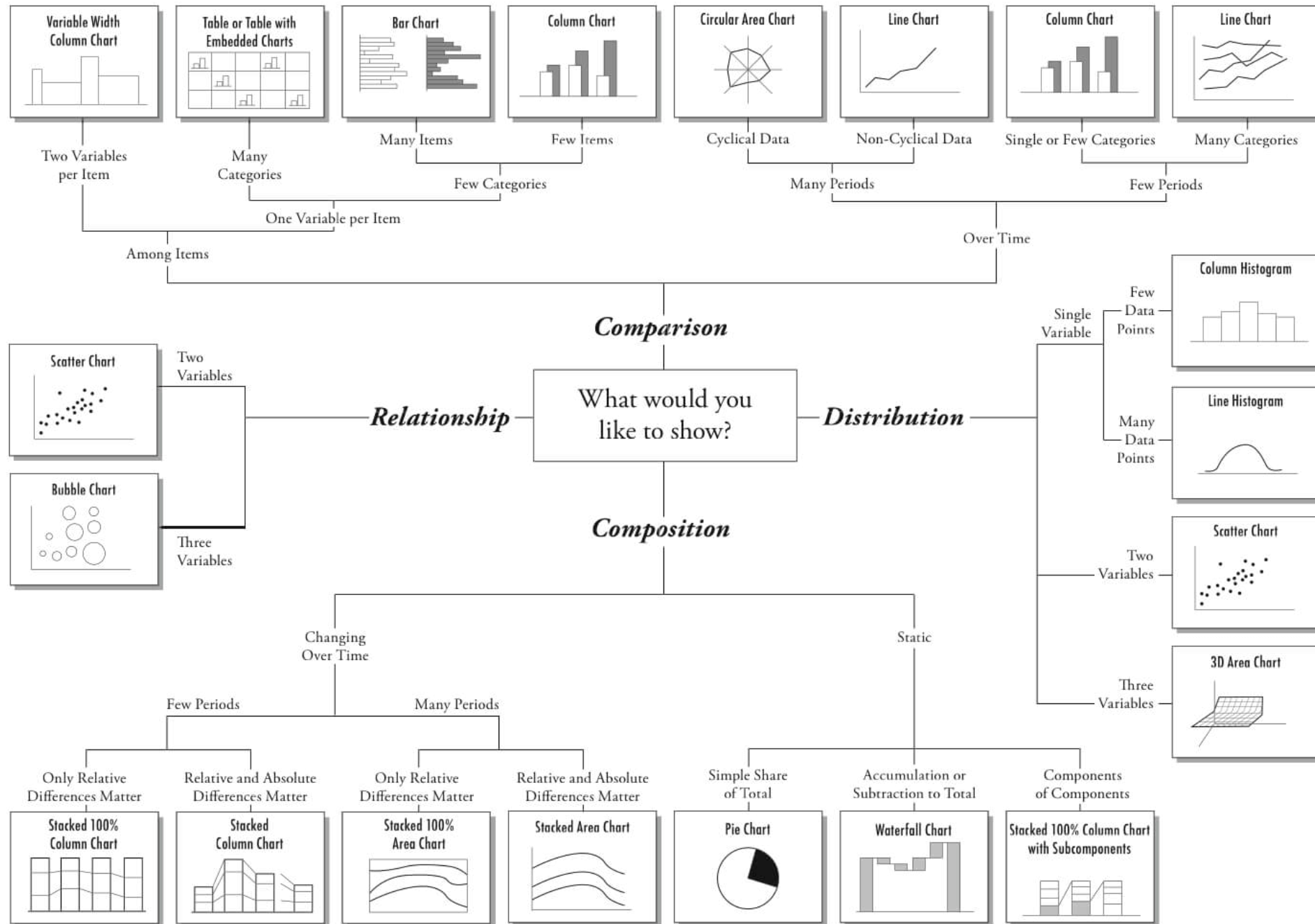


Chart Suggestions—A Thought-Starter

www.ExtremePresentation.com
© 2009 A. Abela — a.v.abela@gmail.com



SELECTING AND DESIGNING A DATA VISUALIZATION



BAR CHART

Shows: Comparison
Data: 2 variables -
categorical & numerical

- Always start axis at zero.
- Turn the entire chart horizontal (like above) if categories are too long or there are too many. *Avoid the tilt & squeeze!**
- Bars should be the same color unless it is a clustered or stacked bar chart.
- Arrange by category order. No order? Arrange bars by size.
- To make a clustered or stacked bar chart: add a 3rd variable that is categorical.



LINE CHART

Shows: Trends
Data: 2 variables -
chronological & numerical

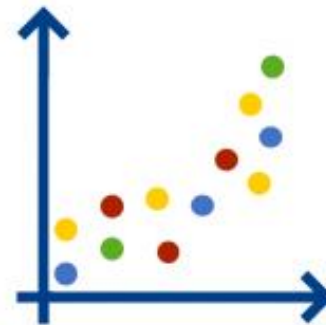
- Line should show a trend across the same sample.
- Axis doesn't have to start at zero, but the line slope should be representative of the trend size.
- To make a chart with multiple lines: add a 3rd variable that is categorical.
- Lines should be solid and easily distinguishable.
- Label lines directly instead of providing a key/legend.
- Avoid more than 4 lines in one chart.



PIE CHART

Shows: Percent to whole
Data: 2 variables -
categorical & percentage

- *Avoid using pie charts at all. If a pie is absolutely needed:
 - All slices must add up to 100% of a variable.
 - Use for only 2-4 slices and when slices have very different sizes.
 - Slices should start at the top and go clockwise largest to smallest, unless there are only two slices.
 - Slices can be different colors, but should be directly labeled and percentages provided.



SCATTER PLOT

Shows: Relationship
Data: 2 variables -
numerical & numerical

- *Great for 1st look at the data.
 - All dots should be the same color, unless you add another variable that is categorical. Then you would distinguish categories by colors.
 - Can also add another variable (ordinal or numerical) that is distinguished by dot size.



Icons adapted from
images by [srip](https://www.flaticon.com) from
[flaticon.com](https://www.flaticon.com)

* the tilt = rotating any text, causing the viewer to tilt their head
the squeeze = trying to fit too many things in a constricted width

FOR ALL VIZZES

Only use a new type of visualization when there is a strong data-related purpose!

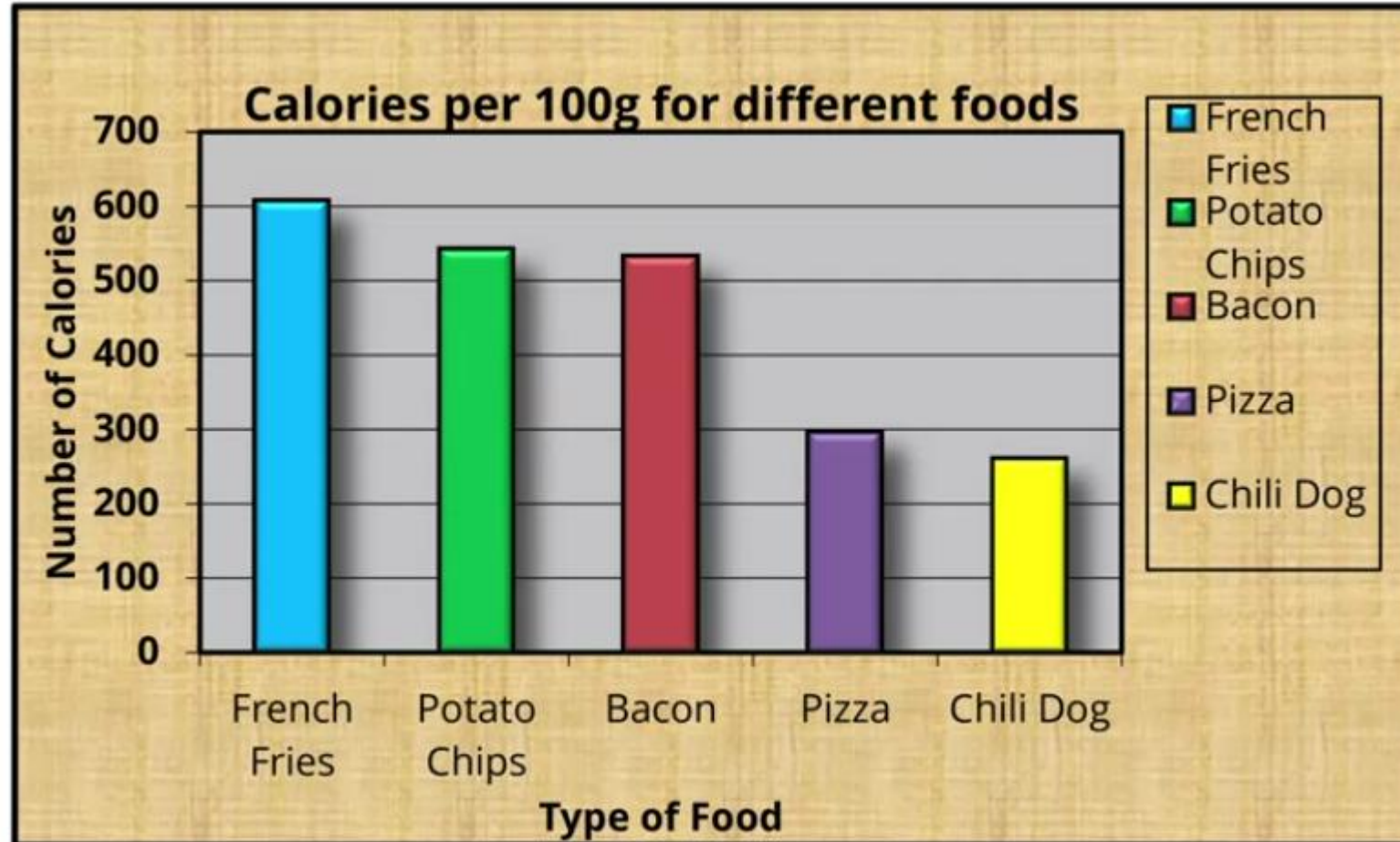
- **Be direct:** Directly label data and categories instead of using a legend.
- **Reduce clutter:** Avoid or minimize extra details, like gridlines, axes & tick marks.
- **Use Gestalt principles:** Use bright color to point something out and greys to push important, but supporting information to the back.
- **Make your point clear:** Use size, color, and borders to draw attention to a subsection of your visualization.
- **Be consistent** with sizes, fonts, and colors.
- **Align meaning:** Use colors that either align with the message and the variables being represented.
- **Be accessible:** Make sure of good contrast & large font. Avoid italics.

There are lots of other charts to choose from, such as area charts, treemaps, butterfly charts, bubble charts, and more! But, it is best to learn the 4 basic charts on the left before determining when to switch to another type.



Data to Ink ratio

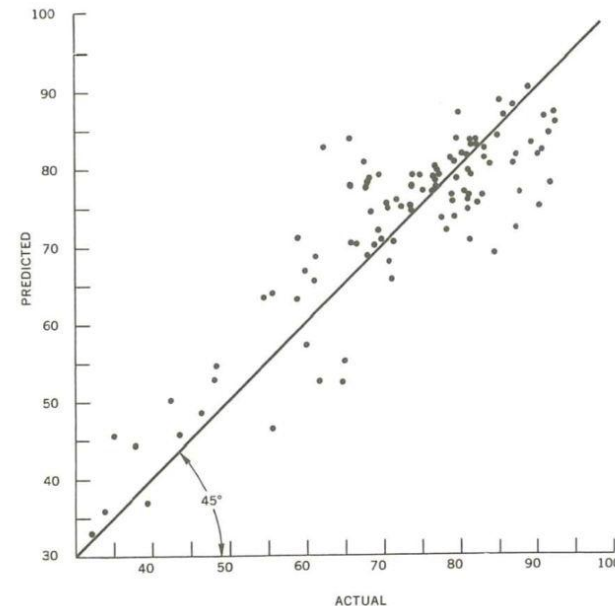
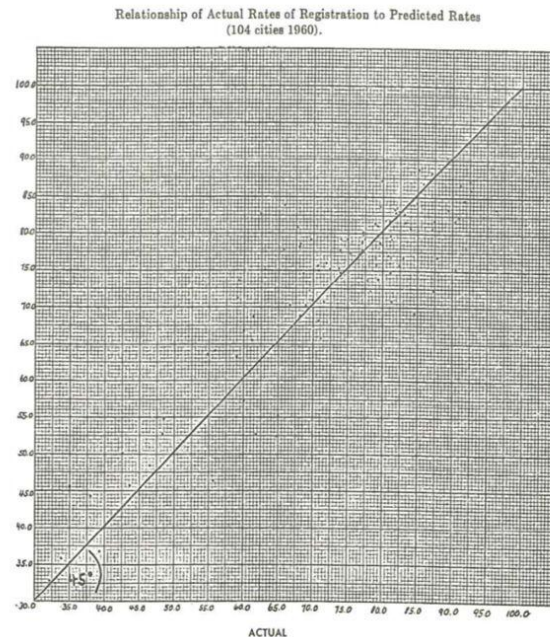
We want to compare calories in bacon with different food.



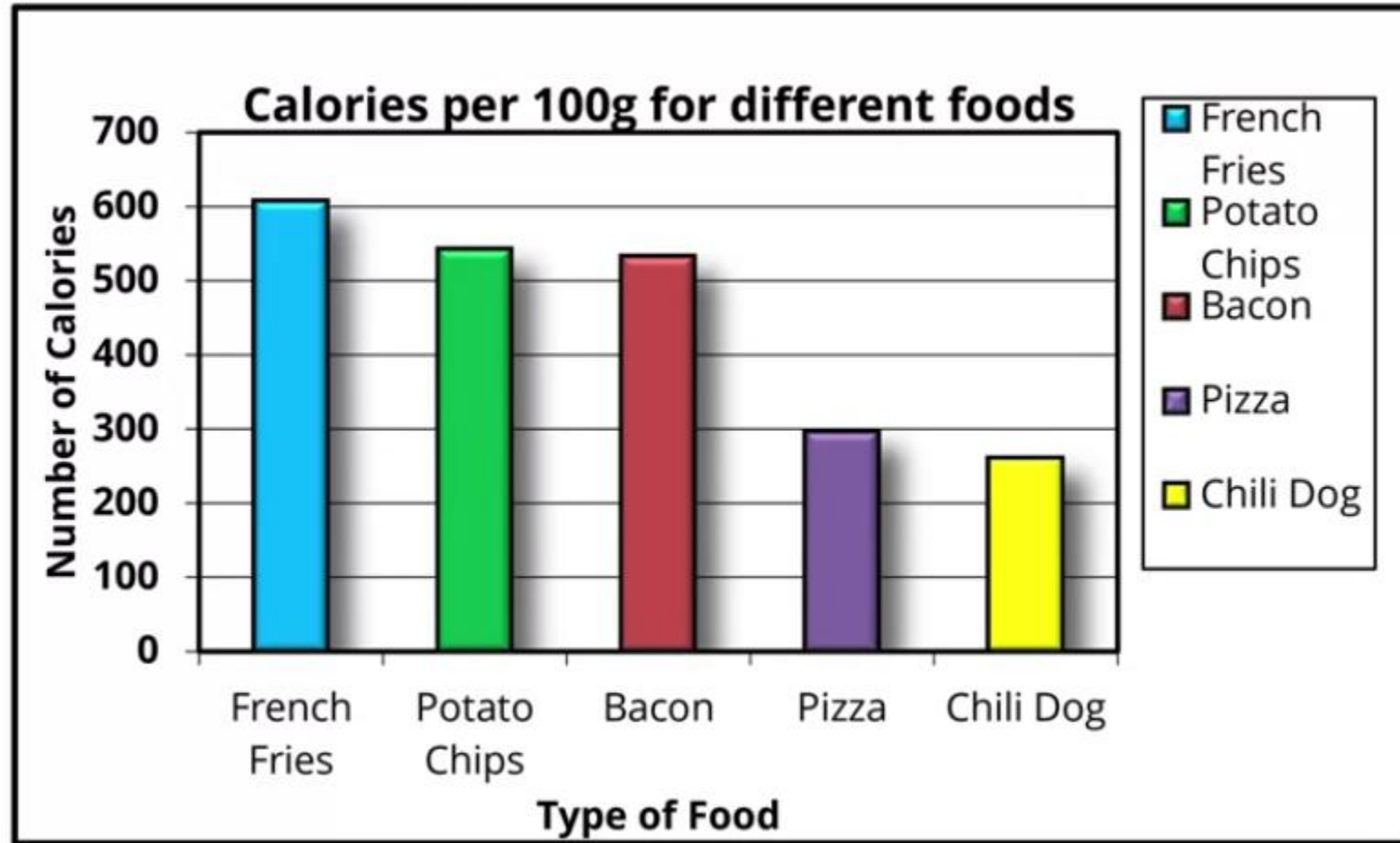
Less is better

Edward Tufte is an author who invented the concept of “**data to ink ratio**”. Overall its concept leads us to identify the **quantity of ink used which really gives us information** in relation to the **total quantity of ink used**.

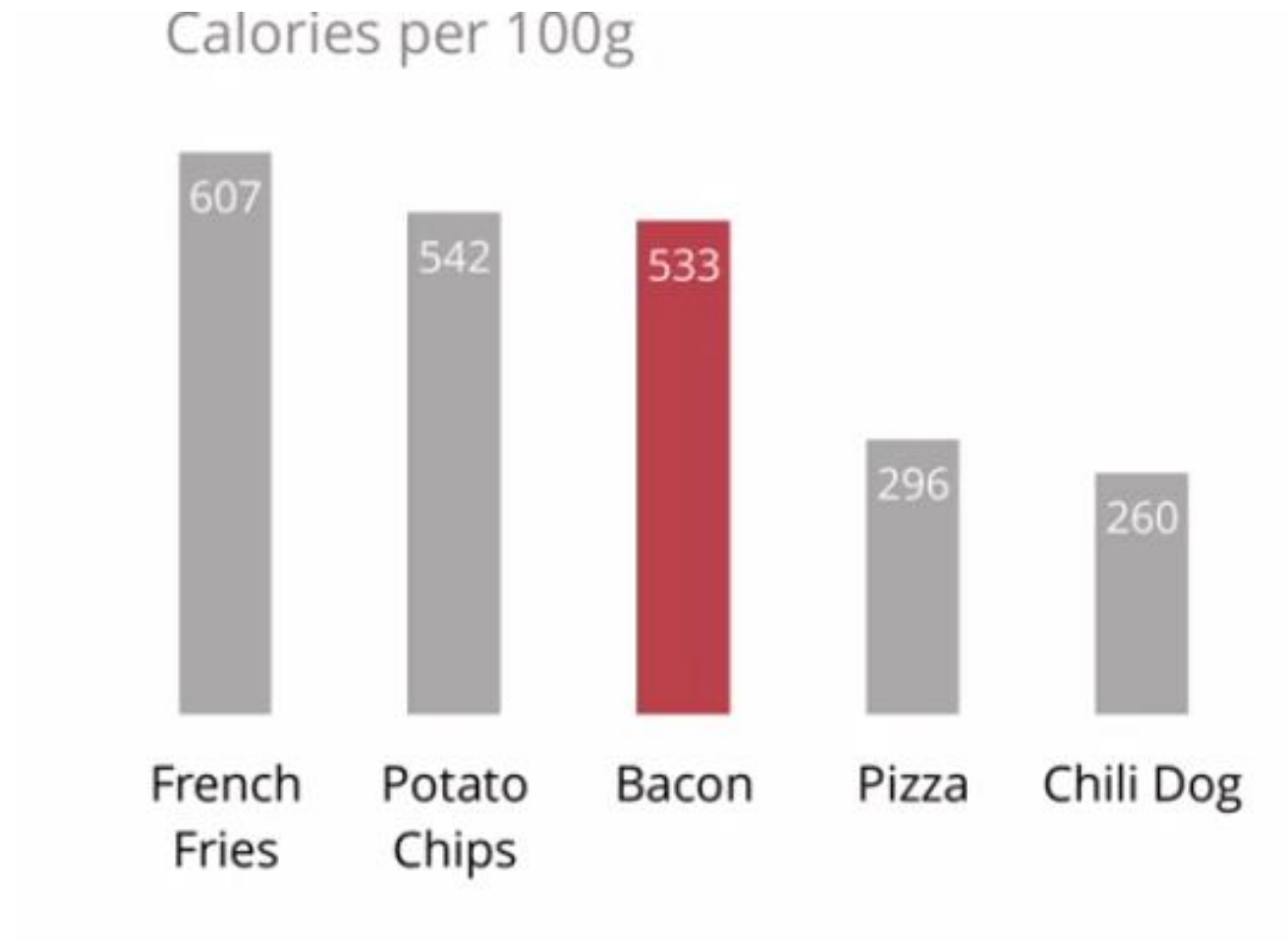
The overall idea is to **reduce redundant and non-essential information** so as not to have visual clutter when reading our graph.



Less is better



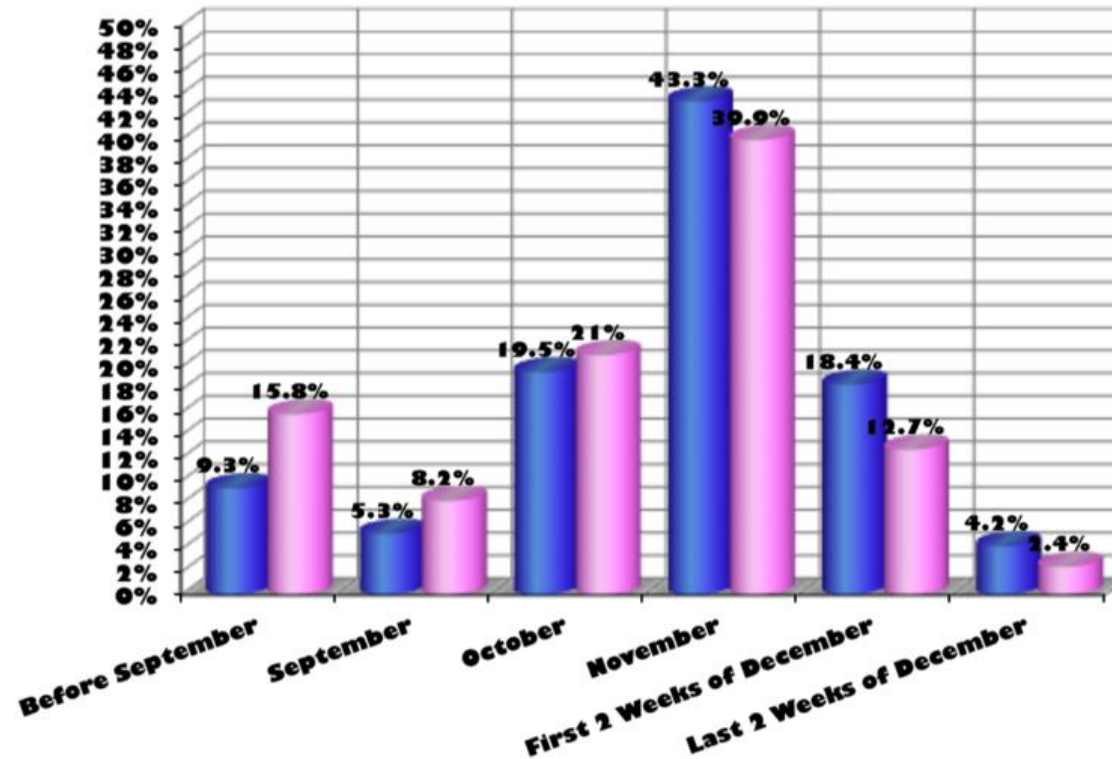
Less is better



Less is better

Shoppers Begins Shopping for Holidays

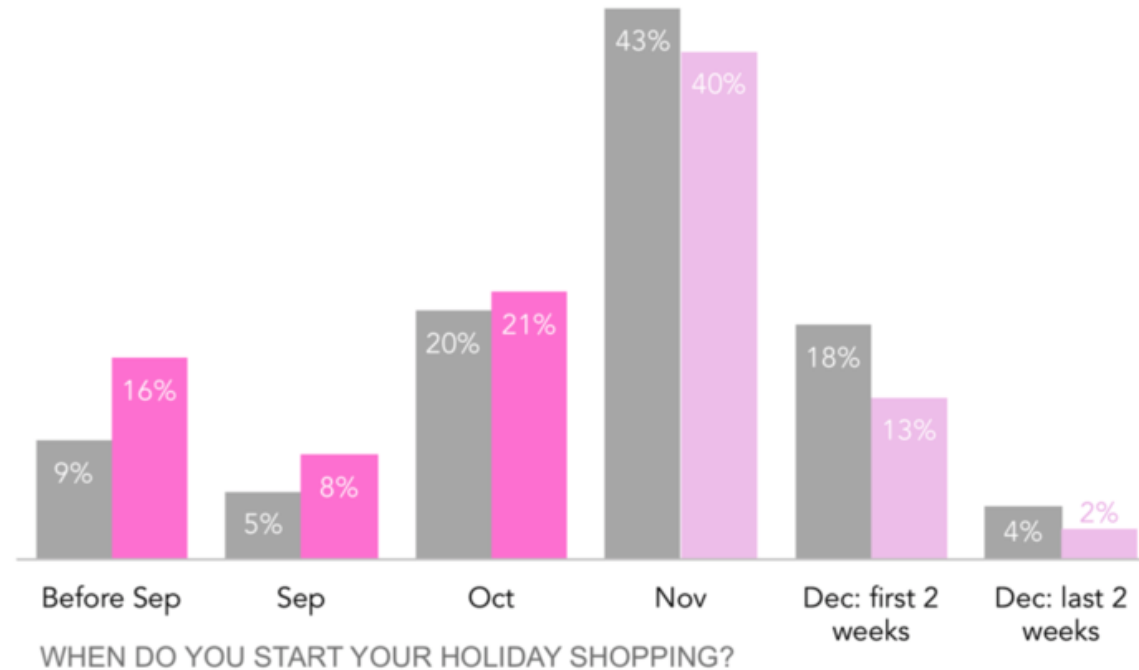
■ Men ■ Women



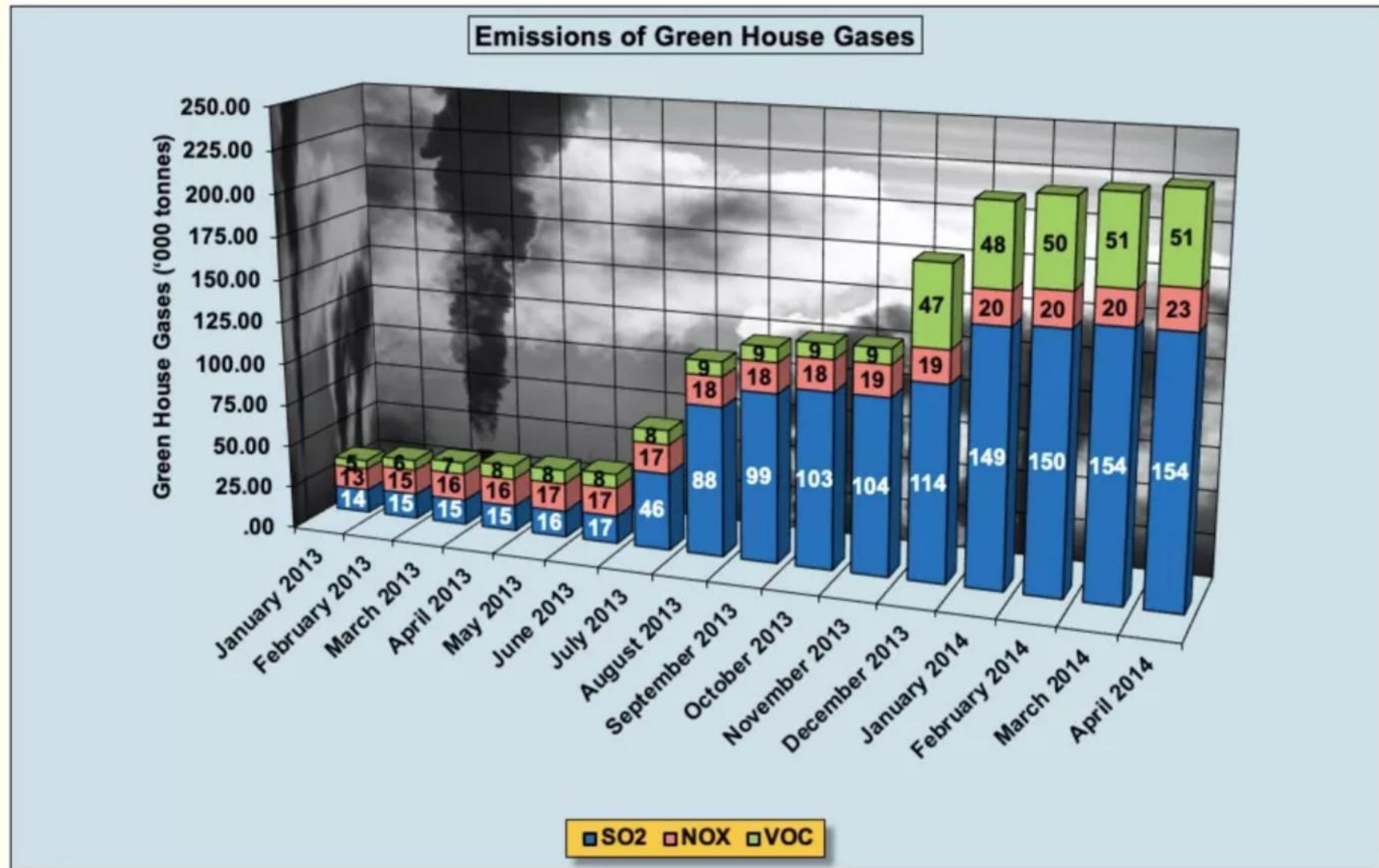
Less is better

More women start their holiday shopping early

■ Men ■ Women
% OF TOTAL



Less is better

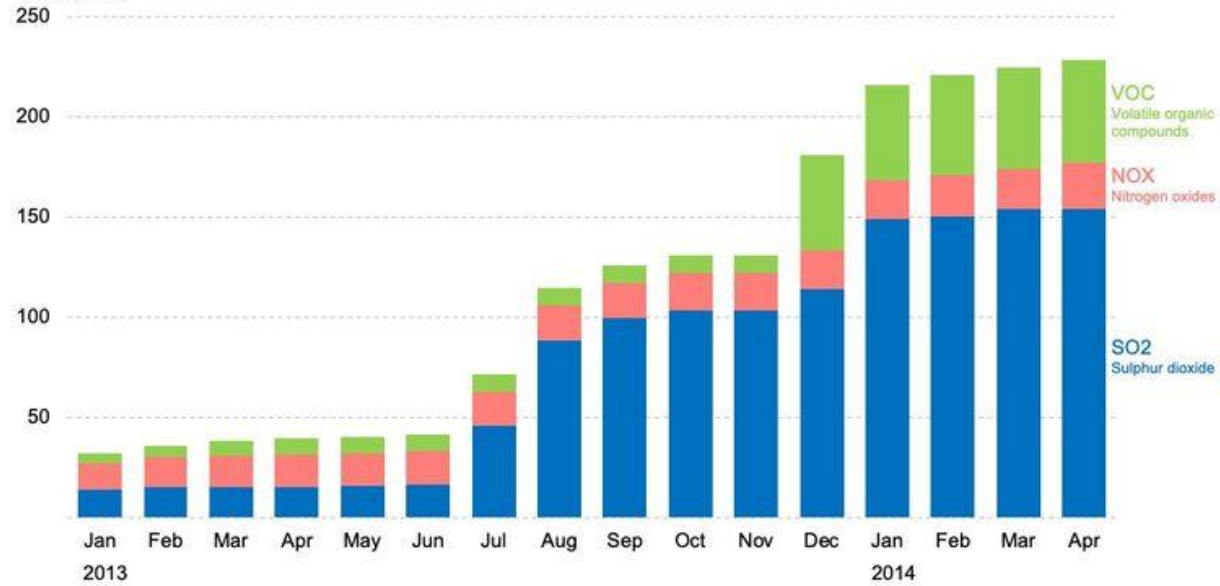


Less is better

Make graph compact

Emissions of Green House Gases

'000 Tonnes



Graph: text

Text in graphs should be chosen thoughtfully.

- The **title** should clearly state what the graph shows — **without interpreting or suggesting a trend**. It should reflect the data shown on the x- and y-axes in a **neutral and objective** manner.
- If you want to comment on patterns or trends, do so in a **subtitle** or in the accompanying **text or figure caption**, not in the main title.
- **legend of the axes and their scale/value is not an option**

Graph: text

Use sans-serif typefaces.

Sparrows
ROBOTO SERIF

↑ Serifs

India: 14.20m
GEORGIA

India: 14.20m
GARAMOND

Sparrows
ROBOTO

India: 14.20m
HELVETICA

India: 14.20m
FUTURA

SERIF FONTS SANS-SERIF FONTS

Use a font with lining and tabular numbers.

124.17
680.90
FP DANCER

124.17
680.90
GEORGIA

124.17
680.90
WORK SANS

124.17
680.90
GARAMOND

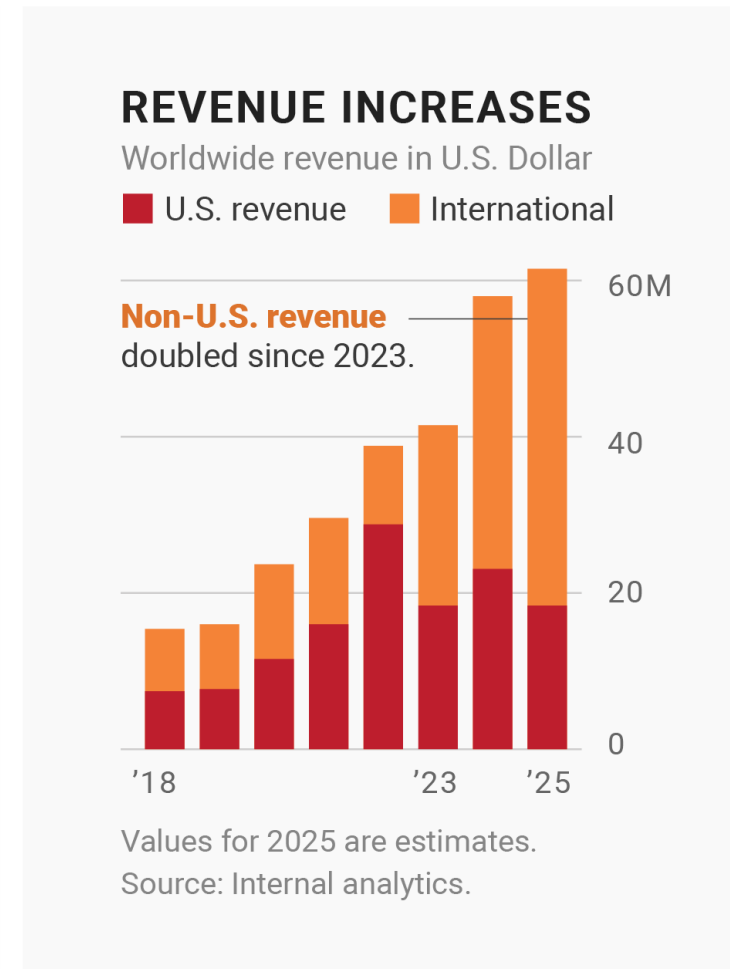
NOT IDEAL: OLDSTYLE BETTER: LINING

Graph: text

Use bold fonts only for emphasis.



NOT IDEAL: EVERYTHING IS BOLD



BETTER: BOLD FOR EMPHASIS

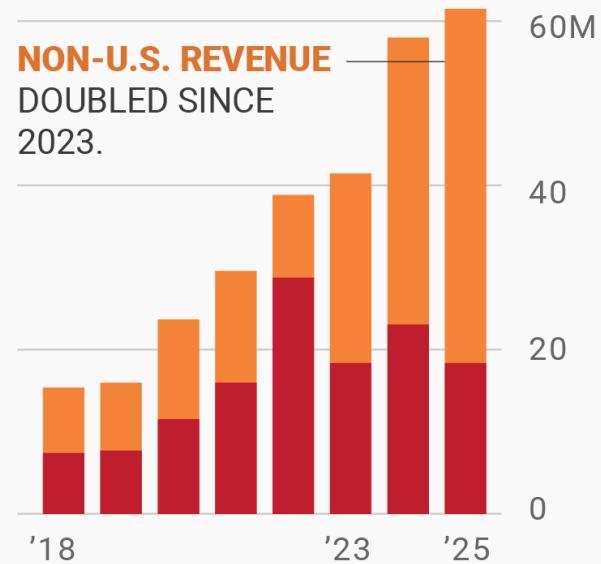
Graph: text

Use uppercase text sparingly.

REVENUE INCREASES

WORLDWIDE REVENUE IN U.S. DOLLAR

■ U.S. REVENUE ■ INTERNATIONAL



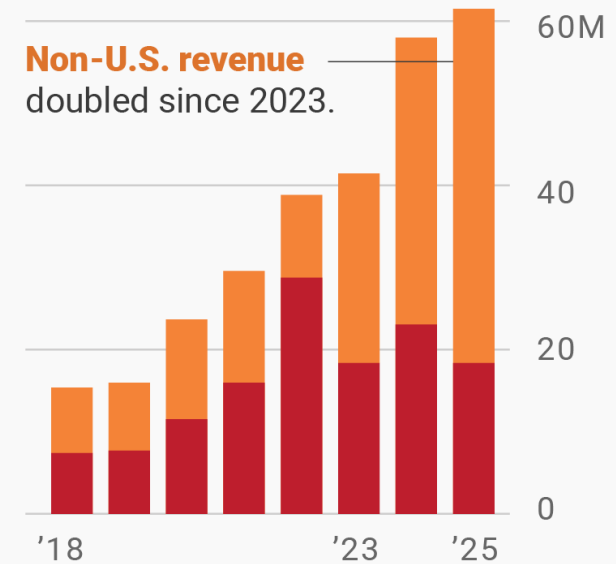
VALUES FOR 2025 ARE ESTIMATES.
SOURCE: INTERNAL ANALYTICS.

NOT IDEAL: UPPERCASE EVERYWHERE

REVENUE INCREASES

Worldwide revenue in U.S. Dollar

■ U.S. revenue ■ International



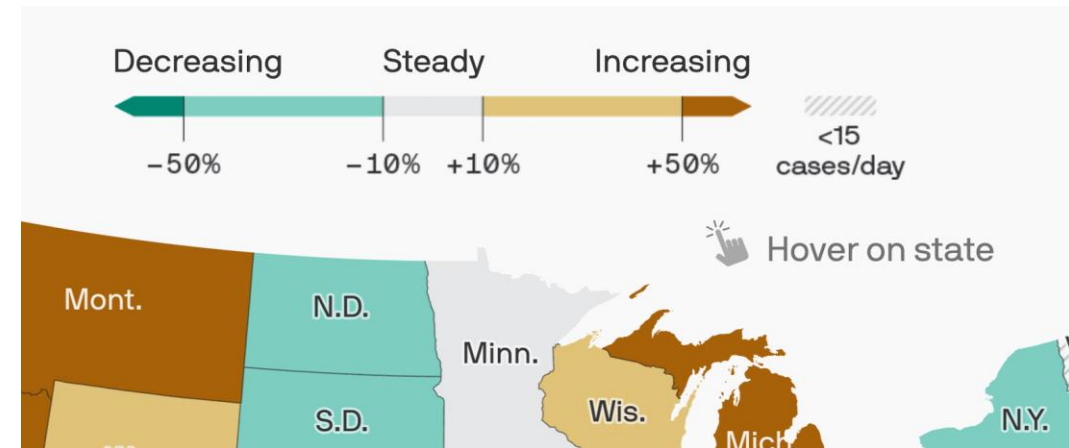
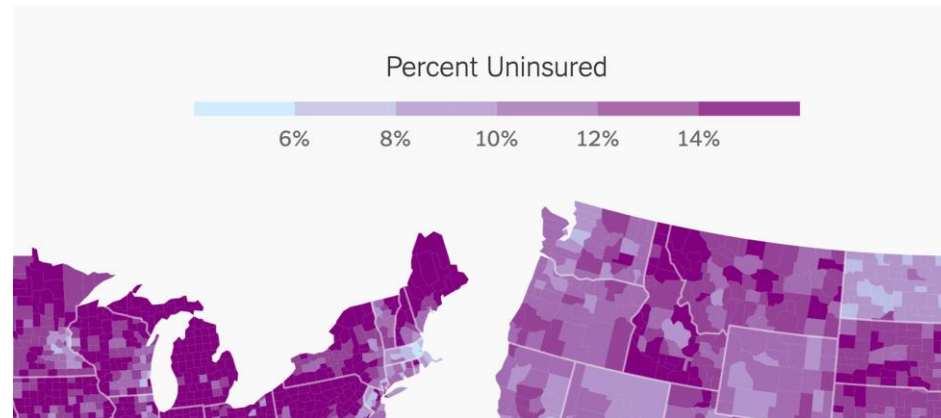
Values for 2025 are estimates.
Source: Internal analytics.

BETTER: UPPERCASE FOR FEW WORDS

Graph: colors

Sequential color scales are gradients that go from bright to dark or the other way round. They're great for visualizing numbers that go from low to high, like income, temperature, or age.

Diverging color scales are the same as sequential color scales – but instead of just going from low to high, they have a bright middle value and then go darker to both ends of the scale in different hues.



Graph: colors

Color gradients can be imprecise and hard to read if not used correctly. The **type of your data** as well as **what you want to show** will help you choose the right gradient shades

- Use a diverging color scale if there's a meaningful middle point
- Use a diverging color scale to emphasize the extremes
- Use a diverging color scale to let readers see more differences in the data
- Use a sequential color scale for a more intuitive reading

QUANTITATIVE COLOR SCALES

SEQUENTIAL & UNCLASSSED



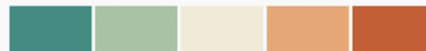
SEQUENTIAL & CLASSSED



DIVERGING & UNCLASSSED

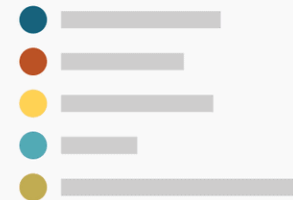


DIVERGING & CLASSSED

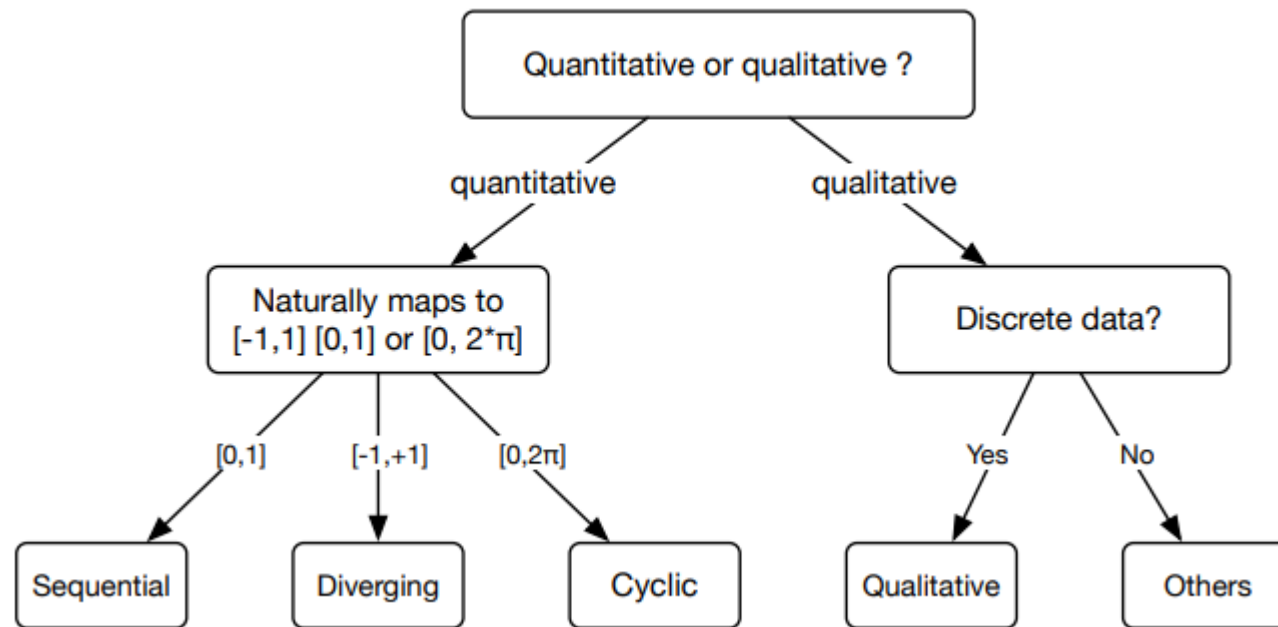


QUALITATIVE COLOR SCALES

CATEGORICAL



Graph: colors



Sequential color palette

- Combination of two sequential palettes with a shared Endpoint sitting at the central value
- Useful when dealing with negative values or ranges that have two extremes with a Baseline in the middle



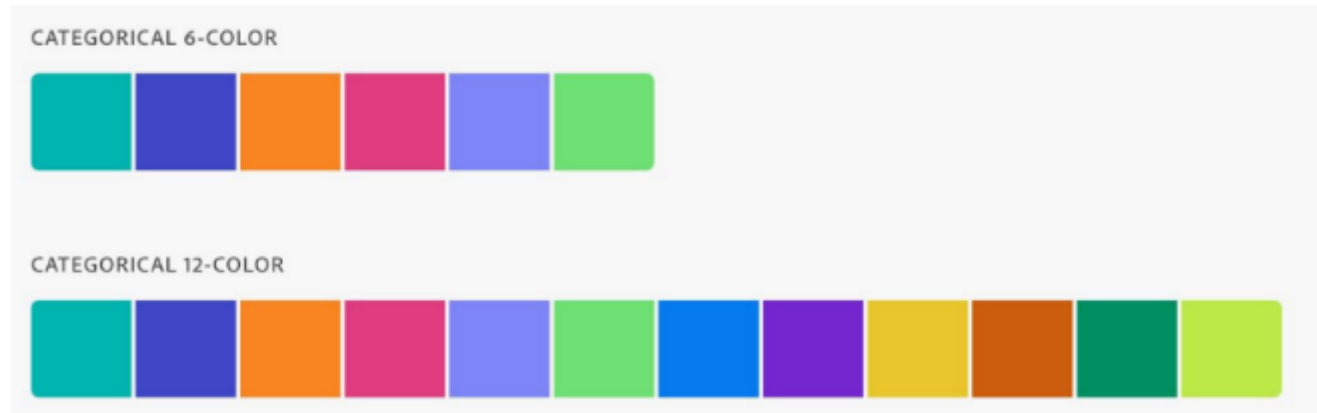
Diverging color palette

- Gradation of colors that go from light to dark
- Used for representing ordered or numeric data



Qualitative color palette

- Variable is categorical
- Suitable for representing data where there is no inherent order or ranking
- Limit the maximum palette size to 10

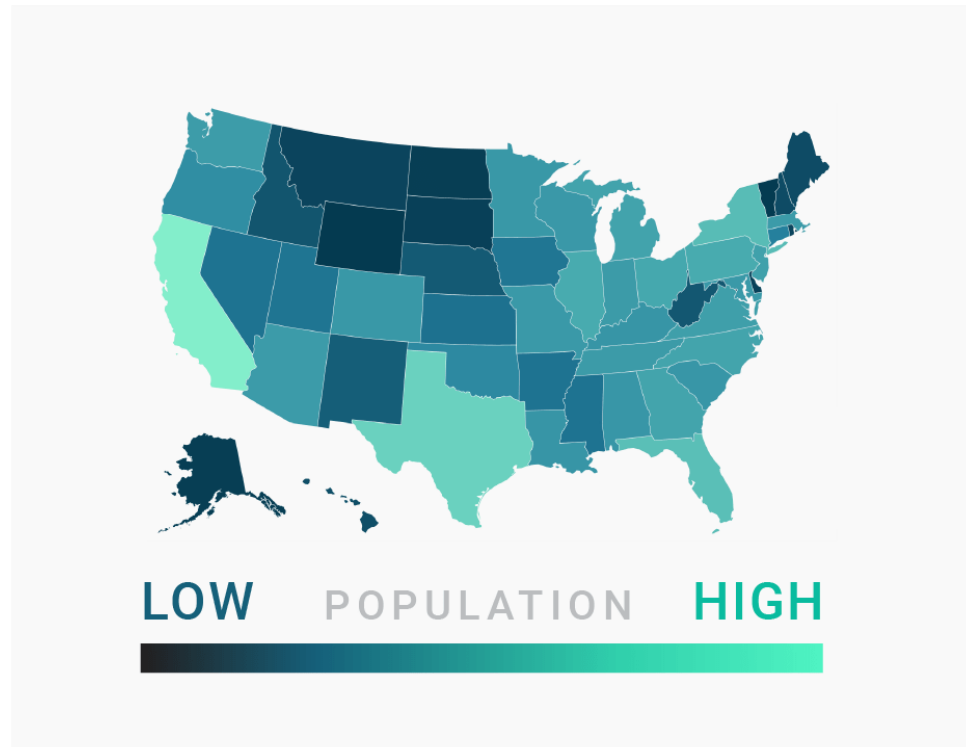


Graph: colors

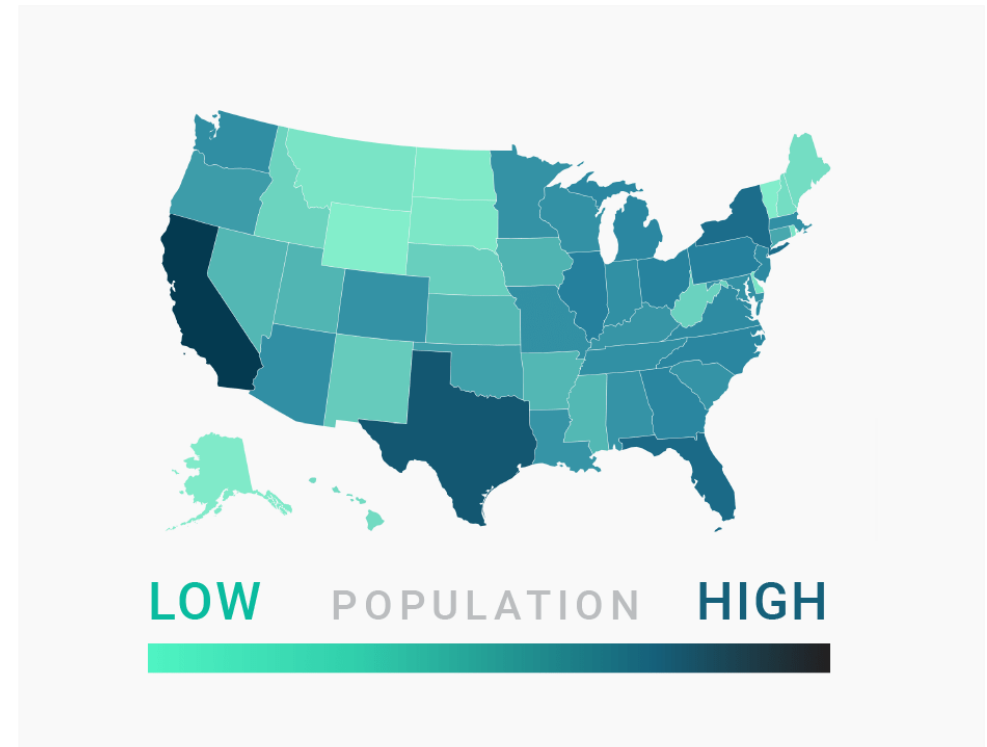


Graph: colors

Use light colors for low values and dark colors for high values



NOT IDEAL



BETTER

Graph: colors

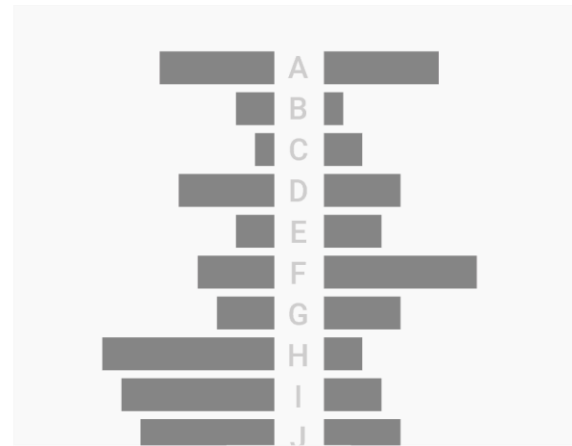


Graph: colors

Avoid to use to much colours



NOT IDEAL



BETTER



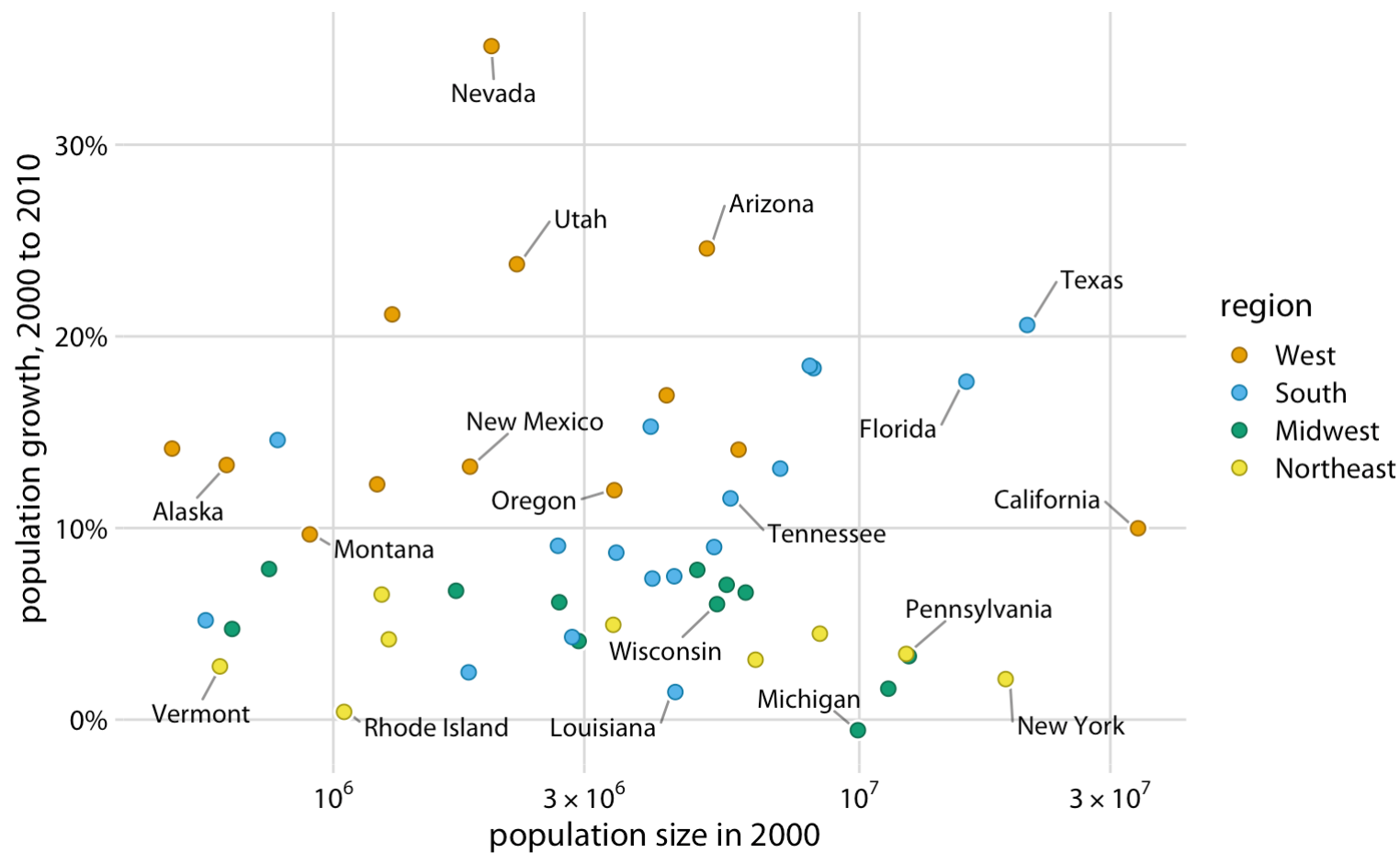
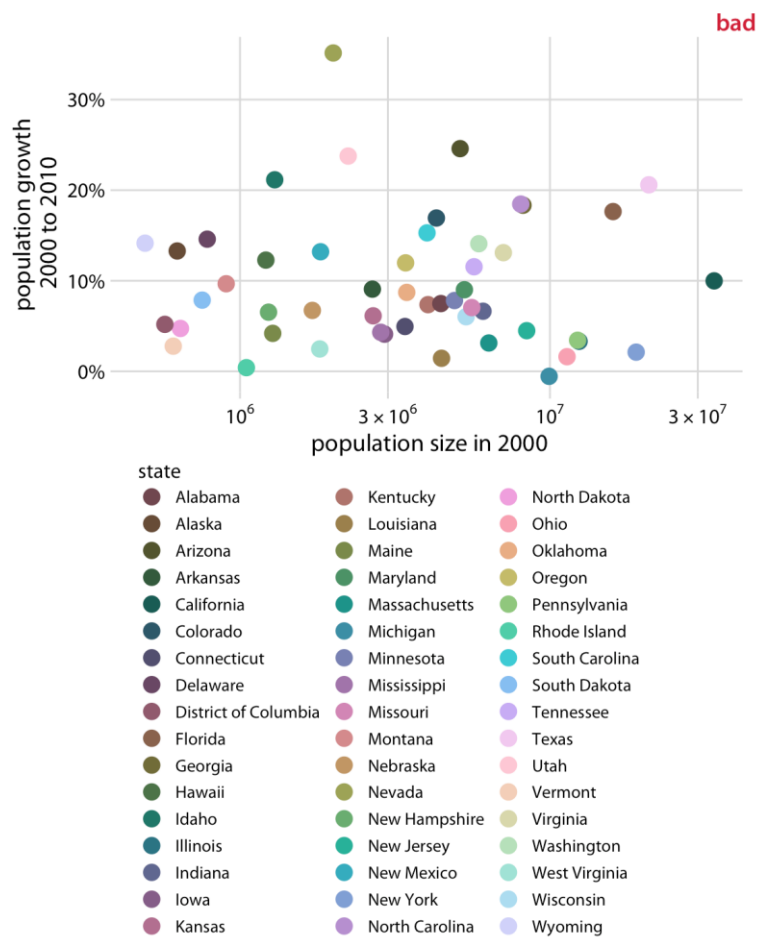
NOT IDEAL



BETTER

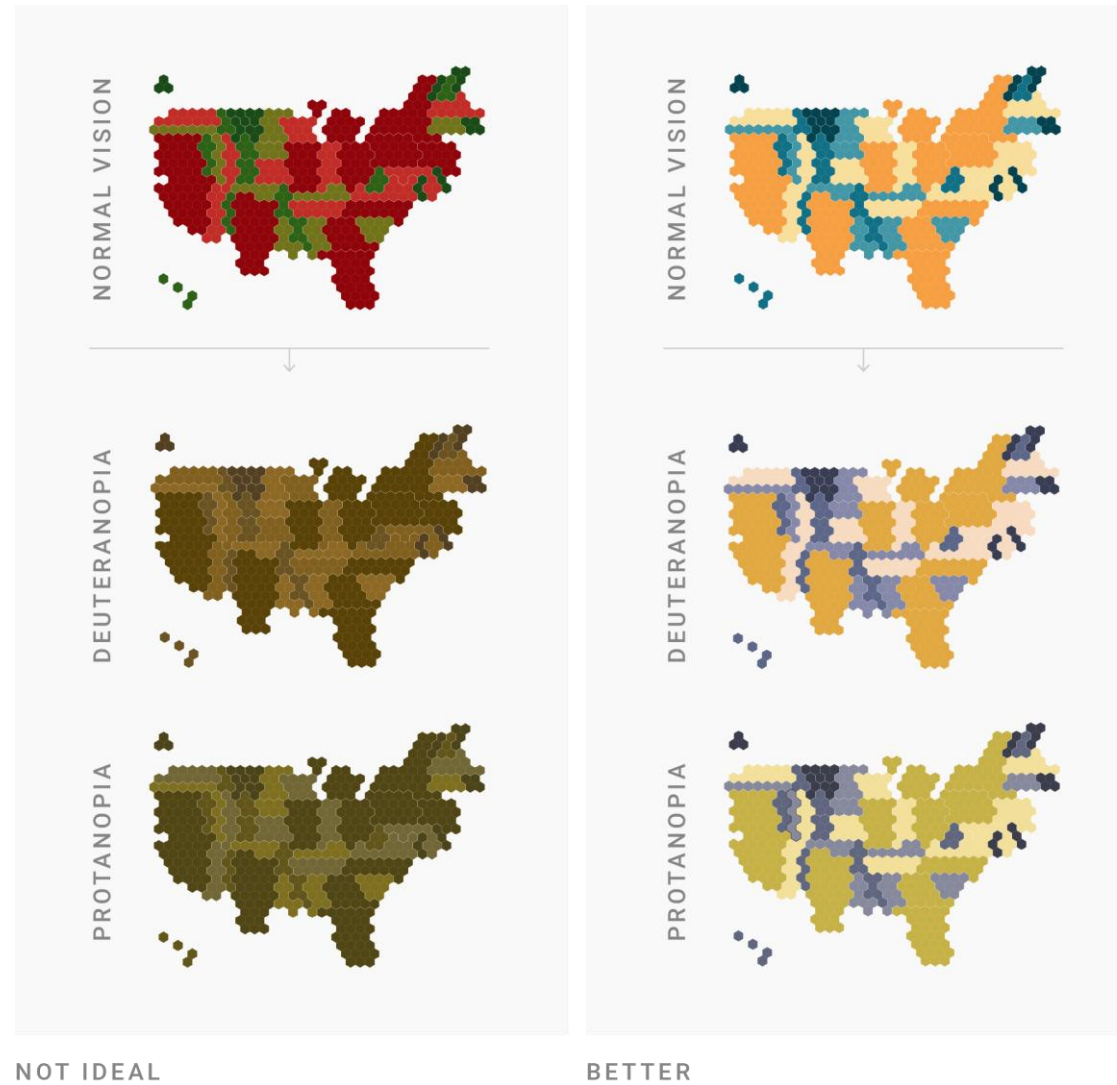
Graph: colors

Avoid to use too much colours



Graph: colors

Consider color-blind people



Information hierarchy

It's essential to consider how the audience will interpret the visuals we create. Knowing the visual hierarchy — which elements will catch attention first — allows us to guide viewers through the data in a logical and meaningful way. This helps ensure a clear message and reduces the risk of unintentional bias.

And you will read this last

You will read this first

And then you will read this

Then this one

Graph: Ethics

Accuracy & Integrity: Visuals must reflect the data truthfully, avoiding distortion or manipulation that could mislead the audience or push a hidden agenda.

Clarity & Simplicity: A good visualization communicates the message clearly, avoiding clutter or overly complex designs that confuse more than they inform.

Objectivity & Fairness: Visualizations should be unbiased and transparent. Clearly stating data sources, methods, and limitations supports fair and responsible interpretation.

Privacy & Responsibility: Always respect the confidentiality of personal or sensitive data. Follow data protection laws and ethical guidelines when sharing information.

Accessibility & Inclusion: Design with everyone in mind — use colorblind-friendly palettes, readable text, and alt text for screen readers. Be aware of cultural and contextual differences in interpretation.

YOUR TURN

- **Select the best graph you find on the web**
- **Explain why you think it is the best**